NFLG 南方路机

TRACKED JAW CRUSHER

NFJ1175

OPERATION MANUAL

Spare Parts Mall

Date of issue： 2015-5-1

Language： Chinese

Revision No： 2015 V1

Serial No：

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NFJ1175 CRUSHER
ENGLISH OPERATION MANUAL

ORIGINAL INSTRUCTIONS

ISSUE NO. 291209/05

OPERATION MANUAL MUST BE WITH THE MACHINE AT ALL TIMES

NFLG

Fujian South Highway Machinery Co.Ltd
No.700, Tiyu Street, High-Tech Industry Zone, Fengze District,
Quanzhou, Fujian,China
Postcode：362000

NFLG reserve the right to change specification without prior notice.

Owners, Users, and Operators:

NFLG appreciates your choice of our product for your application. Our number one priority is user safety which is best achieved by our joint efforts. We feel that you can make a major contribution to safety if you as the equipment users and operators:

1. Comply with EC,OSHA, and Local Regulations.
2. Read, Understand, and Follow the instructions in this and other manuals supplied with this product.
3. Use Good, Safe Work Practices in a common sense way.
4. Only have trained operators — directed by informed and knowledgeable supervision — operating this product.

If there is anything in this manual that is not clear or which you believe should be added, please send your comments to Customer Service, (Tel:0595-22906815)

The safety alert symbol is used to alert you to potential personal injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.

Intended use:
This product and its approved attachments are designed to used in Material Processing applications such as crushing, screening and recycling. Use of this product in any other way is prohibited and contrary to its intended use.

Contents

SECTION 1 : SAFETY ..... SA - 2

1.1 SAFETY INTRODUCTION ..... SA - 2
1.2 HYDRAULIC SAFETY ..... SA - 9
1.3 ELECTRICAL AND ENGINE SAFETY ..... SA - 10
1.4 SAFETY BEFORE AND DURING OPERATION ..... SA - 11
1.5 SAFETY BEFORE AND DURING MAINTENANCE ..... SA - 12
1.6 SAFETY BEFORE AND DURING TRANSPORT ..... SA - 14
1.7 SAFETY SIGNS IN THIS MANUAL ..... SA - 15
1.8 SAFETY SIGNS ON YOUR MACHINE ..... SA - 16
1.9 REMOTE STOPS ON YOUR MACHINE ..... SA - 17
1.10 SAFETY GUARDS ON YOUR MACHINE ..... SA - 18
1.11 NOISE LEVELS AROUND YOUR MACHINE ..... SA - 19

SECTION 2 : TECHNICAL INFORMATION ..... TE - 3

2.1 FAMILIARISATION ..... TE - 3
2.2 KEY COMPONENTS ..... TE - 4
2.3 OPTIONS ..... TE - 6
2.4 TECHNICAL DATA ..... TE - 7
2.5 TRANSPORT POSITION DIMENSIONS ..... TE - 10
2.6 WORKING POSITION DIMENSIONS ..... TE - 11
2.7 CORRECT USE OF MACHINE ..... TE - 12

SECTION 3: INSTALLATION ..... IN - 2

3.1 SAFETY INTRODUCTION ..... IN - 2
3.2 STARTING ENGINE ..... IN - 11
3.3 ENGINE INFORMATION DISPLAY ..... IN - 12
3.5 DETACHABLE WHEELED UNIT (OPTIONAL) ..... IN - 17
3.6 PRIOR TO WELDING ..... IN - 19
3.7 HOPPER EXTENSION FLAPS ..... IN - 20
3.8 PLACING DIRT CONVEYOR IN WORKING POSITION(Optional) ..... IN - 21
3.9 UNFOLDING MAIN CONVEYOR ..... IN - 24
3.10 FITTING ACCESS LADDER ..... IN - 25
3.11 CRUSHER PROTECTION GUARD ..... IN - 26
3.12 DUST SUPPRESSION ..... IN - 27

SECTION 4 : OPERATION ..... OP - 2

4.1 SAFETY INTRODUCTION ..... OP - 2
4.2 SAFETY BEFORE AND DURING OPERATION ..... OP - 11
4.3 PRE-OPERATION CHECK ..... OP - 12
4.4 INITIAL START-UP AND RUNNING IN ..... OP - 13
4.5 BLOCKED CRUSHER ..... OP - 14

SECTION 5 : MAINTENANCE ..... MA - 2

5.1 SAFETY INTRODUCTION ..... MA - 2

5.2 SAFETY BEFORE AND DURING MAINTENANCE ..... MA - 11

5.3 HYDRAULIC SAFETY ..... MA - 13

5.4 ELECTRICAL AND ENGINE SAFETY ..... MA - 14

5.5 HYDRAULIC SYSTEM ..... MA - 15

5.6 HYDRAULIC FLUIDS ..... MA - 16

5.7 FILTRATION ..... MA - 17

5.8 OILTANK, PIPES & FITTINGS ..... MA - 19

5.9 DIESEL ENGINE ..... MA - 20

5.10 CRUSHER ..... MA - 20

5.11 FEEDER ..... MA - 20

5.12 BELT TRACKING ..... MA - 21

5.13 BELT TRACKING MAIN CONVEYOR ..... MA - 22

5.14 BELT TENSIONING ON MAIN CONVEYOR ..... MA - 22

5.15 BELT TRACKING DIRT CONVEYOR (OPTIONAL) ..... MA - 23

5.16 BELT TENSIONING DIRT CONVEYOR (OPTIONAL) ..... MA - 23

5.16 CHANGING HYDRAULIC FILTERS & OIL ..... MA - 24

5.17 CHANGING ENGINE FILTERS & OIL ..... MA - 29

5.18 CHANGING FUEL FILTER ..... MA - 32

5.19 CHANGING GRIZZLY FEEDER GEARBOX OIL ..... MA - 33

5.20 CHANGING GEARBOX OIL IN TRACKS ..... MA - 34

5.21 TRACK ADJUSTMENT ..... MA - 35

5.22 CRUSHER MAINTENANCE ..... MA - 36

5.23 CHANGING FEEDER MESH ..... MA - 39

5.24 CHANGING MESH ON PRE-SCREEN (OPTION) ..... MA - 41

5.25 RECOMMENDED LUBRICANTS ..... MA - 43

5.26 GREASING SCHEDULE ..... MA - 45

5.27 HYDRAULIC RAM MAINTENANCE ..... MA - 46

5.28 SAFELY REMOVE & INSTALL BATTERY ..... MA - 47

5.29 MAINTENANCE SCHEDULES ..... MA - 50

SECTION 6 : TRANSPORT ..... TR - 2
6.1 SAFETY INTRODUCTION ..... TR - 2
6.2 SAFETY BEFORE AND DURING TRANSPORT ..... TR - 11
6.3 STOPPING THE MACHINE ..... TR - 13
6.4 PREPARING FOR TRANSPORT ..... TR - 13
6.5 CRUSHER PROTECTION GUARD ..... TR - 14
6.6 ACCESS LADDER ..... TR - 15
6.7 FOLDING MAIN CONVEYOR ..... TR - 16
6.8 PLACING DIRT CONVEYOR IN TRANSPORT POSITION (Optional) ..... TR - 17
6.9 PLACING EXTENDED DIRT CONV IN TRANSPORT POSITION ..... TR - 18
6.10 HOPPER EXTENSION FLAPS ..... TR - 20
6.11 DETACHABLE WHEELED UNIT (OPTIONAL) ..... TR - 21
6.12 READY FOR TRANSPORT ..... TR - 23
6.13 PRIOR TO HAULAGE: ..... TR - 23

SECTION 7 : Tracked Jaw Crusher Control system.

CONTENTS:

1.1 SAFETY INTRODUCTION ..... SA - 2

1.2 SAFETY SIGNS ON YOUR MACHINE ..... SA - 19

1.3 WORK AREA SAFETY ..... SA - 24

1.4 HYDRAULIC SAFETY ..... SA - 25

1.5 ELECTRICAL AND ENGINE SAFETY ..... SA - 26

1.6 SAFETY BEFORE AND DURING OPERATION ..... SA - 27

1.7 SAFETY BEFORE AND DURING MAINTENANCE ..... SA - 28

1.8 SAFETY BEFORE AND DURING TRANSPORT ..... SA - 30

1.9 INFORMATION SIGNS ..... SA - 31

1.10 REMOTE STOPS ON YOUR MACHINE ..... SA - 37

1.11 SAFETY GUARDS ON YOUR MACHINE ..... SA - 38

1.12 NOISE LEVELS AROUND YOUR MACHINE ..... SA - 39

SECTION 1 : SAFETY

1.1 SAFETY INTRODUCTION

1.1.1 Personal Safety

Operators should be trained in the correct and safe use of all equipment.

Before operation the operator(s) must:-

(i) Have received specific and adequate training in the task to be carried out.

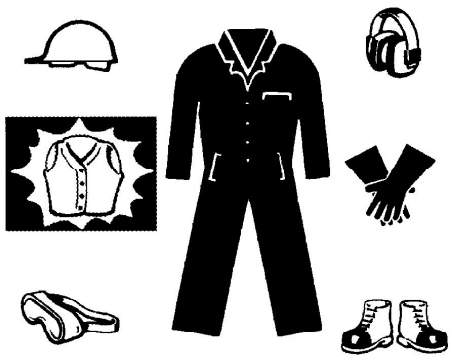
(ii) Have read and understood the Operators Manual and the Safety Signs in this manual and around the machine(s).

(iii) Knows the location and function of controls and safety features such as emergency stop buttons and safety guards.

(iv) Be aware of all moving parts of the machine.

(v) Always wear correctly fitting (EN/ANSI approved) protective clothing.

WARNING



1. Loose or baggy clothing can get caught in running machinery.

2. Protective clothing includes Hard Hat, Safety Glasses, Ear Defenders, Overalls, Gloves, Steel Toed Boots and a High Visibilty Vest.

1.1.2 General Safety

Overleaf are the safety symbols used on your machine, a detailed explanation of each symbol is outlined. Take time to familiarise yourself with the symbols and explanations. The safety symbols in this section cover the complete *NFLG* product range, there-fore every symbol explained may not apply to the your machine.

Safety signs throughout this manual are identified by the safety alert symbol. When you see this symbol be alert to the possibility of personal injury or death.

Please take time to familiarise yourself with each of these signs and what they mean.
Always follow the instructions in the Safety message.

⚠ DANGER

This indicates an imminently hazardous situation which if not avoided, **will** result in death or serious injury.

⚠ WARNING

This indicates a potentially hazardous situation which, if not avoided, **could** result in death or serious injury.

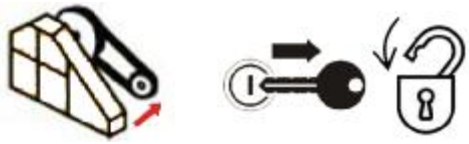
⚠ CAUTION

This indicates a potentially hazardous situation which, if not avoided, **may** result in minor or moderate injury.
It may also be used to alert against unsafe practices.

⚠ IMPORTANT

This indicates a statement of company policy, which if not adhered to **will** result in damage to your machine and/or other peoples property.

DANGER



1. Entanglement Hazard.
   Contact with moving belt will result in serious injury or death.
2. Stay clear of moving belt.
3. Do not operate this machine without all guards and covers in place.
4. Switch off and lockout machine before adjusting or servicing .

DANGER

1. Injection Hazard.
   Beware of escaping Hydraulic oil. High pressure hydraulic oil can penetrate the skin causing serious injuries.
2. Always relieve pressure from the hydraulic system before disconnecting hydraulic lines, attempting maintenance or adjustments. Use a piece of cardboard to check for oil leaks.
3. If oil is injected under the skin, get medical help immediately.
4. Read manual prior to operation

DANGER

1. Electrocution Hazard.
   Risk of serious injury or death will result from contact with electrical power lines.

2. Beware of overhead cables when folding main conveyor

3. Maintain required clearance

Line Voltage	Required Clearance
0 - 50kW	10 ft (3.0M)
50 - 200kW	15ft (4.6M)
200 - 350kW	20ft (6.1M)
350 - 500kW	25ft (7.6M)
500 - 750kW	35ft (10.7M)
750 - 1000kW	45ft (13.7M)

4. Contact electric powerline owner, Electric power shall be disconnected or powerline moved or insulated before machine operation

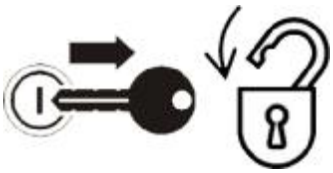
DANGER

1. Electrocution Hazard
   Contact with hazardous voltage will result in serious injury or death.

2. Disconnect, Lockout power before servicing.
   Only authorized personnel may service this equipment.

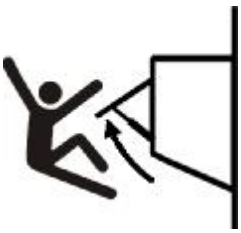
3. Read all manuals prior to operation.

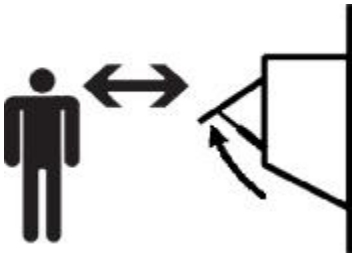
DANGER



1. Entanglement Hazard.
   Do not reach into an unguarded machine.Contact with shredder will result in death or serious injury.
2. In-running nip points can cause serious injury or even death. Your arm could be pulled in and amputated.
3. Switch off and lockout machine before opening or removing guard doors.

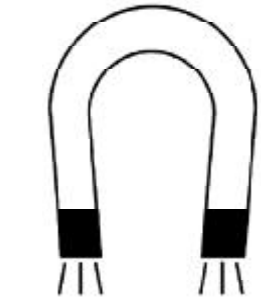
WARNING





1. Impact Hazard.
   Contact with spring loaded doors can result in serious injury.
2. Keep head and face away from door. Keep hand firmly on door until it is completly open.

DANGER



1. Magnetic Hazard.
   Risk of serious injury or death.

2. Magnetic fields can disturb pacemaker operation if within 3 metres of the magnet and result in serious injury or death to pacemaker wearer.

3. Keep Clear from Magnet.

DANGER

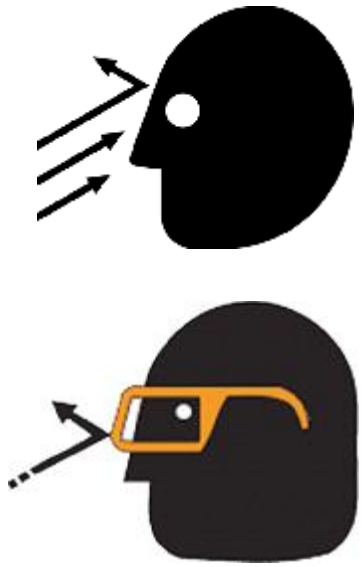
1. Flying material Hazard.
   Risk of serious injury or death.

2. Platform is for Maintenance purposes only.

3. If Flying material hits the head or body it can cause serious injury or death .

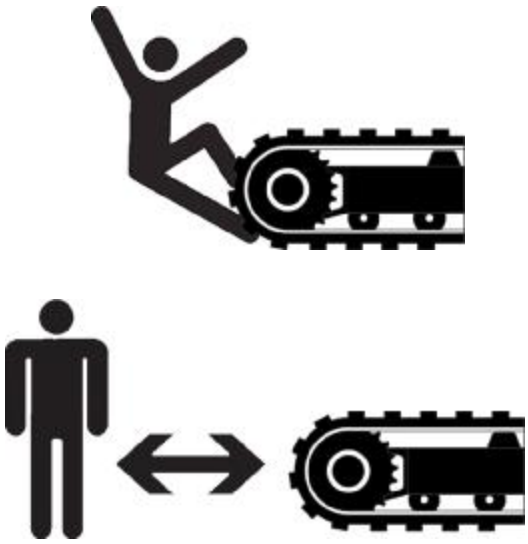
4. Keep Clear. Do not use platform when plant is in operation.

DANGER



1. Flying material Hazard.
   Risk of serious injury or death.
2. Flying material can cause loss or degradation of eye sight.
3. Wear proper eye personal protective equipment.

DANGER



1. Crush Hazard.
   Risk of serious injury or death.
2. Death or serious injury can result from contact with the Track frames.
3. Keep clear of Track frames when the unit is in operation.

WARNING

1. Improper operation or maintenance can result in serious injury or death.

2. Read and understand the operators manual before using or maintaining machine.

2. Read and understand safety signs before using or maintaining machine .

3. If you do not understand the information in the manuals, consult your supervisor, the owner or the manufacturer.

WARNING

1. Burn Hazard.
   Contact with hot surfaces can result in serious injury or death

2. KEEP CLEAR . Allow surfaces to cool before servicing

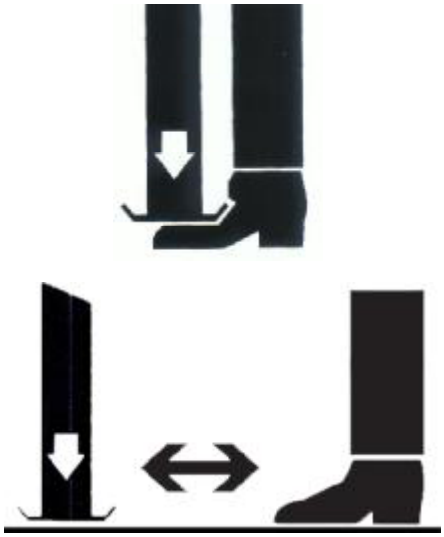
WARNING

1. Falling Material Hazard.

2. Do not walk near material discharge areas.
   You are at risk of serious injury or death.

3. Keep Clear of hopper during operation

WARNING

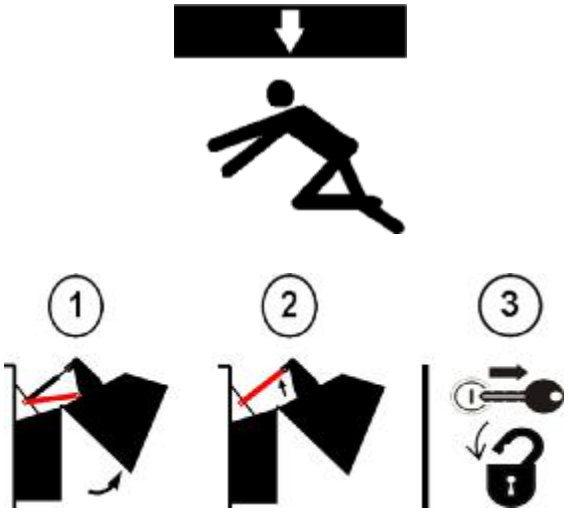


1. Crush Hazard.
   Beware of dangerous Hand and Foot crush points.

2. Do not use any body parts e.g. hands or feet to check hole alignment or support machinery.

3. Always use pins and/or mechanical supports provided.

WARNING



Crushing Hazard.
Unsupported crusher can crush you.
Death or serious injury can result.

1. Raise crusher.
2. Connect red safety bar.
3. Switch off and Lockout machine before carrying out any maintenance work under the shredder.

WARNING

1. Crush Hazard.
Engine inspection cover can crush..

2. death or serious injury can result.

3. Install support before working under or around engine inspection cover.

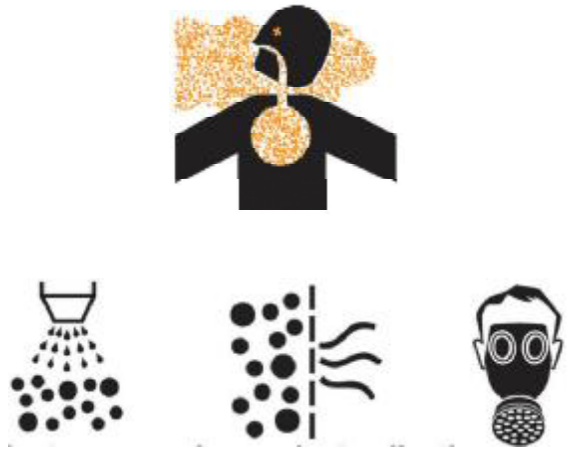
WARNING

Automatic Startup
Conveyors and related equipment may start at any time without warning.
SERIOUS INJURY or DEATH could result.

Unauthorized persons prohibited.

Switch off, and lockout all energy sources prior to performing maintenance or servicing.

WARNING

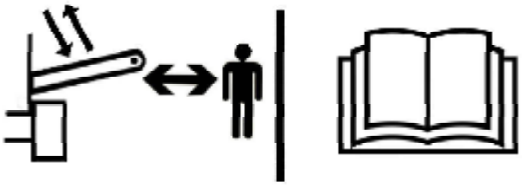


INHALATION HAZARD
Death, serious injury or delayed lung disease may result from breathing hazardous dust.

Use dust suppression or dust collection equipment to minimise the dust exposure during operation of this machine. Use approved respiratory protection to avoid inhalation of dusts, when required by Federal, State, and local safety health regulations. Contact your employer to establish whether these regulations require that you use respiratory protection.

WARNING

Crushing Hazard.
Contract with folding conveyors can result in death or serious injury.



Keep clear of conveyors when folding for transport or opening to working position.

Read all manuals prior to operation.

WARNING

Explosion Hazard.
May cause severe injury or death due to ignition of explosive gases.

Keep welding sparks away from the battery area. Remove battery if required.

WARNING

Electrocution Hazard.
Contact with hazardous voltage will result in serious injury or death.

Disconnect, lockout power before servicing.

Only authorizd personnel may service this equipment.

Read all manuals prior to operation.

WARNING

Hearing Hazard.
Exposure to loud noise could result in serious hearing loss.

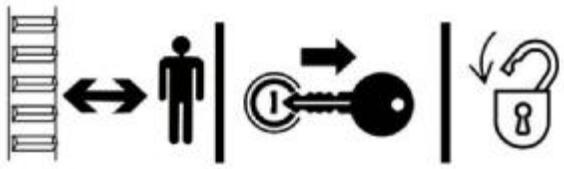
Wear proper hearing personal protective equipment.

WARNING

Explosion / Burn Hazard.
Death, burns, blindness or serious injury can result from release of pressurized liquids or due to ignition of explosive gases and corrosive acid.

Keep all open flames and sparks away. Wear personal protective equipment, including shield, gloves and long sleeve shirt. Consult operation and maintenance manual for jump starting instructions. Read all manuals prior to operation.

WARNING



Entanglement Hazard.
Exposed buckets can cause severe injury or death.

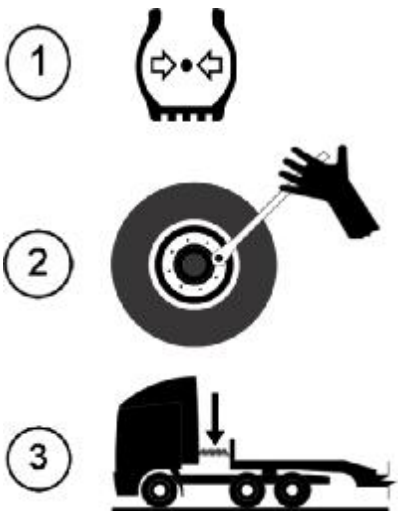
Keep clear of exposed buckets.

Switch off and lockout machine before opening, removing guards or servicing.

WARNING

1. Fall Hazard.
   Falling from this machine can result in serious injury or even death.
2. Do not Climb on Machine.
3. Always use a suitable lifting platform to service the machine.

WARNING



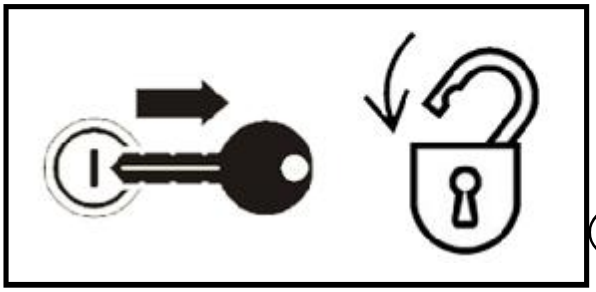
Vehicle Hazard.
Vehicle control can be affected by improper setup of vehicle.

1. Check tyre pressures.
2. Check wheel nut torque. Recheck wheel nut torque every 150 miles (200km).
3. Connect and check braking system.

Lockout procedure.

Lockout is a procedure that is designed to prevent the unexpected or accidental startup of equipment and to alert all workers whenever it is unsafe to operate any piece of equipment. When used as intended, Lockout also protects personnel from energy stored in devices such as springs, accumulators, batteries, hydraulic systems, etc.
When the symbols below appear on a safety sign it indicates that the machine must be switched off and locked out before maintenance can be carried out.

Starter Panel



Isolator Switch "ON" Machine Operational

The Isolator Switch *(ItemB)* is located beside the battery box, the switch cuts the electric supply from the battery ensuring the machine cannot be operated. The machine must always be isolated prior to any maintenance work being carried out.

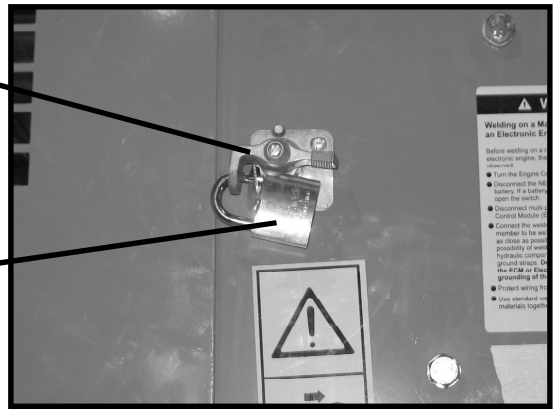
To Lockout:

(i) Turn off the Engine at Starter Panel and remove key *(Item A)*

(i) Rotate the switch *(item B)* to the "OFF" position *(Item C).*

(ii) Attach a padlock *(Item D)* to secure in the lockout position.

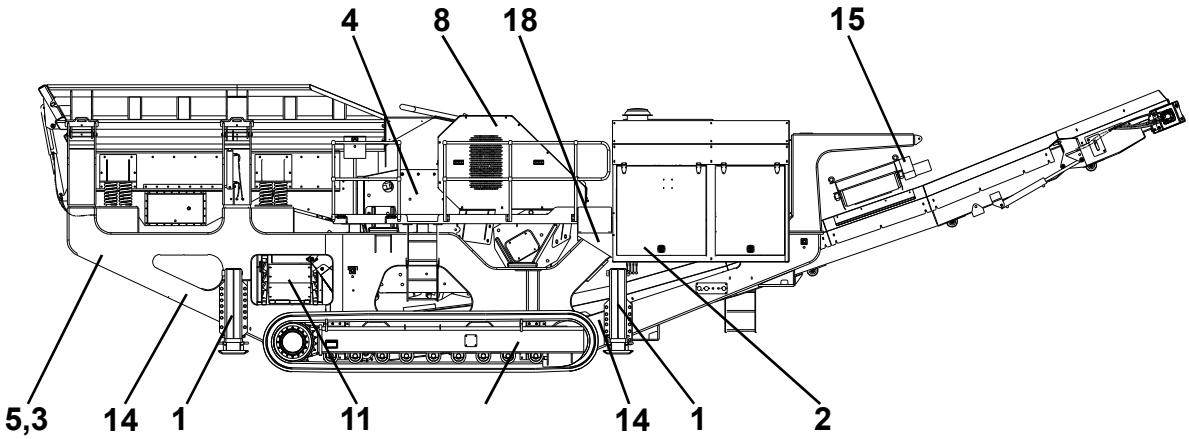
(iii) During maintenance work always ensure the isolator is locked so the machine cannot be started accidently.

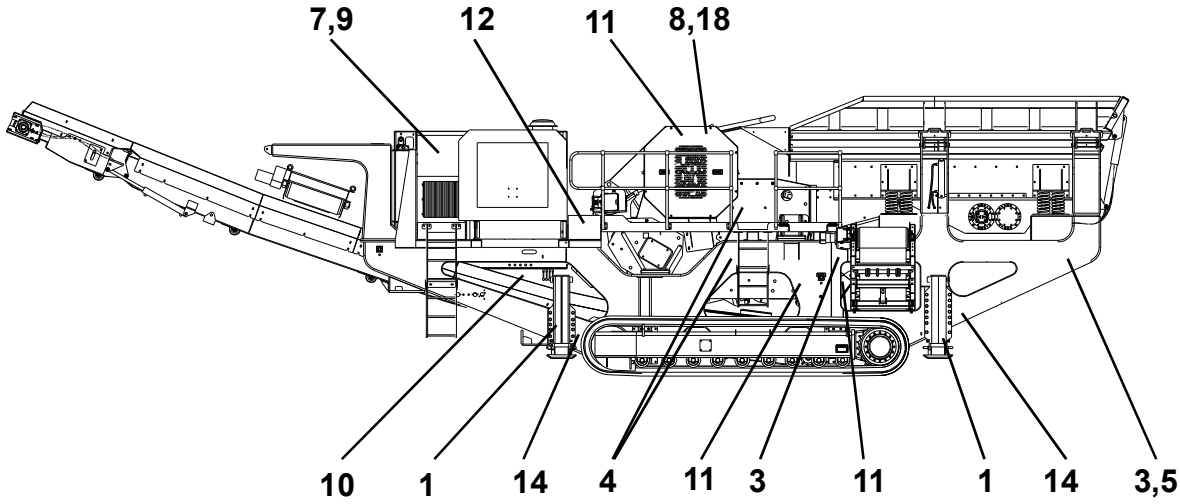


Isolator Switch "OFF" and Locked out Machine cannot operate

1.2 SAFETY SIGNS ON YOUR MACHINE

Safety Signs are important features of your *NFLG* machine. Please take time to familiarise yourself with their meaning and locations on the machine as seen below.



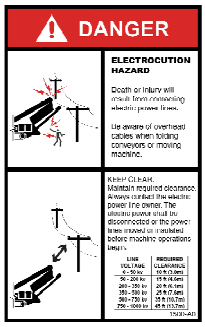


IMPORTANT

ANY DAMAGED, MISSING OR PAINTED OVER SAFETY SIGNS MUST BE REPLACED IMMEDIATELY

ANSI Safety Signs (used in USA, AUSTRALIA, N. ZEALAND CANADA)

1.



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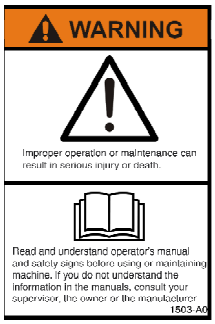

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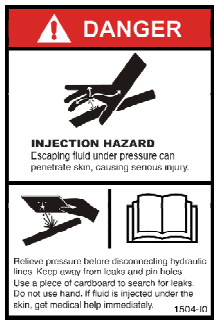

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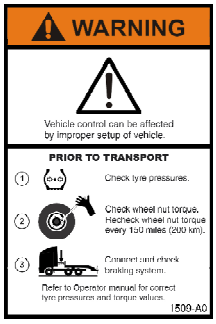

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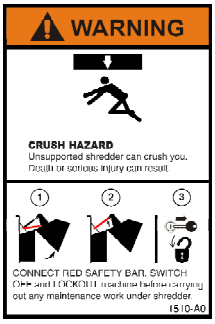

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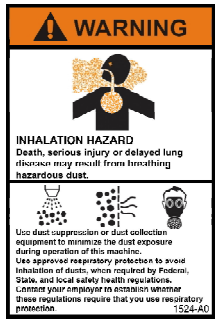

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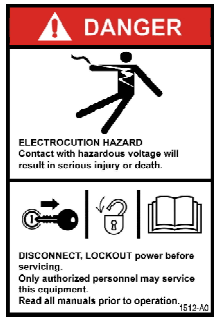

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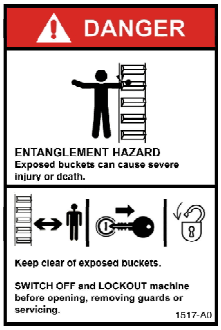

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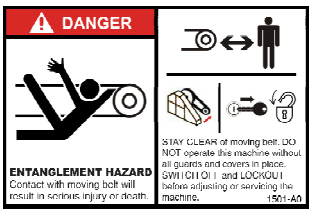

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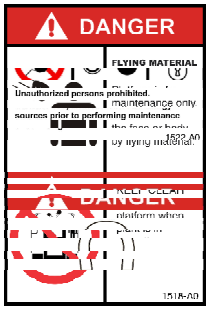

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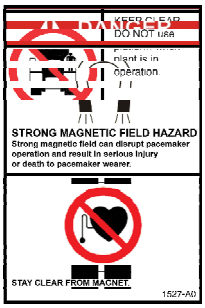

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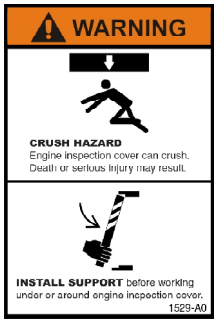

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ISO Safety Signs- (Yellow Background, Black Graphics)(INTERNATIONAL)

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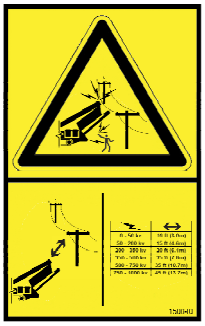

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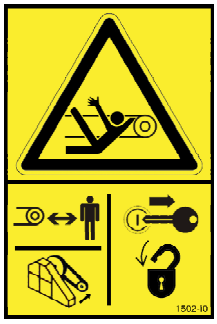

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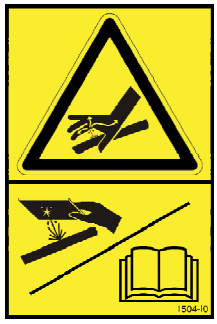

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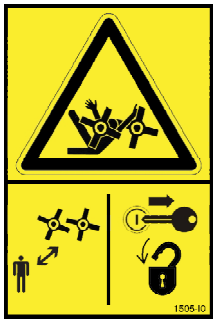

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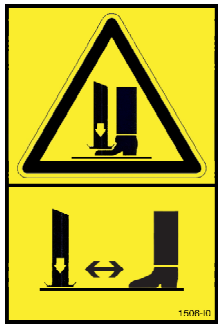

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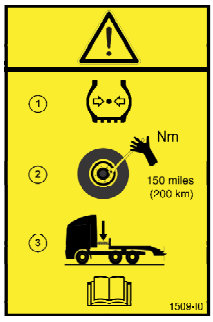

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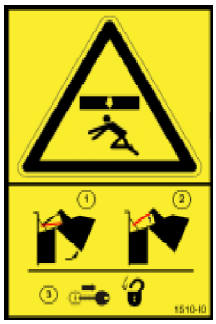

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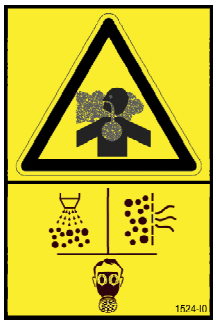

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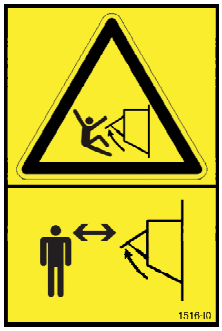

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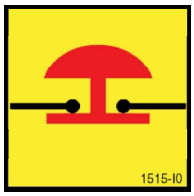

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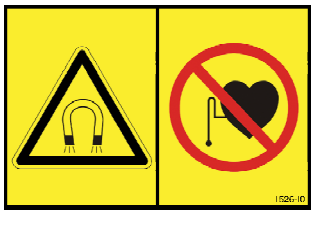

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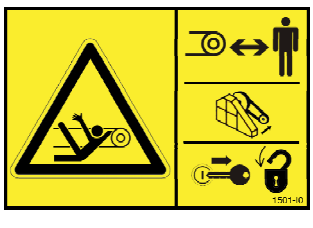
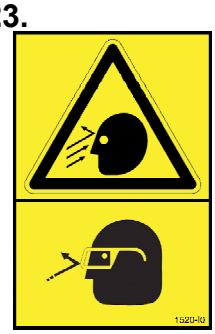


20.	21.	22.	23. 
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1.3 WORK AREA SAFETY

1. Keep the work area as neat and as clean as practical.

2. Keep all warning and caution signs clear and up to date

3. Make certain all electrical equipment is properly grounded.

4. Store hazardous materials in restricted access areas and mark them clearly.

5. Do not smoke or allow smoking near flammable fuels or solvents. Use non-flammable solvents for cleaning parts and equipment.

6. Make certain enough ventilation is present to run engines safely. Do not start an engine in an enclosed space without properly vented exhaust. Exhaust fumes from gasoline or diesel engines can kill!

7. Avoid electrical and static sparks and any open flame while handling, storing, moving or pouring fuels, electrolytes for batteries, hydraulic fluids or coolants

8. Check for leaks in tanks or tubing with flashlights or other proper equipment. Never use an open flame to check for leaks.

9. Know where fire extinguishers and other fire suppression equipment are located. Learn how to use them.

10. Be alert and wary around any pressurized system: hydraulic or air. High pressure gases and oils can be very dangerous. Know your equipment and operate it properly.

11. Be alert and watch for pinch points, closing mechanisms, and falling parts when working on or around any machinery. Keep hands and tools clear

12. Clean and properly labeled tools are safer tools. Keep your equipment and tools in order.

13. Always use hoisting equipment for heavy loads. Regularly check hooks, cables, shackles and chains for stretch and wear. Never overload hoists, cranes or other lifting devices.

14. Keep your equipment clean and free of dirt and grease so it can be checked for loose, cracked or broken parts. Replace defective parts as soon as they are discovered.

15. For questions concerning accumulators, which are pressure-containment vessels, contact your *NFLG* distributor. Malfunctioning valves or poor maintenance practices can result in build-up of extremely high hydraulic and/or pneumatic pressures inside the accumulator.

1.4 HYDRAULIC SAFETY

1. Never disable or alter any hydraulic circuit or component without consulting with NFLG or your local dealer.

2. Any work carried out on a hydraulic system must be performed by an engineer with special knowledge and experience in hydraulics.

3. Relieve all pressure in hydraulic system by returning controls to the neutral position. Turn machine off and isolate power supply before any pipes, filter caps, filters or hydrau lic fittings are disconnected or removed.

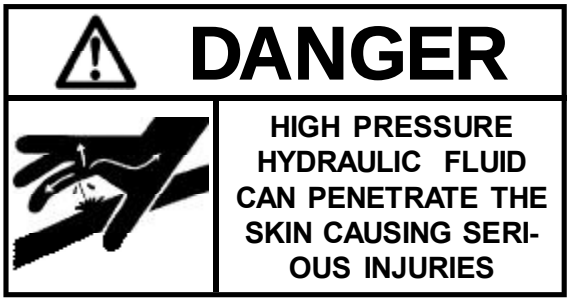
4. Ensure replacement hydraulic components are the same standard and quality as the original parts. Ensure they are fitted correctly.

5. Beware of hydraulic oil leaks. Hydraulic fluid under pressure can penetrate the skin or damage eyes. Fluid leaks under pressure may not be visible. Use a piece of cardboard to find leaks but do not use bare hand.

6. Wear safety goggles for eye protection. If fluid enters skin or eye, get immediate medical attention.

7. Replace any damaged hydraulic pipe. Leaks can cause fires. Always ensure a replacement pipe has the same specification as the original pipe, length, fittings, orientation and diameter. Ensure pipes are laid and fitted correctly.

8. Do not exceed safe limits. Never set a pressure relief valve to a pressure higher than that set at the factory. 175 Bar *(2500 PSI).*



1.5 ELECTRICAL AND ENGINE SAFETY

1. Never work on the electrical system of any equipment unless you are thoroughly familiar with system details.

2. Work on the electrical system must only be carried out by a qualified electrician.

3. If machine is electric-hydraulic or direct electric drive, the 'mains' electrical supply to the machine should always be isolated by unplugging/uncoupling the 'mains' power socket.

4. The electrical equipment of the machine should be inspected at regular intervals. Damaged cables or loose connections must be corrected immediately. Use only original fuses with the specified current rating.

5. It is recommended that an earth leakage safety switch be fitted in the supply line to the power point on site. Special care should be taken to ensure that earth wires are correctly connected.

6. Disconnect the battery ground leads whenever performing any electrical maintenance or servicing to your engine.

7. Battery posts, terminals and related accessories contain lead and lead compounds, chemicals known to the State of California to cause cancer and reproductive harm. Wash hands after handling.

8. Never short across the starter terminals of a battery as this can cause a fire and could also damage the electrical system.

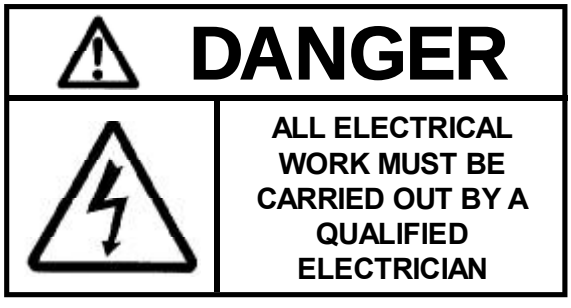
9. Battery electrolyte contains acid and can cause serious injury. Avoid contact with skin and eyes. Use gloves and protective glasses.

10. Diesel engine exhaust emissions contain products of combustion which may be harmful to your health. Always operate the machine in a well ventilated area and if operating in an enclosed area, vent the emissions outside.

11. Do not touch any part of the engine or exhaust system. Allow the engine and exhaust to cool before performing any repair or maintenance.

12. Never fill the fuel tank with the engine running, while smoking or when near an open flame.

13. Never overfill the tank or spill fuel. If fuel is spilled, clean it up immediately.



1.6 SAFETY BEFORE AND DURING OPERATION

Only qualified and trained personnel should attempt to operate this machinery.

1. Read and understand the Operators Manual before you start the machine.
Study the DANGER, WARNING, CAUTION and IMPORTANT safety signs on your machine.

2. Check your machine and have all systems in good operational condition.

- Check for broken, missing or damaged parts.Replace and make necessary repairs.
- Ensure all safety guards and emergency stops are fitted and in working order.
- See 'Daily Maintenance' for more detailed checklist before start-up.

3. Before starting, walk completely around the machine. Make sure no one is under it, on it or close to it. Let other workmen and bystanders know you are starting up and do not start until everyone is clear of the machine.

4. During operation, do not climb onto the machine, over or under moving conveyor belts and rollers. Always use ladders, steps and walkways when mounting and dismounting.

5. Hole alignment on mechanical supports must be checked and secured with pins provided and in accordance with safety signs.

6. Never check the tension of 'V' belts, drive chains and conveyors when machine is running.

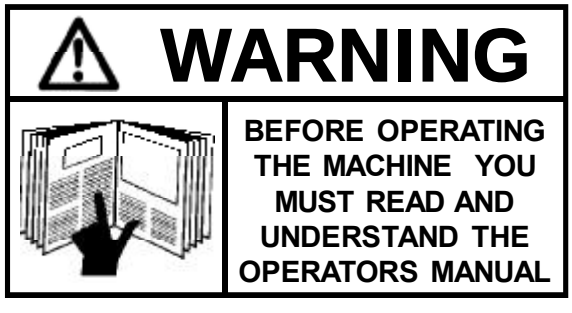
7. Never work or stand beneath machinery or attachments as it is raised or lowered.

8. Never work or stand beneath machinery as they are being loaded with and/or discharging material.

9. Follow safe operating practices. Operate the machine controls smoothly. Avoid sudden stops, starts or changes in direction.

10. Only use emergency stop buttons or emergency stop lines *(if fitted)* in emergency situations or during safety drills.

11. After each day's operation, always run the machine dry; never leave material in the beltfeeder on conveyor belts or screenbox. Starting a machine with a full load will cause strain problems in your machine.



1.7 SAFETY BEFORE AND DURING MAINTENANCE

Maintenance should only be carried out by trained and qualified personnel.

1. Whenever maintenance or service is being carried out a minimum of two *(2)* persons should be present at all times. NEVER WORK ALONE.

2. Prepare yourself. Wear a hard hat, safety goggles, hearing protection and other protective equipment as required by job conditions. Do not wear loose clothing or jewellery that can catch on controls or moving parts. Long hair must be tied back.

3. Prepare the machine. Move the machine onto a level surface and apply parking brakes and/or wheel chocks. Shut off the engine and remove the key. Relieve all hydraulic pressure by returning controls to neutral. Secure all hydraulically operated attachments with pins provided.

4. Isolate all electrical supplies to the machine before starting any maintenance work.

5. Never attempt repairs or adjustments to the machine while it is running. *(Exempt to this rule: Belt tracking adjustments are only possible during working process - see Maintenance Section Belt Tracking).*

6. Remove only guards or covers that provide access. Wipe away excess grease and oil.

7. Never leave guards off or access doors open when unattended. Keep bystanders away if access doors are open.

8. When working beneath raised equipment, always use blocks, jack-stands or other rigid and stable supports. Never work under unsupported equipment.

9. Performing work above 6ft-6” *(2m)* always use an approved *(EN/ANSI)* safety harness. Always use walkways and platforms provided or a safe secure platform approved by the machine operators management. Do not use any unauthorised or unsafe platforms.

10. Never operate any type of engine without proper ventilation – EXHAUST FUMES CAN KILL. *(See Electrical and Engine Safety for more detailed checklist).*

11. Checking for hydraulic leaks. Beware hydraulic fluid under pressure can penetrate the skin or damage eyes. Fluid leaks under pressure may not be visible. Use a piece of cardboard to find leaks but do not use bare hand.

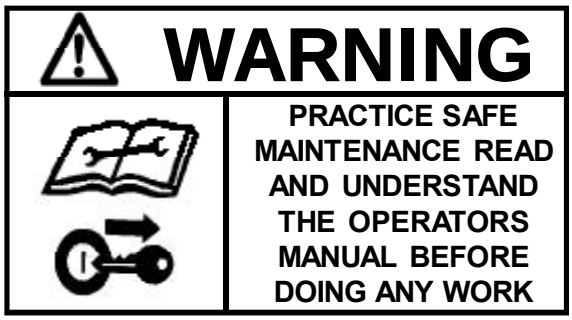
12. Wear safety goggles for eye protection. If fluid enters skin or eye, get immediate medical attention. *(See Hydraulic Safety for more detailed checklist).*

13. Clean or replace damaged, missing or painted over safety signs that cannot be read.

14. Rotating and moving parts must be inspected during maintenance and replaced if cracked or damaged. Excessively worn or damaged parts can fail and cause injury or death.

15. After maintenance, tighten all bolts, fittings and connections.
Install all guards, covers and shields. Replace or repair any damaged ones. Refill and recharge pressure systems with recommended fluids.
Start the engine and check for leaks. Operate all controls and make sure the machine is functioning properly. After testing, shut down, check the work you performed *(any missing cotter pins, washers, locknuts, etc.)*?
Recheck all fluid levels before releasing machine for operation.

16. NEVER make any modifications, additions or conversions which might affect safety without the supplier's approval.



1.8 SAFETY BEFORE AND DURING TRANSPORT

1. The preparations to move equipment by articulated lorry should be supervised by a minimum of two *(2)* persons.

2. Ensure persons transporting the machine adhere to all Safety Signs and procedures.

3. Before transportation on public roads, ensure the machine has been properly secured with no loose material left in or on the machine. *(See Section 6 – Transport).*

On Wheeled Machine

4. Ensure wheel nuts are torqued between 500 to 550 ft.lb *(69 to 76 kg.m)* prior to transport.

5. Check your tyres for:-
- correct pressure
- cuts or bulges
- nails or spikes
- uneven or excessive wear
- missing valve caps

6. Check your wheels for:-
- damaged rims
- missing or loose wheel nuts or bolts
- obvious misalignment

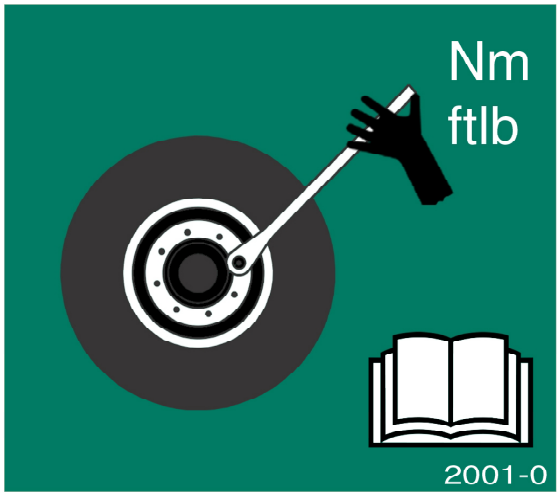
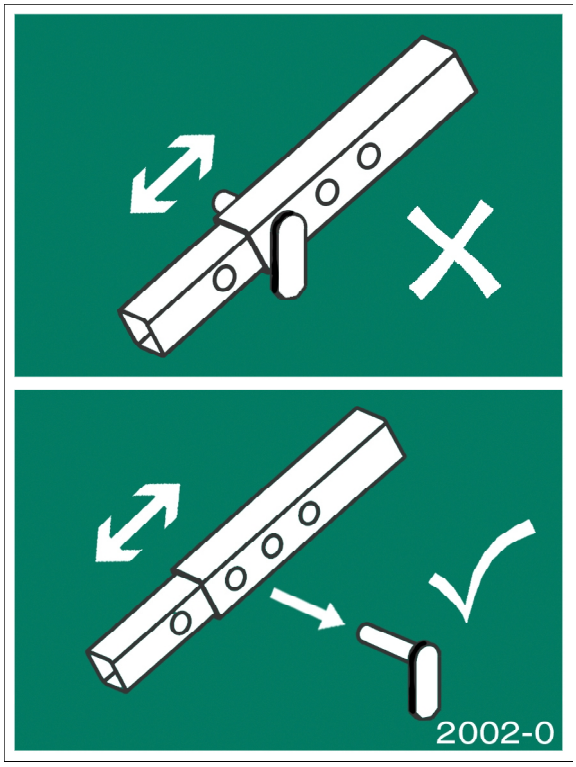
7. Have cuts or punctures repaired by authorised personnel before adding air. Beware an over-inflated tyre can explode and cause serious injury or death.

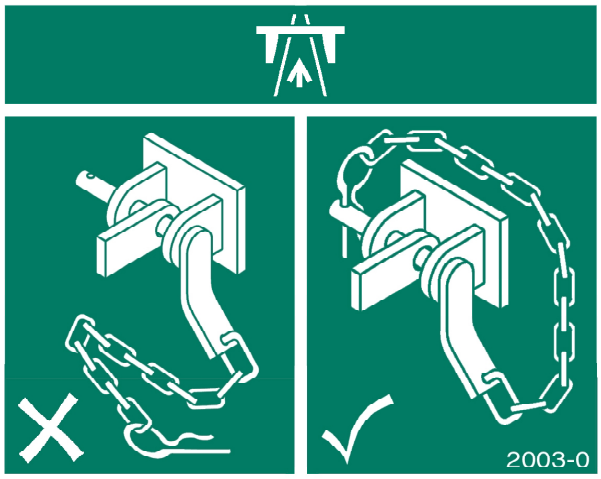
8. Extreme caution is required when transporting machinery on site. Soft or uneven ground may cause accidents.

9. When transporting a machine on a truck or trailer, know the overall height to avoid contacting overhead obstructions such as bridges, power lines etc. *(See Technical Section)*

1.9 INFORMATION SIGNS

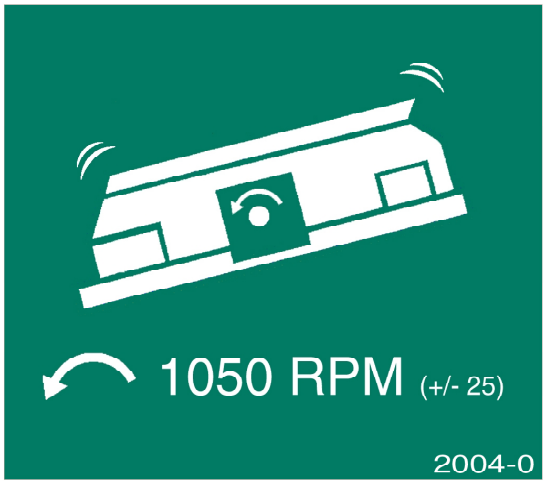
A number of information signs are located on your machine, these signs give information that must be followed to prevent damage to the machine.
The information outlined in the next few pages covers the entire NFLG product, so some of these signs do not necessarily refer to the specific product you use.

	Tighten all whell nuts before towing, Read Manual for details
	Do not elevate with pins in position Remove pins prior to elevation.



Prior to Transport

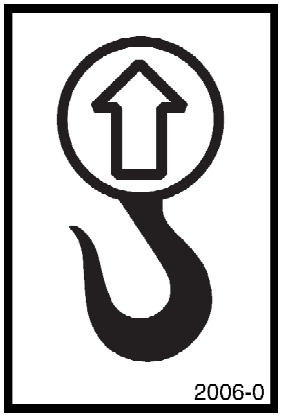
Ensure transport pins & chains are secure.



Maximum Grid Speed 1050 RPM

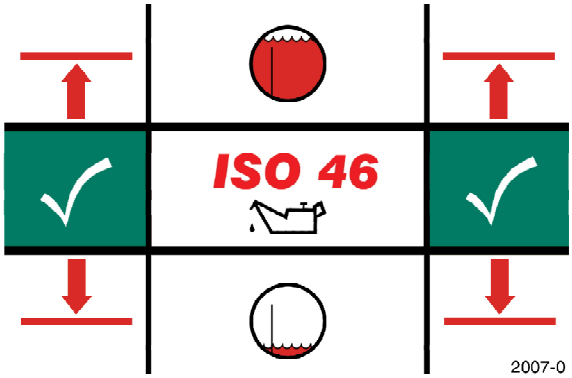
Change reduction gear oil every 1000 hours or at least every 6 months.

Use oil type : DARMEX 9140 NMNND



Machine Lift point.

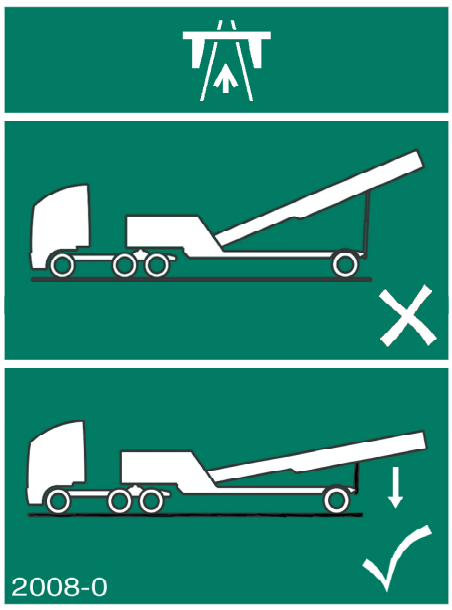
Machine should only be lifted using all lift point locations



Recommended Hydraulic Oil ISO 46.

Oil must be between Minimum and Maximum levels (Red).

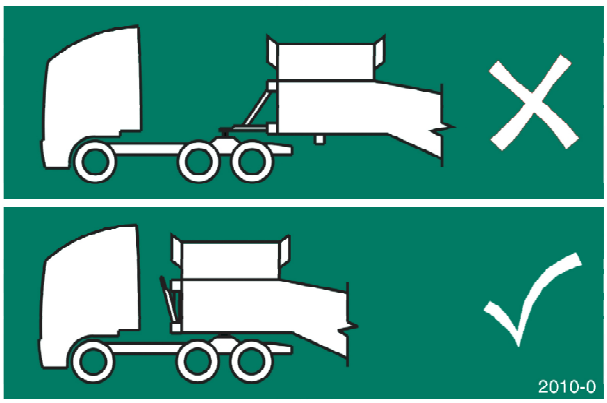
It is preferable to have the oil level in Green area (üarea)
.



When Transporting

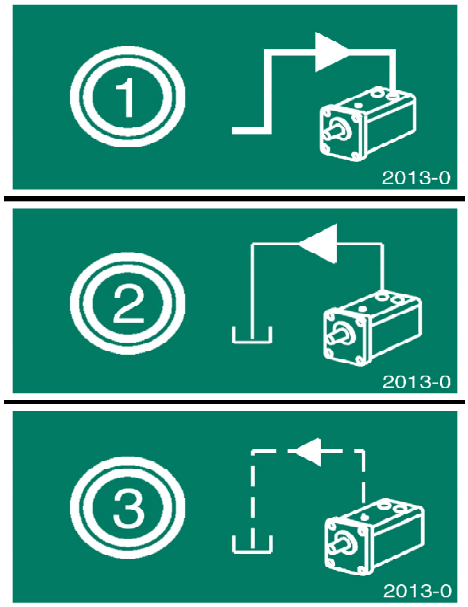
Do not tow machine in elevated position.

Lower the telescopic.



Site drawbar must not be used on public roads, for site use only.

Use 5th wheel coupling on public roads.

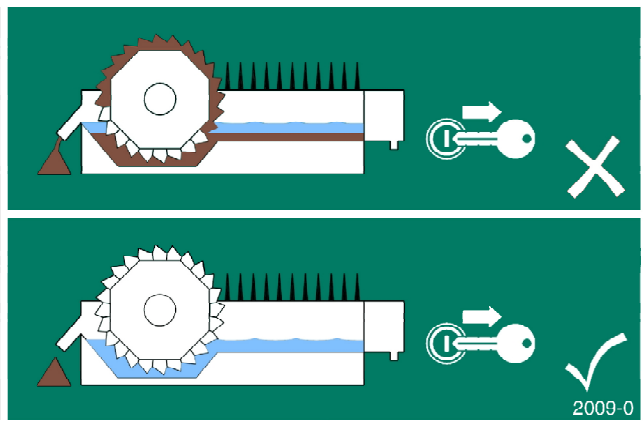


Hydraulic power connections.

1. Pressure Line

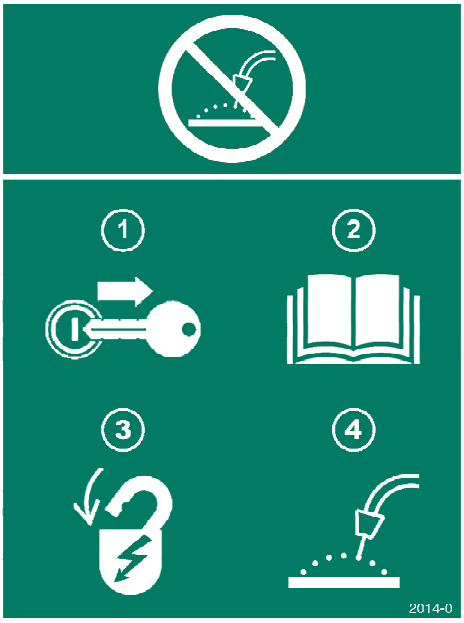
2. Return Line

3. Bleed Line



Do not Stop the Hydrasander with material in the machine.

Ensure Hydrasander has discharged all sand before stopping..



Welding can destroy the electrical components on your machine.

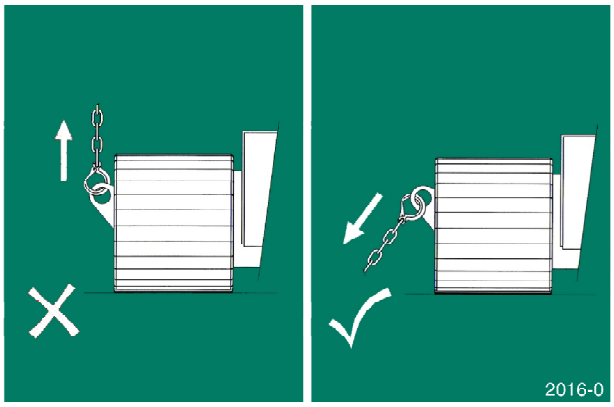
1. Switch off machine.

2. Refer to Operation manual,ie Disconnecting remote receivers, Electronic control Modules if fitted.*(see Installation section)*

3. Isolate the machine using Isolator switch located on the powerunit.

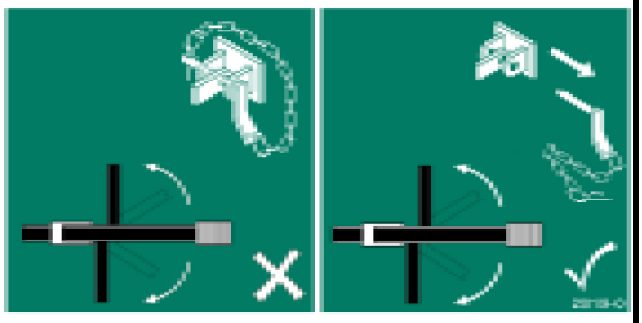
4. Welding can now take place

Protect wiring, hoses and other exposed components from welding debris and splatter.



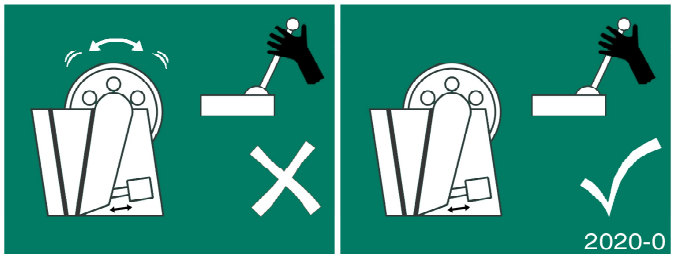
The track hooks must only be used to Tie down the macine.

The track hooks must not be used as a lifting hook.



Do not try to unfold the conveyors with the Transport pins in position.

Remove the Transport securing pins before and unfold the conveyor.

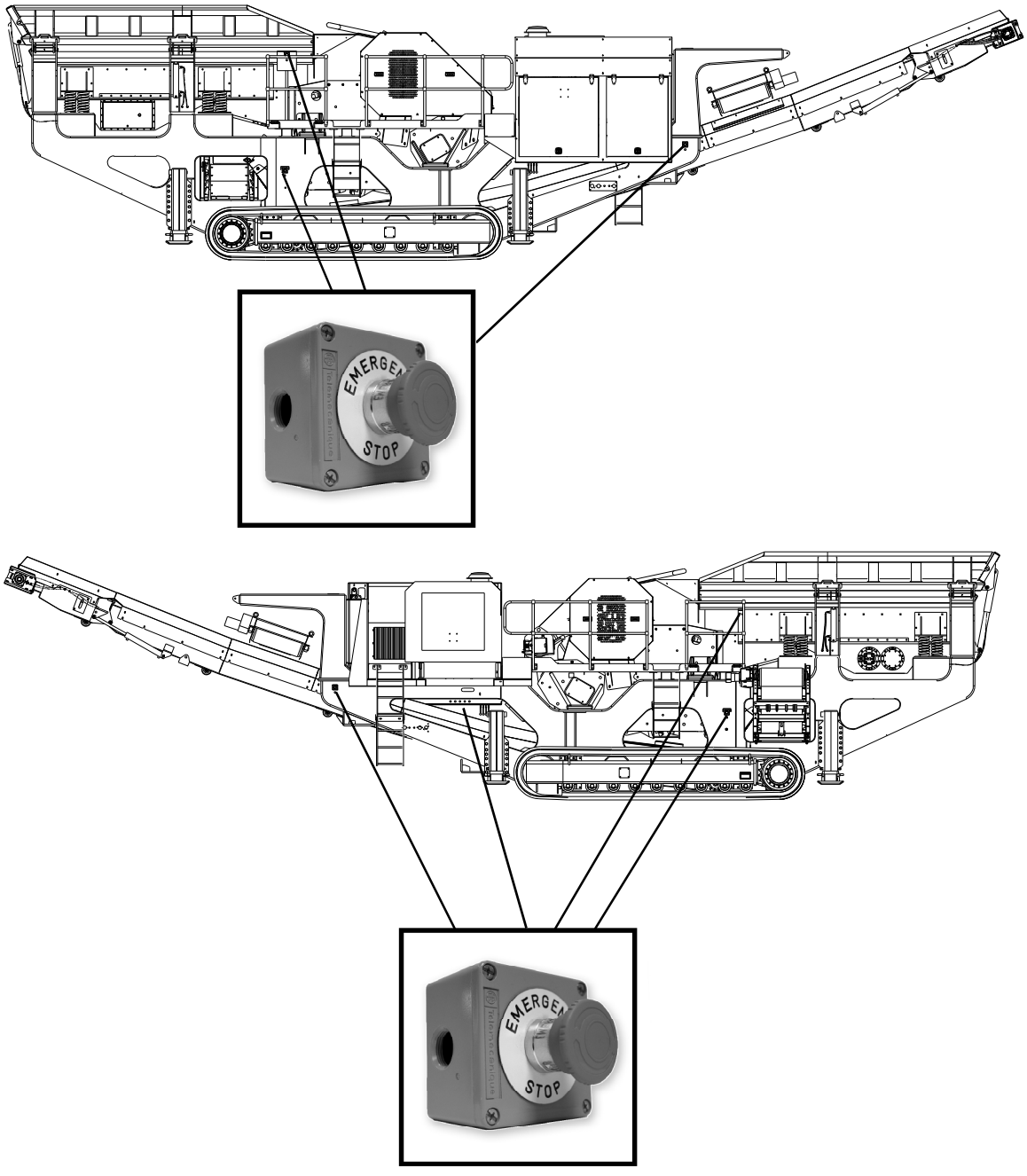


Do not operate the adjust control when the crusher is operating.

Stop the crusher and then adjust.

1.10 REMOTE STOPS ON YOUR MACHINE

There are 6 off Remote Stop Buttons on your machine*(Plus 1 on remote Handset and Dog lead).*
Please take time to familiarise yourself with their locations on the machine as seen below.

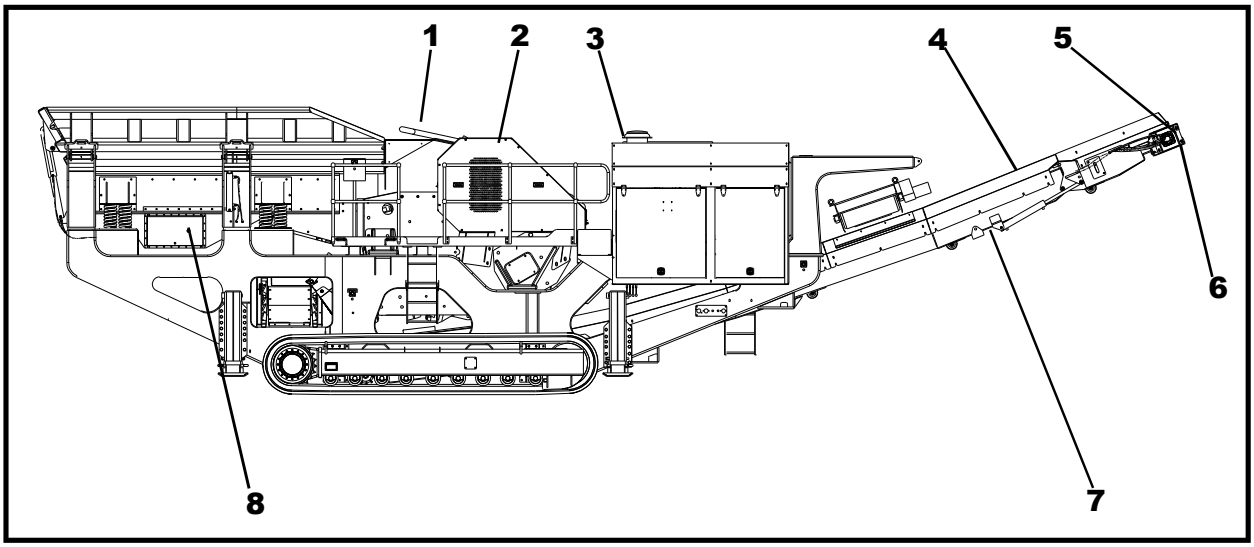


IMPORTANT

REMOTE STOP BUTTONS ARE AN IMPORTANT SAFETY FEATURE IF DAMAGED OR BROKEN THEY MUST BE REPLACED IMMEDIATELY

1.11SAFETY GUARDS ON YOUR MACHINE

Safety Guards are important features of the *NFLG* machine. Please take time to familiarise yourself with their locations on the machine as seen below.



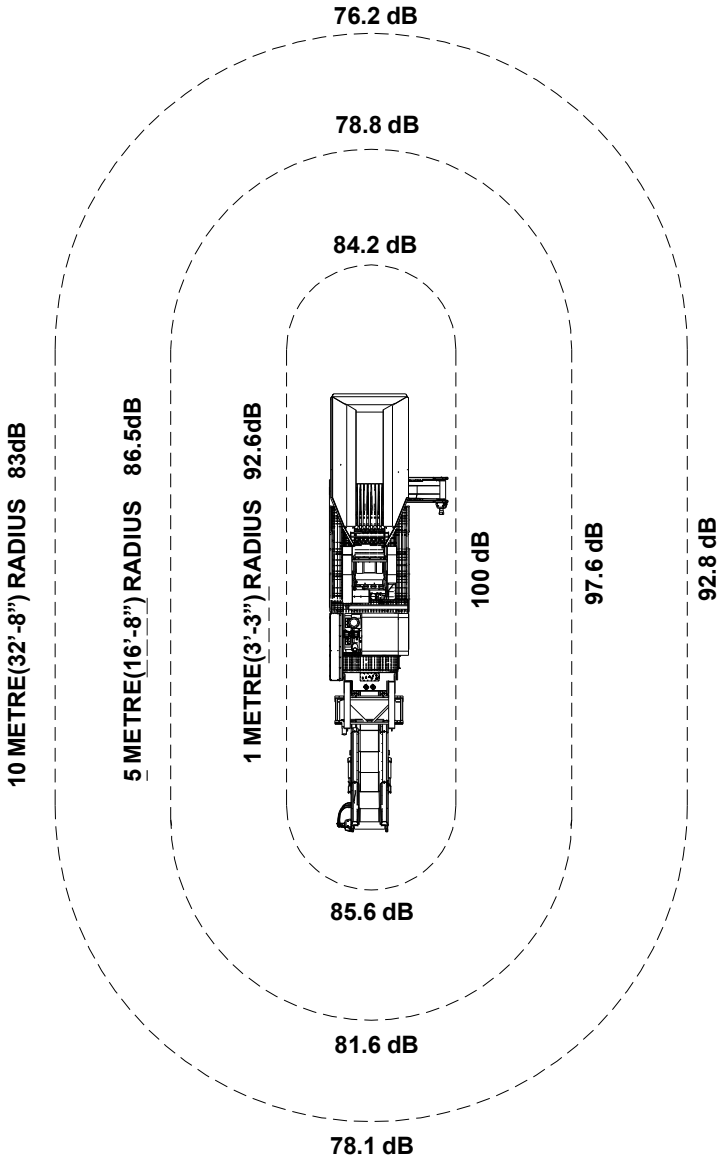
1. Crusher Protection Grid
2. Crusher Pulley Guards
3. Powerunit Guards
4. Main Conveyor Skirting Guards
5. Main Conveyor Drive Coupling Guard
6. Main Conveyor Cheeseguard
7. Main Conveyor Underside Guard
8. Feeder Drive Cover Guard *(opposite side, guard not in view)*

⚠ WARNING

ALWAYS REPLACE SAFETY GUARDS AFTER MAINTENANCE.
NEVER START THE MACHINE WITH FAULTY OR MISSING GUARDS.

1.12 NOISE LEVELS AROUND YOUR MACHINE

Noise Levels for the *NFLG* JW1175 Crusher fitted with CAT C9



Contents

2.1 FAMILIARISATION ..... TE - 3

2.2 KEY COMPONENTS ..... TE - 4

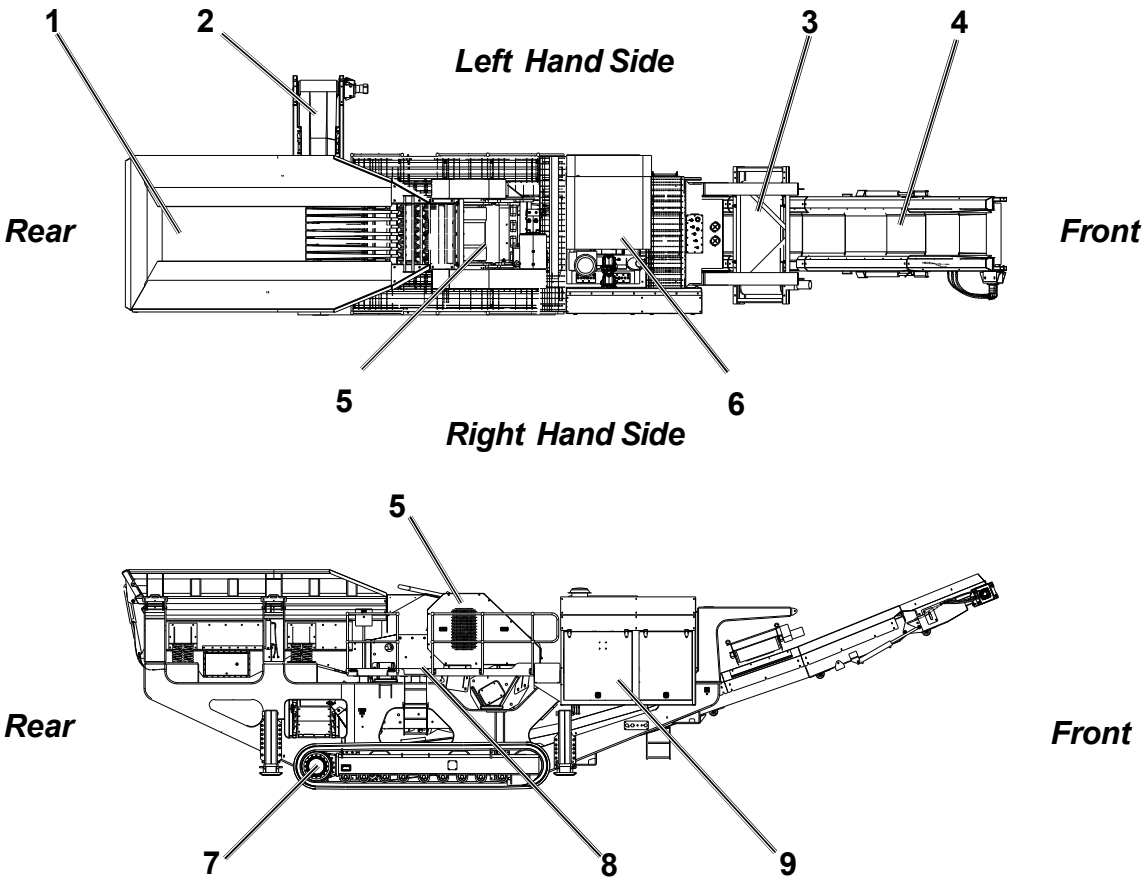
2.3 TECHNICAL DATA ..... TE - 6

2.4 TRANSPORT POSITION DIMENSIONS ..... TE - 10

2.5 WORKING POSITION DIMENSIONS ..... TE - 11

2.6 CORRECT USE OF MACHINE ..... TE - 12

SECTION 2 : TECHNICAL INFORMATION
2.1 FAMILIARISATION



1. Hopper and Feeder
2. Bypass Conveyor (Optional)
3. Magnetic Separator (optional)
4. Main Conveyor
5. Crusher unit
6. Engine Compartment.
7. Track Assembly
8. Maintenance Platform .
9. Pumps Access Door

Right and Left hand sides:-

As viewed from the Rear of the machine.

Front:-

The Main conveyor end of the machine.

Rear:-

The Hopper end of the machine.

2.2 KEY COMPONENTS

2.2.1 Engine Compartment

This plant is powered by a Caterpillar C9 261kW *(350hp@1900rpm)* water cooled diesel engine coupled to hydraulic pumps. Fitted in a purpose built compartment it is situated next to the hydraulic tank providing easy access to all filters and gauges.
Access to the engine compartment area is gained through hinged doors on either side of the compartment an access platform is fitted around the engine.

2.2.2 Jaw Crusher

The Crusher unit is a JW1175 Single Toggle (1070x 762mm). The unit is Hydrostatic driven and has hydraulic adjustment. The crusher has the facility to run in reverse to clear blockages.

2.2.3 Twin Track Undercarriage

The machine is mounted on and driven by a high specification tracked undercarriage fitted with transmittal gearboxes with integrated brake and overcentre valve.
The movement of the machine is controlled by two alternative methods either by a hand-held radio control transmitter (optional or a hand held control connected to the receiver by a dog lead.

2.2.4 Hopper & Grizzly Feeder

The Loading Hopper is designed with a 9 cu metre *(11.8 cu yds)* capacity, the sides are hydraulically folded for quick set up time. The side walls are fitted with 15mm Hardox to increase durability. The feeder is mounted on heavy duty springs which cushion against shock, provides flexible support and prevents vibration being transferred to rest of the plant.The vibrator unit with its counter-rotating eccentric shafts produces the vibrating action to move material along the feeder.

2.2.5 Main Conveyor

The main conveyor can be hydraulically folded from Transport to working position. The conveyor has a 1000mm wide belt and has impact bars at the feed point to prevent damage. The conveyor has a high specification scraper on the head drum and has dust suppression fitted as standard. There is also the option of a extended main conveyor.

2.2.6 BypassConveyor(Optional)

The Bypass conveyor can be hydraulically folded from Transport to working position. The conveyor has a 650mm wide belt and can be fitted on either the left or right hand side of the machine. There is also the option of an extended dirt conveyor.

2.2.7 Magnetic Separator (Optional)

The magnet is suspended above the Main conveyor and is fitted with a 750mm belt.

2.2.8 Other Options

- Hydraulically driven waterpump for dust supression.
- Belt weighter
- Tracked and wheeled version
- Stockpiler drive
- Radio remote control of tracks and machine operation.
- Rebar deflector Plate.
- Dust Covers.
- Fully Hydraulic adjustable
- Independant Pre-screen.

2.3 TECHNICAL DATA

2.3.1 Conveyor Speeds

Feeder (Variable Speed)	450-850 RPM(motor)
Main Conveyor	100 metres/minute
Bypass Conveyor	86 metres/minute

2.3.2 Engine Speed

Running	Idle Speed
1900RPM	700RPM

2.3.3 Crusher Speed

220- 280 RPM**(approx)**

2.3.4 Hydraulic Pressures

		BAR	PSI
Maximum Working Pressure on Tracks		300	4351
High Pressure point on Crusher		250	3626
Maximum Working Pressure on rest of machine		210	3046
Pressure Rating on Hydraulic Hoses	**3/8”**	330	4800
	1/2”	275	4000
	3/4”	215	3125

2.3.5 Machine Weights(Approximates)

	TONNES	US TONS
JW1175 Crusher	52.5	57.9

2.3.6 Noise Levels

Sound Level at Operators Position 100 dB(A)

2.3.7 Tank Capacities

	LITRES	GALLONS
Diesel Tank Capacity	431	95
Hydraulic Tank Capacity	720	158

2.3.8 Crusher Capacities

To obtain the optimum output from your crusher it's feed should be continuous and regulated.

Approximate PlantThroughputs (tph) at closed side settings

mm	50	60	75	90	100	115	125	140	150	165	175
Inch	2"	2.5"	3"	3.5"	4"	4.5"	5"	5.5"	6"	6.5"	7"
Mtph	113-132	138-157	166-186	195-220	226-251	264-289	289-314	327-364	346-383	377-427	402-452
Stph	122-142	152-172	182-205	215-242	249-276	290-318	318-347	361-302	377-422	415-449	444-499

Notes

(1) Are listed in metric tonnes & short TPH are to be used as a guide only figures are based on a dry free flowing, Limestone material of standard feed grade weighing 1600kg/m³. Material finer than the closed side settings is to be removed from the crusher feed.

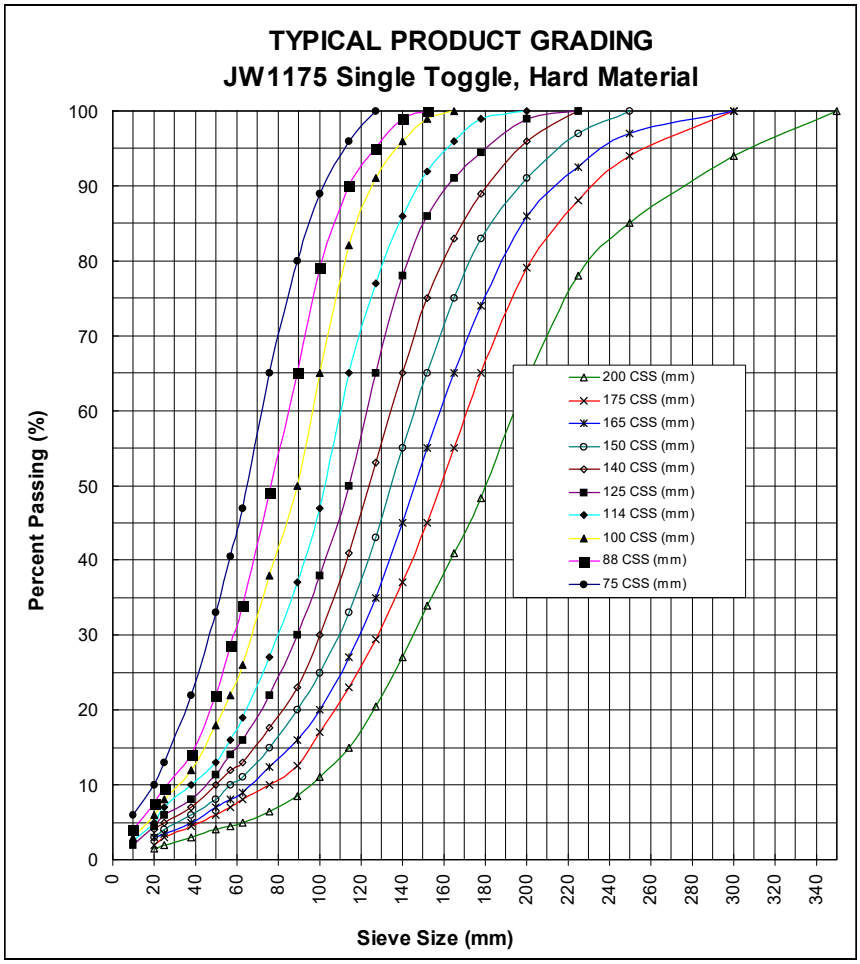
(2) The crushabilty of material can vary greatly dependent upon the work index value, moisture & the feed grading all these factors influence the crusher performance, grading plus output.

(3) All information included herein is subject to change without prior notice.

(4) The capacities and settings shown are a guide only and require confirmation by NFLG for correct selection and capacity to suit your application.

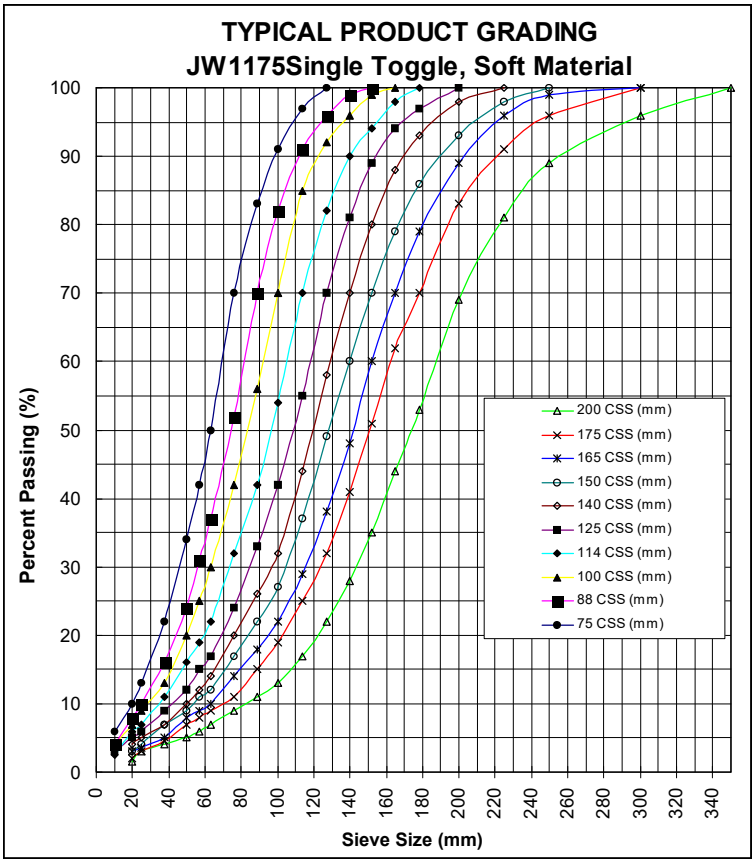
GRADATION CURVES FOR JW1175, IN A HARD ROCK APPLICATION.

Sieve Size (mm)	200 CSS (mm)	175 CSS (mm)	165 CSS (mm)	150 CSS (mm)	140 CSS (mm)	125 CSS (mm)	114 CSS (mm)	100 CSS (mm)	88 CSS (mm)	75 CSS (mm)
400										
350	100									
300	94	100	100							
250	85	94	97	100						
225	78	88	93	97	100	100				
200	64	79	86	91	96	99	100			
178	48	65	74	83	89	95	99			
165	41	55	65	75	83	91	96	100		
152	34	45	55	65	75	86	92	99	100	
140	27	37	45	55	65	78	86	96	99	
127	21	30	35	43	53	65	77	91	95	100
114	15	23	27	33	41	50	65	82	90	96
100	11	17	20	25	30	38	47	65	79	89
89	9	13	16	20	23	30	37	50	65	80
76	7	10	12	15	18	22	27	38	49	65
63	5	8	9	11	13	16	19	26	34	47
57	5	7	8	10	12	14	16	22	29	41
50	4	6	7	8	10	11	13	18	22	33
38	3	5	5	6	7	8	10	12	14	22
25	2	3	4	4	5	6	7	8	10	13
20	2	2	3	3	4	5	5	6	8	10
10						2	3	3	4	6



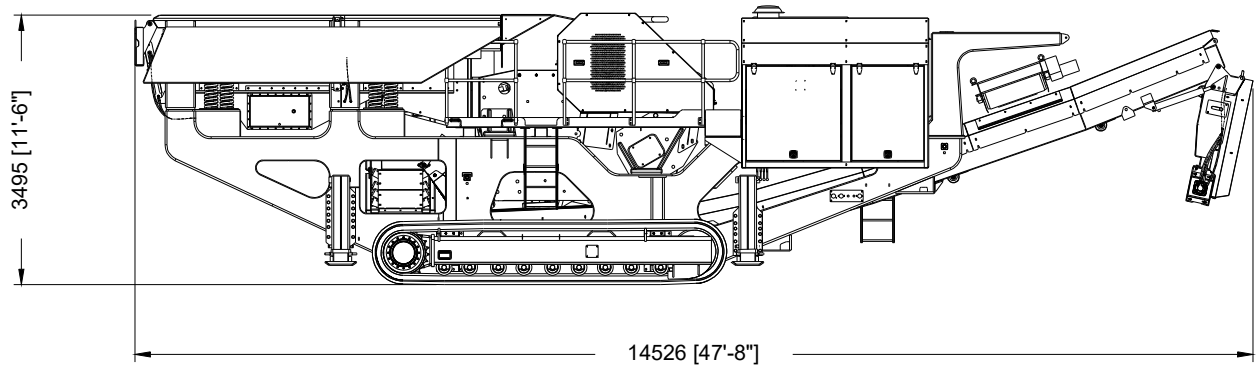
GRADATION CURVES FOR JW1175, IN A SOFT ROCK APPLICATION

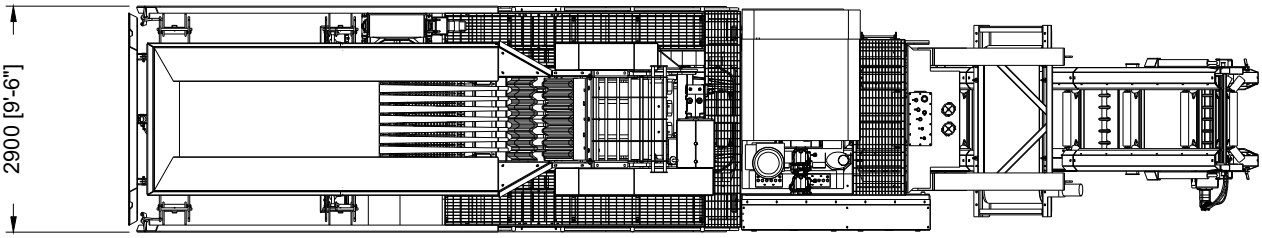
Sieve Size (mm)	200 CSS (mm)	175 CSS (mm)	165 CSS (mm)	150 CSS (mm)	140 CSS (mm)	125 CSS (mm)	114 CSS (mm)	100 CSS (mm)	88 CSS (mm)	75 CSS (mm)
400										
350	100									
300	96	100	100							
250	89	96	99	100						
225	81	91	96	98	100					
200	69	83	89	93	98	100				
178	53	70	79	86	93	97	100			
165	44	62	70	79	88	94	98	100		
152	35	51	60	70	80	89	94	99	100	
140	28	41	48	60	70	81	90	96	99	
127	22	32	38	49	58	70	82	92	96	100
114	17	25	29	37	44	55	70	85	91	97
100	13	19	22	27	32	42	54	70	82	91
89	11	15	18	22	26	33	42	56	70	83
76	9	11	14	17	20	24	32	42	52	70
63	7	9	10	12	14	17	22	30	37	50
57	6	8	9	11	12	15	19	25	31	42
50	5	7	8	9	10	12	16	20	24	34
38	4	5	5	7	7	9	11	13	16	22
25	3	3	4	4	5	6	7	9	10	13
20	2	2	3	3	4	5	6	7	8	10
10						3	3	4	4	6



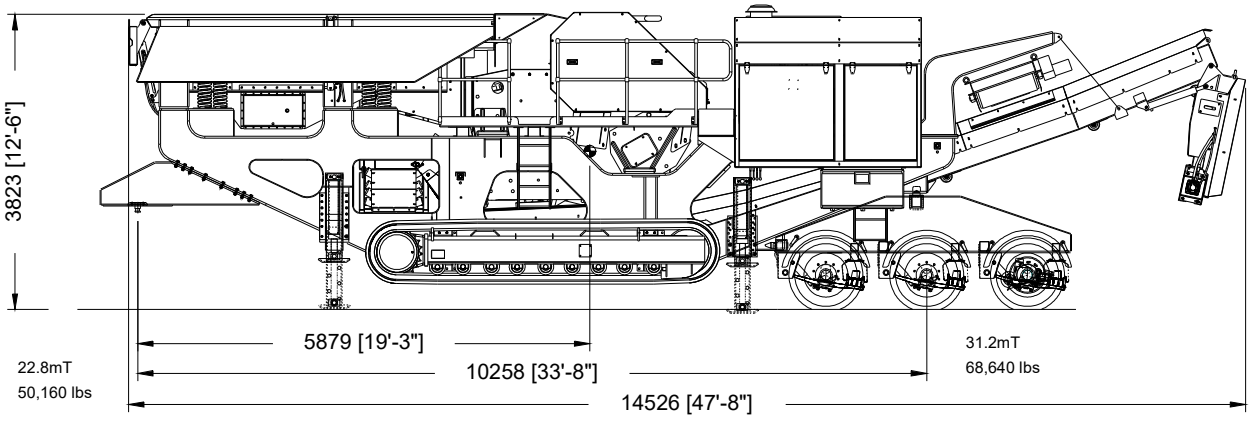
2.4 TRANSPORT POSITION DIMENSIONS

2.4.1 NFJ1175 CRUSHER -TRANSPORT POSITION DIMENSIONS



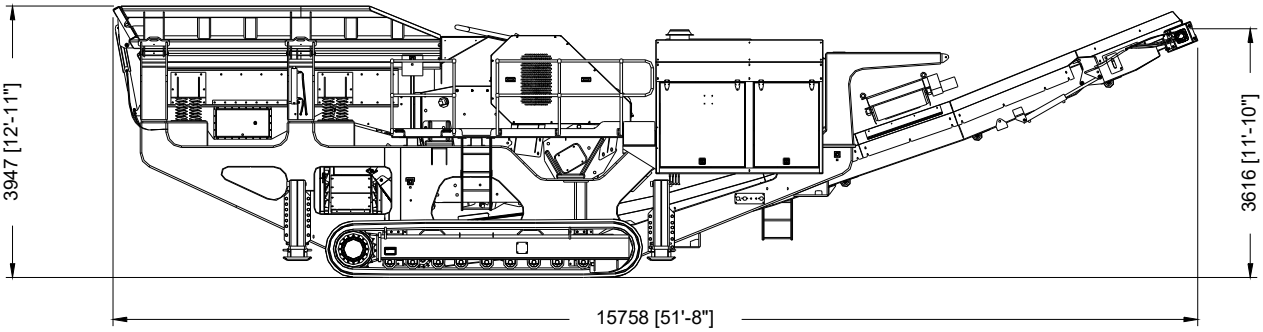


NFJ1175 TRACKED & WHEELED -TRANSPORT POSITION DIMENSIONS

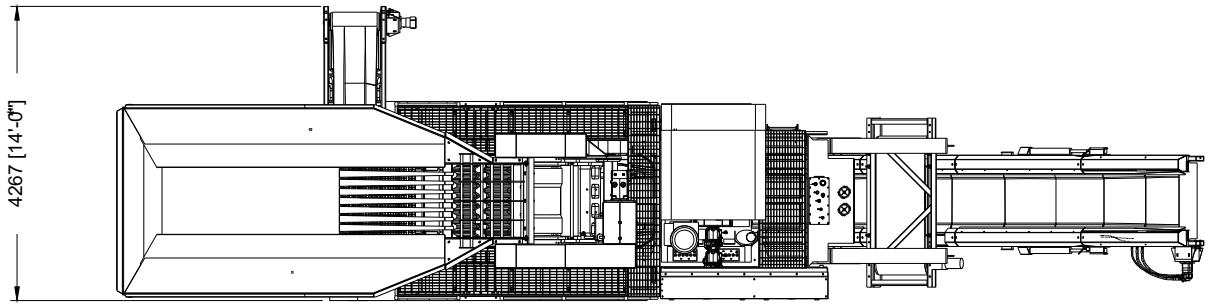


2.5 WORKING POSITION DIMENSIONS

2.5.1 J1175-CRUSHER- WORKING POSITION DIMENSIONS



* Extended main conveyor increases machine length to 16.7m(54’ 9”). the discharge height increases to 4m(13’ 3”)



* Extended bypass conveyor increases machine width *to 6.2 m (20’ 4”).*

2.6 CORRECT USE OF MACHINE

1. This machine should be used, for the loading, conveying, crushing and stockpiling of Quarry, Demolition,Recyclable and Aggregate type materials. This does not include hot materials, which will damage your machine.

2. Machine should be run in well ventilated areas.

3. Appropriate space should be allowed for material stockpiles and loading of machine.

4. This machine should only be loaded via its Loading hopper and beltfeeder. Loading materials on the main belt or directly onto the screen may cause surge blockages and eventual damage to the machine.

5. Before operating, ensure the machine has been properly installed. *(See Section 3.Installation)*

6. Before operating, the machine should be levelled across the axles or undercarriage and along the chassis . Ensuring that the weight of the machine is evenly distributed. (*See Section 3 Installation)*

7. Before operation the operator(s) must:
(i) Have received specific and adequate training in the task to be carried out.
(ii) Have read and understood the safety instructions in Section 1
(iii) Knows the location and function of controls and safety features
(iv) Be aware of all moving parts of the machine

8. Before operating the operator(s) must ensure that they are properly equipped with hard hat, ear and eye defenders.

9. Before operation, all safety guards must be in place

10. Before operation, ensure the machine has been properly lubricated. Always adher to the lubrication schedule provided. *(See Maintenance Section 5)*

11. Before transportation on public roads, ensure the machine has been properly secured with no loose material left in or on the machine. *(See Section 6 - Preparing for Transport)*

12. After each day's operation, always run the machine dry, never leave material in the beltfeeder, crusher or on conveyor belts. Starting a machine with a full load will cause strain problems in your machine.

Contents

3.1 SAFETY INTRODUCTION ..... IN - 2

3.2 STARTING ENGINE ..... IN - 11

3.3 ENGINE INFORMATION DISPLAY ..... IN - 12

3.5 DETACHABLE WHEELED UNIT (OPTIONAL) ..... IN - 17

3.6 PRIOR TO WELDING ..... IN - 19

3.7 HOPPER EXTENSION FLAPS ..... IN - 20

3.8 PLACING DIRT CONVEYOR IN WORKING POSITION(Optional) ..... IN - 21

3.9 UNFOLDING MAIN CONVEYOR ..... IN - 24

3.10 FITTING ACCESS LADDER ..... IN - 25

3.11 CRUSHER PROTECTION GUARD ..... IN - 26

3.12 DUST SUPPRESSION ..... IN - 27

SECTION 3 : INSTALLATION
3.1 SAFETY INTRODUCTION

DANGER

1. Entanglement Hazard.
   Do not reach into an unguarded machine you can be pulled in.
2. In-running nip points can cause serious injury or even death.
   Your arm could be pulled in and amputated.
3. Switch off and lockout machine before opening or removing guard doors.

DANGER

1. Beware of escaping Hydraulic oil. High pressure hydraulic oil can penetrate the skin causing serious injuries.
2. Always relieve pressure from the hydraulic system before attempting maintenance or adjustment. Use a piece of cardboard to check for oil leaks.
3. If oil is injected under the skin, it must be surgically removed or gangrene will result. Get medical help immediately.

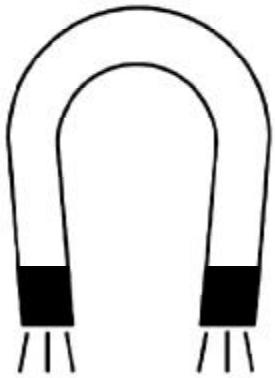
DANGER

1. Electrical Hazard.
   Risk of serious injury or death.
2. Never work on the electrical system of any equipment unless you are thoroughly familiar with system details.
3. Work on the electrical system must only be carried out by a qualified electrician.

DANGER

1. Nip Hazard. Do not reach into an unguarded machine.
2. In-running nip points can cause serious injury or even death. Your arm could be pulled in and amputated.
3. Switch off and lockout machine before opening or removing guard doors.

DANGER



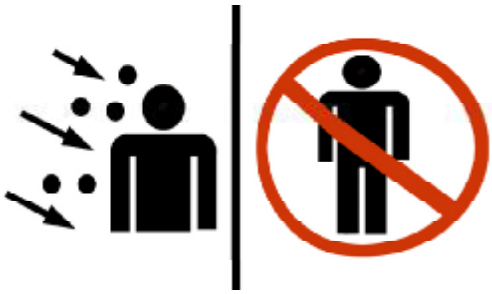
1. Electrical Hazard.
   Risk of serious injury or death.

2. Magnetic fields will affect personnel with pacemakers (within 3 metres), medical electronic implants or metallic prostheses.

3. Ferrous tools and materials could be attracted to the magnet with considerable force.

4. Do not place credit cards or watches near the magnet, they could be permanently damaged

DANGER

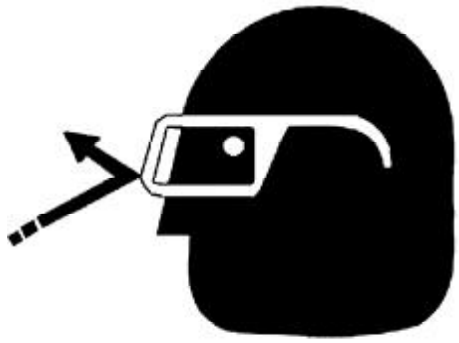


1. Flying material Hazard.
   Risk of serious injury or death.

2. Platform is for Maintenance purposes only. Do not use platform when the plant is in operation

3. If Flying material hits the head or body it can cause serious injury or death .

DANGER

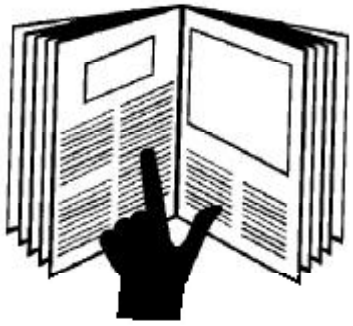


1. Flying material Hazard.
   Risk of serious injury or death.
2. Wear proper eye personalprotective equipment.
3. .Flying material can cause loss or degradation of eye sight.

DANGER

1. Crush Hazard.
   Risk of serious injury or death.
2. Keep clear of Track frames when the unit is in operation.
3. Death or serious injury can result from contact with the Track frames.

⚠ WARNING



1. Read and understand the operators manual before you work at this machine.
2. Study the different safety signs on this machine.
3. Only qualified and trained personnel should attempt to work at this machine.

⚠ WARNING

1. Loose or baggy clothing can get caught in running machinery.
2. Always wear correctly fitting (EN/ANSI approved) protective clothing.
3. Protective clothing includes Hard Hat, Safety Glasses, Ear Defenders, Overalls, Gloves, Steel Toed Boots and a High Visibilty Vest.

WARNING

1. Falling Rock Hazard.
   Do not walk near material discharge areas.
   You are at risk of serious injury or death.

2. Wear a hard hat at all times when working in the vicinity of the machine.

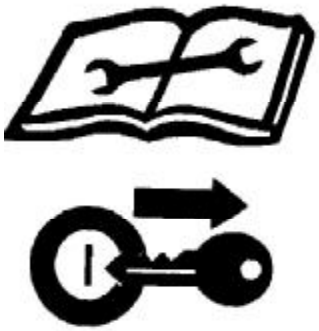
WARNING

1. Crush Hazard.
   Beware of dangerous Hand and Foot crush points.

2. Do not use any body parts e.g. hands or feet to check hole alignment or support machinery.

3. Always use pins and/or mechanical supports provided.

WARNING



1. When performing maintenance or adjustment to this machine always switch off engine and remove the ignition key, (if electric isolate power supply at source).
2. Keep ignition key on person during maintenance.
3. Never work alone.

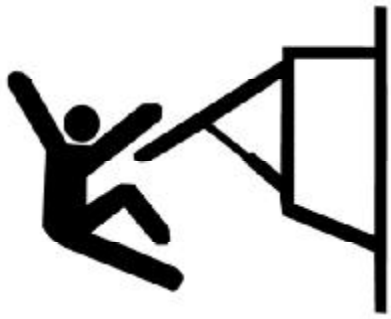
WARNING

1. Fall Hazard.
   Do not climb onto moving or working machinery. You could fall and be seriously injured.
2. Switch off and lockout machine before attempting maintenance or adjustment to this machine.
3. Always use a suitable lifting platform before attempting any maintenance work above 2 metres(6-6”).

WARNING

1. Burn or scald Hazard.
   Beware of hot surfaces at engine and exhaust areas. Be careful when draining hot coolants there is a risk of scalds.

WARNING

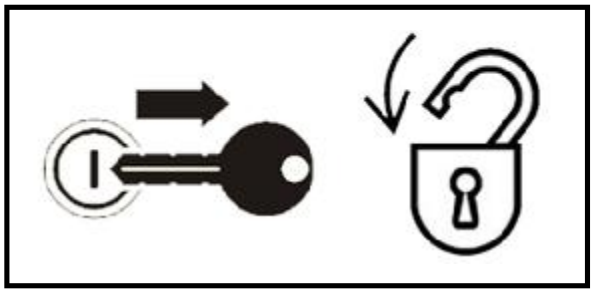


1. Impact Hazard.
   Beware of spring loaded doors opening unrestrained, facial injuries could result.

Lockout procedure.

Lockout is a procedure that is designed to prevent the unexpected or accidental startup of equipment and to alert all workers whenever it is unsafe to operate any piece of equipment. When used as intended, Lockout also protects personnel from energy stored in devices such as springs, accumulators, batteries, hydraulic systems, etc.
When the symbols below appear on a safety sign it indicates that the machine must be switched off and locked out before maintenance can be carried out.

Starter Panel



The Isolator Switch *(ItemB)* is located beside the battery box, the switch cuts the electric supply from the battery ensuring the machine cannot be operated. The machine must always be isolated prior to any maintenance work being carried out.

Isolator Switch "ON" Machine Operational

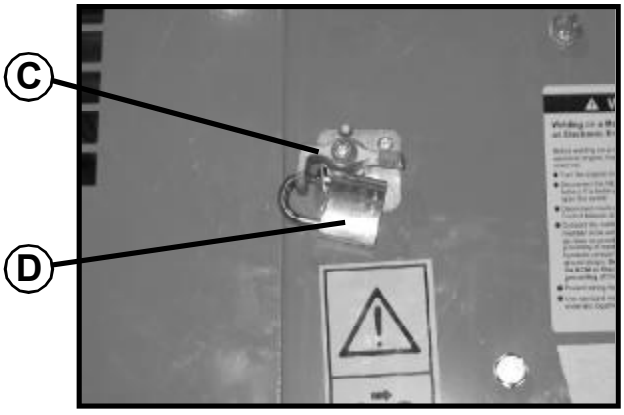
To Lockout:

(i) Turn off the Engine at Starter Panel and remove key *(Item A)*

(i) Rotate the switch *(item B)* to the "OFF" position *(Item C).*

(ii) Attach a padlock *(Item D)* to secure in the lockout position.

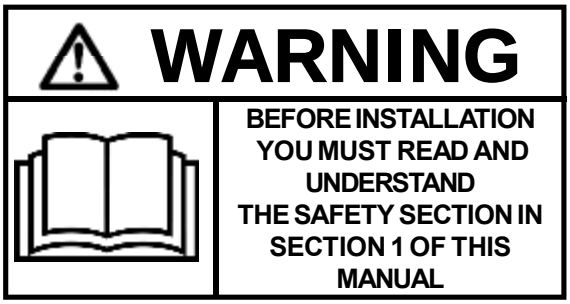
(iii) During maintenance work always ensure the isolator is locked so the machine cannot be started accidently.



Isolator Switch "OFF" and Locked out Machine cannot operate

3.2 STARTING ENGINE

The Engine must be started to provide hydraulic power to all functions. *(For more details read the Engine 'Operation and Maintenance Manual')*



Before Starting:

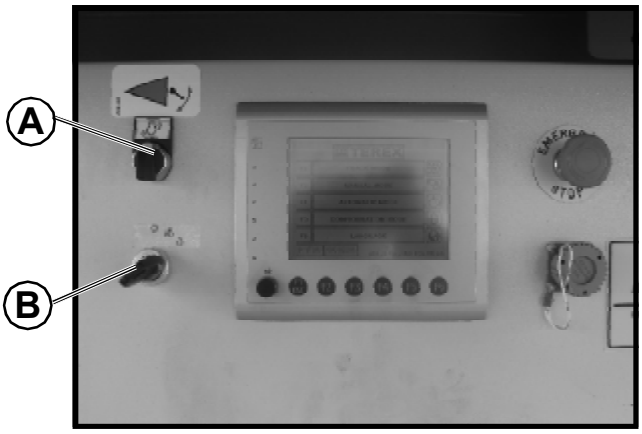
(i) Check the Engine oil level,coolant level and the Fuel level and replenish if necessary.

(ii) Turn the ignition key *(Item B)* to the "ON" position *(Item C)*. The display screen will illuminate then a warning beeper will sound for a 7 seconds. The engine will not start if the key is not kept at position *(Item C)* for the required period.

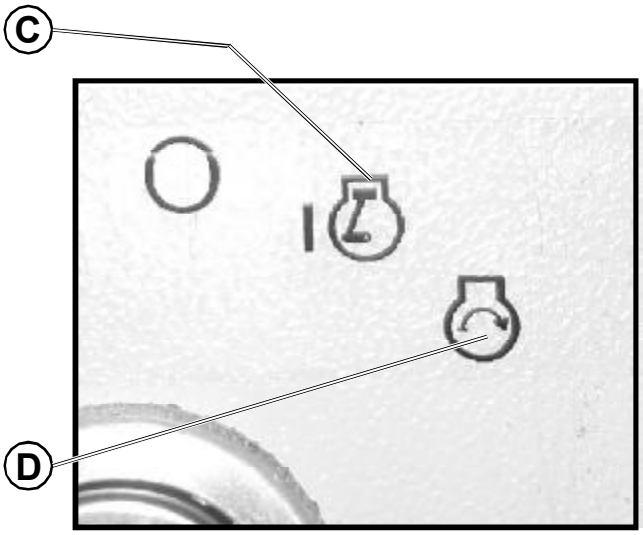
(iii) Turn the ignition key*(Item B)* further to position *(Item D)* to start the engine.

(iv) If the engine fails to start within 30 seconds, release the starter switch and allow the starter motor to cool for 2 minutes before attempting again.

(v) Once the engine has been started allow to idle for approximately 3 minutes before increasing engine speed using the speed control *(Item A).*



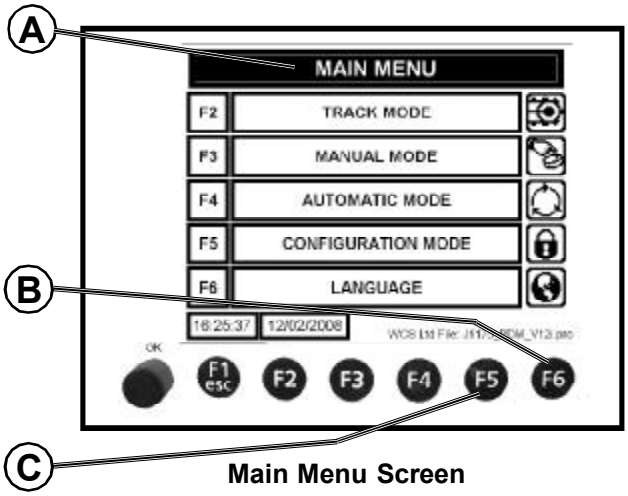
Control Panel



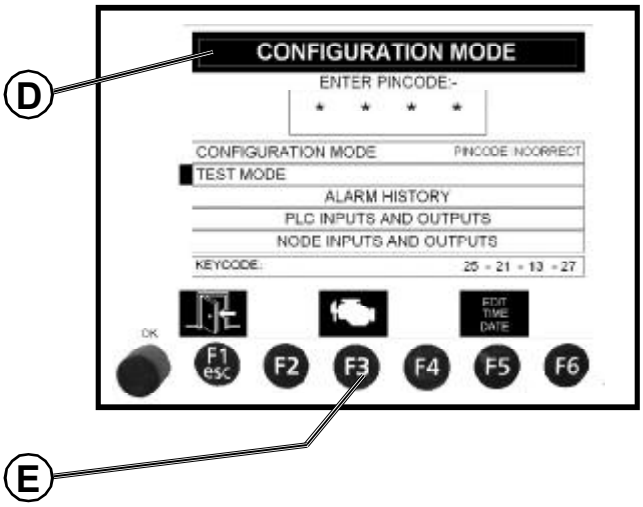
Key Positions

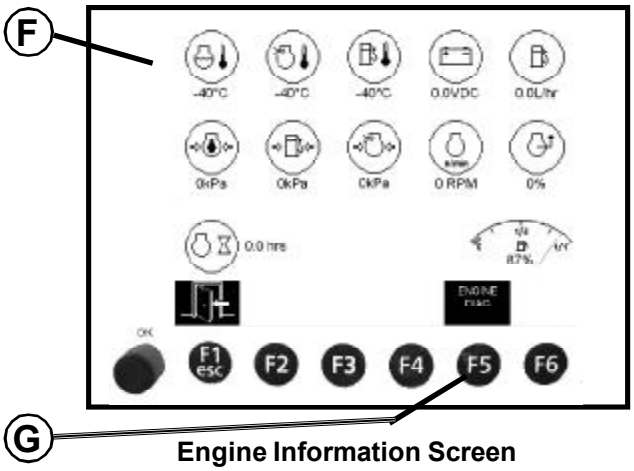
3.3 ENGINE INFORMATION DISPLAY

The Engine information is display on the NFJ1175 control panel screen, this is intended to allow the operator to monitor the engine information while the engine is being operated.



Main Menu Screen





Engine Information Screen

Instantaneous Data

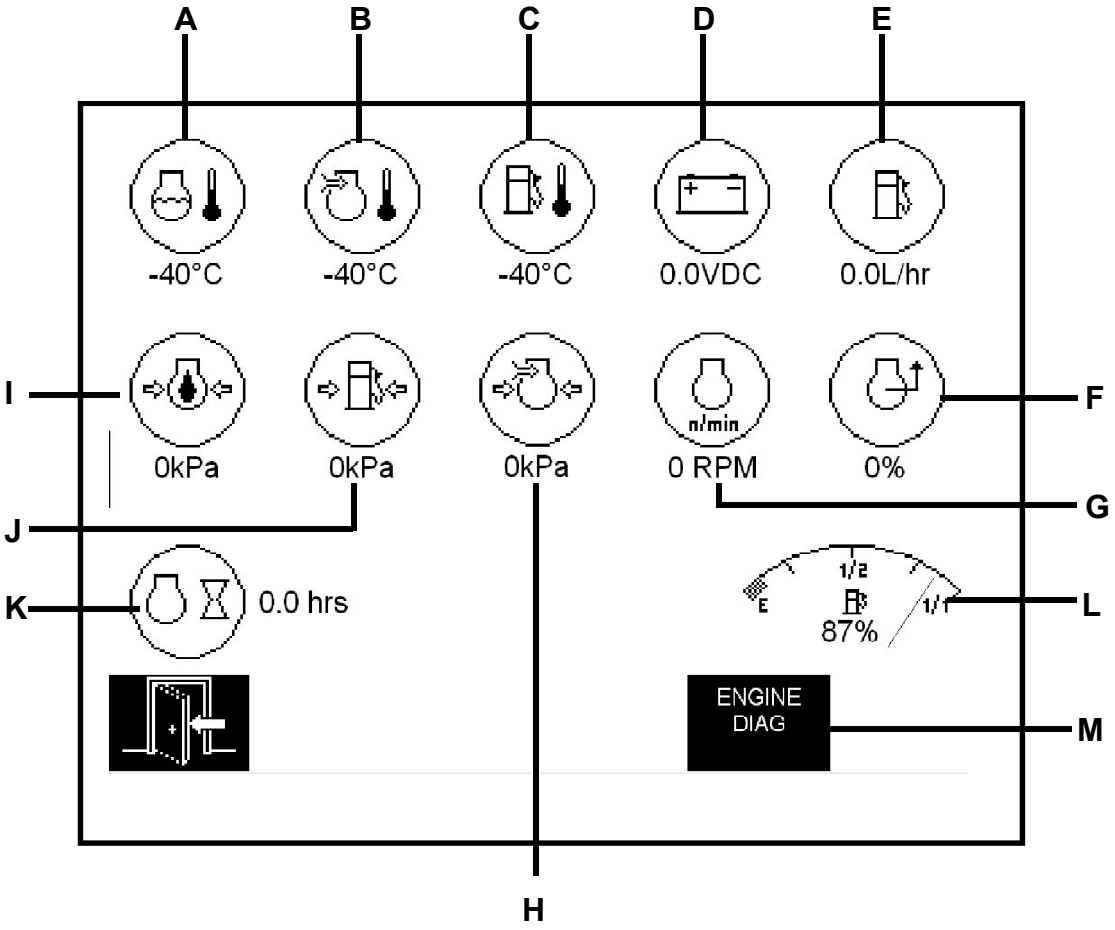
The display provides information on engine speed, oil pressure, engine load, fuel temperature, boost pressure, coolant temperature, fuel rate, intake manifold temperature, auxiliary pressure, auxiliary temperature, desired engine speed, battery voltage, fuel pressure, and engine hours. On the Main Menu *(Item A)* enter the Configuration mode *(Item D)* by pressing F5 *(Item C).* Press the button *F3 (Item E)* this will display the Engine information screen *(Item F).* On the next page a detailed outline of the Diplay screen *(Item D)* is given.

Diagnostic Data

Engine diagnostic codes may be displayed for the operator*(Item D)*. The control panel screen will automatically display potentially serious engine problems. The operator should shut down the engine whenever a code appears. The engine should be repaired as soon as possible. After the display of a serious diagnostic code, the engine may shut down within 20 seconds. Refer to the Diagnostic Code Chart for a list of these codes.

Language Selection

The Language can be changed on the Main Menu screen by Selecting the Language Icon *(Item B)*, and scroll to the required language. The majority of the Display Screens are Pictorial therefore reducing the dependancy on translated text.



Engine Information

A	**Coolant Temperature**
B	**Intake Manifold Temperature**
C	**Fuel Temperature**
D	**Battery Voltage**
E	**Fuel Consumption**
F	**Percentage Load on the Engine**
G	**Engine Speed**
H	**Boost Pressure**
I	**Oil Pressure**
J	**Fuel Pressure**
K	**Engine Hours**
L	**Fuel Level**
M	**Engine Fault Diagnostic**

The diagnostic codes may alert the operator to conditions that may damage engine components.

Procedure to Address the Occurrence of Codes

Whenever a diagnostic code occurs, try to note all operating conditions of the engine. It is especially critical to take note of the operating conditions for intermittent codes. This information provides the technicians with the operating conditions at the time of the intermittent code. This will enhance the technician's ability to produce the code again. This will also enhance the technician's ability to diagnose the problem. The codes are recorded in the ECM and the codes may be recovered with the electronic service tool.

Diagnostic Code and Event Code Chart

The following "Suggested Operator Action for Diagnostic Codes" chart is provided in order to help determine the action that should be taken if a particular diagnostic code is active. The chart is intended as a general guideline. The current operating conditions will determine the reaction of the operator to the codes. As an example, only intermittent service codes that are a persistent problem should be serviced.

Diagnostic Codes

Code	Description	Shutdown Engine(1)	Service ASAP(2)	Schedule Service(3)
1-11	Cylinder 1 fault		X	
2-12	Cylinder 2 fault		X	
3-11	Cylinder 3 fault		X	
4-11	Cylinder 4 fault		X	
5-11	Cylinder 5 fault		X	
6-11	Cylinder 6 fault		X	
22-13	Check Timing Sensor Calibration	X		
41-03	+8 Volt Supply Above Normal			X
41-04	+8 Volt Supply Below Normal			X
91-08	Throttle Signal abnormal		X	
91-13	Throttle Sensor Calibration		X	
94-03	Fuel Press Signal Shrt High			
94-04	Fuel Press Signal Shrt Low			
100-03	Oil Pressure Open Ckt			X
100-04	Oil Pressure Shrt Ckt			X
110-03	Engine Coolant Temperature Open/Short to +Batt			X

Diagnostic Codes

Code	Description	Shutdown Engine(1)	Service ASAP(2)	Schedule Service(3)
110-04	Engine Coolant Temperature Open/Short to -Batt			X
168-02	Intermittent Battery			X
172-03	Intake Manifold Air Temp Open/Short to +Batt			X
172-04	Intake Manifold Air Temperature Short to -Batt			X
174-03	Fuel Temperature Sensor Open/Short to +Batt			X
174-04	Fuel Temperature Sensor Open/Short to -Batt			X
247-00	Data Valid but Above Normal Range			
247-01	Data Valid but Below Normal Range			
247-02	Data Erratic, Intermittent, or Incorrect			
247-03	J-1939 Data Link Communication			
252-11	Incorrect Engine Software	X		
253-02	Personality Module Mismatch	X		
254-12	Electronic Control Module Error	X		
261-13	Engine Timing Calibration	X		
262-03	+5 Volt Supply Open/Short to +Batt		X	
262-04	Fuel Temperature Sensor Open/Short to -Batt			X
263-03	Digital Sensor Supply Open/Short to +Batt			X
263-04	Digital Sensor Supply Open/Short to +Batt			X
268-02	Check Customer or System Parameters			X
274-03	Atmospheric Pressure Open/Short to +Batt			X

Diagnostic Codes

Code	Description	Shutdown Engine(1)	Service ASAP(2)	Schedule Service(3)
274-04	Atmospheric Pressure Open/Short to -Batt			X
320-02	Speed Timing Sensor Loss of Signal			X
320-11	Speed Timing Sensor Mechanical Failure			X
342-02	Loss of Secondary Speed Timing Signal			X
342-11	Speed Timing Sensor Mechanical Failure			
545-05	Ether Start Relay open/short to +batt			X
545-06	Ether Start Relay short to EUC line			X
1835-03	Auxiliary Pressure open/short to +bat			X
1835-04	Auxiliary Pressure short to ground			X
1836-03	Auxiliary Temperature open/short to +batt			X
1836-04	Auxiliary Temperature short to ground			X

(1) The “Shutdown Engine” code indicates the presence of a condition that could potentially damage the engine.
(2) The “Service ASAP” code indicates that the operator should contact a qualified technician for service if engine
performance is adversely affected.
(3) The “Schedule Service” code should be addressed at the next convenient opportunity if engine operation is adversely affected.

3.5 DETACHABLE WHEELED UNIT (OPTIONAL)

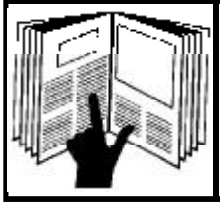
For ease of transport the NFJ1175 can be sup-plied with a detachable wheeled unit. When the NFJ1175 is removed from the lorry wheel chocks are placed behind the wheels of the wheeled unit to prevent accidental move-ment.

To place the Tracks in the operational posi-tion it is necessary to remove the wheeled unit.

To remove:

(i) Disconnect the Air Brake Connections and the Light Connection *(Item A).*

⚠ WARNING

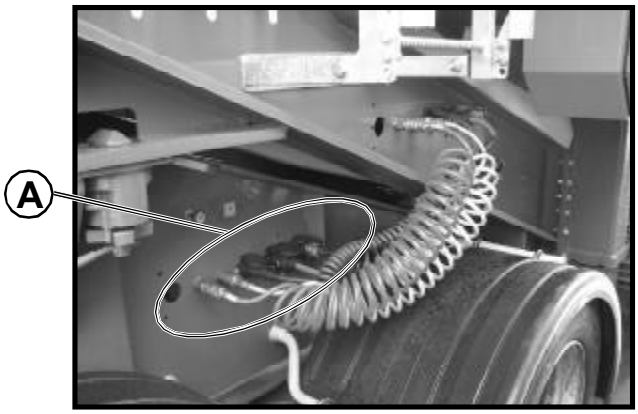


BEFORE INSTALLATION YOU MUST READ AND UNDERSTAND THE SAFETY SECTION IN SECTION 1 OF THIS MANUAL

(ii) Lock the Rear Axle by operating the valve *(Item C)*,(Ensure the axle is aligned correctly before locking)

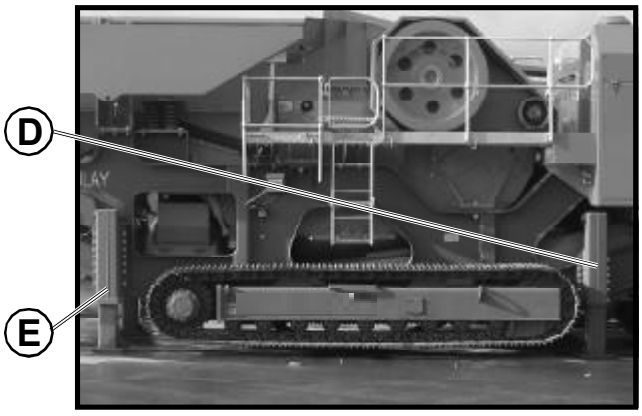
(iii) Push the control *(Item B).* This re-leases the brakes so the bogie can be removed.

(iv) Lower the jacklegs *(Item D)* by oper-ating the control levers *(Item G and H).* Raise the Jackleg *(Item E)* . The chassis should how be raised at the wheeled unit end.

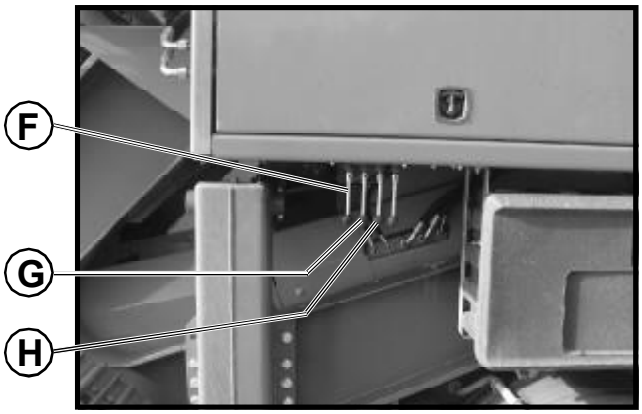


Air Brake Connections

Air Brake Controls



Jack Legs



Jack Leg Controls

(v) Disconnect the lightboard connection *(Item A).*

(vi) Remove the pins *(Item B)* securing the lightboard in the transport position.

(vii) Push the lightboard *(Item D)* in and secure in position.

(viii) Remove the pin *(Item C)* holding the towbar *(Item D)* in the transport position. Pull our the towbar and secure with pin *(Item C).*

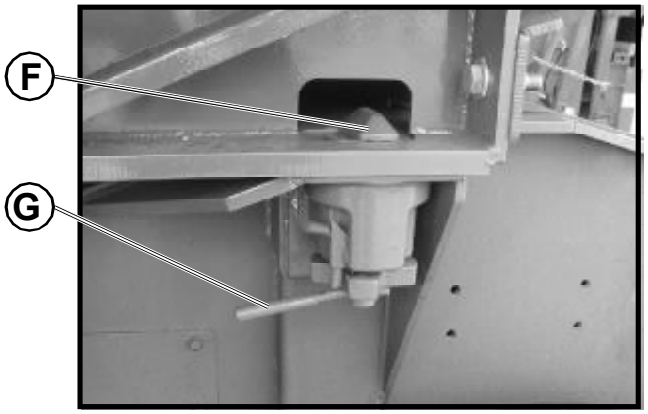
(ix) Rotate the lever *(Item G)* until the locator *(Item F)* detaches from the crusher chassis. Repeat for opposite side of the machine.

(x) When the machine has been raised sufficiently, use a mechanical aid to tow the wheeled unit away using the towbar *(Item D)*.

(xi) The front jack legs can now be raised allowing the tracks to be lowered into the operational position.

Wheeled Unit Lightboard

Wheeled Unit Towbar



WARNING

HAZARDOUS NIP AND CRUSH POINTS IN THIS AREA

3.6 PRIOR TO WELDING

Before any welding work can be carried out at this machine a number of electrical components must be disconnected to prevent potential damage.

To Disconnect it:

(i) Ensure the machine has been switched off.

(ii) Behind the Guard *(Item A)* the Hectronic receiver is positioned.

Disconnect the connector *(Item C)*.

(iii) Disconnect the Aerial *(Item B)*

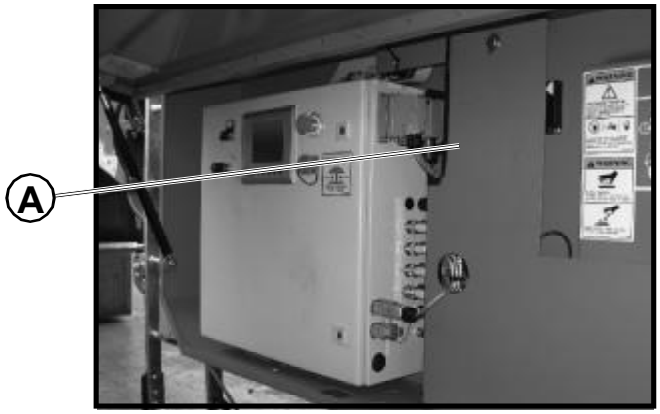
(iv) Turn the Isolator switch *(item D)* to 'off' and lock.

(v) Disconnent the two ECU's *(Item E)* from the engine *(You will need an Allen key to open)*

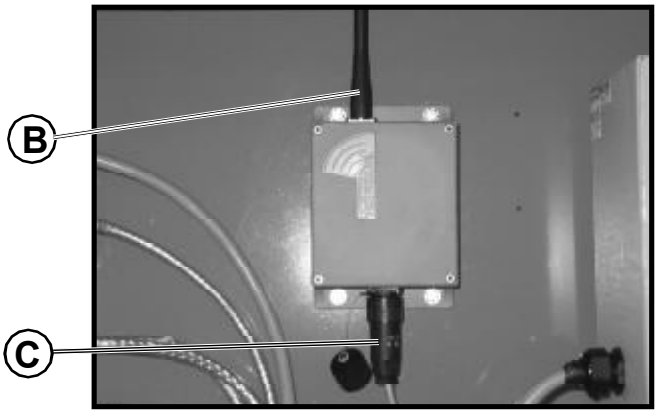
(iv) Reconnect all connections after welding work is completed.

⚠ WARNING

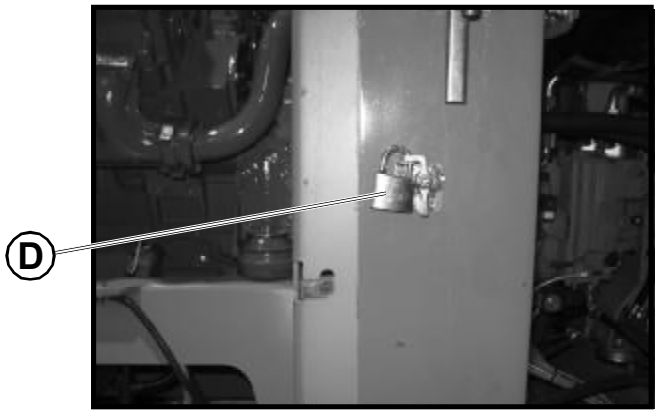
THE RADIO CONTROL RECEIVER UNIT WILL BE PERMANENTLY DESTROYED IF NOT DISCONNECTED PRIOR TO WELDING AT THIS MACHINE

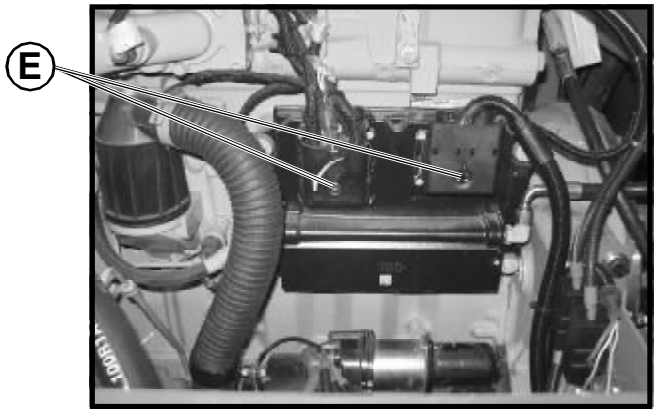


Radio Control Receiver in Main Control Panel



Radio Control Receiver Unit.(guard removed for





3.7 HOPPER EXTENSION FLAPS

When you receive your Machine it will arrive with the Extension flaps folded in the transport position *(Items A, B & C)*. To place into the working position:

(i) Locate the control Bank *(Item D)* .

(ii) Place hopper extension *(Item C)* into the working position by operating the control *(Item G)*.

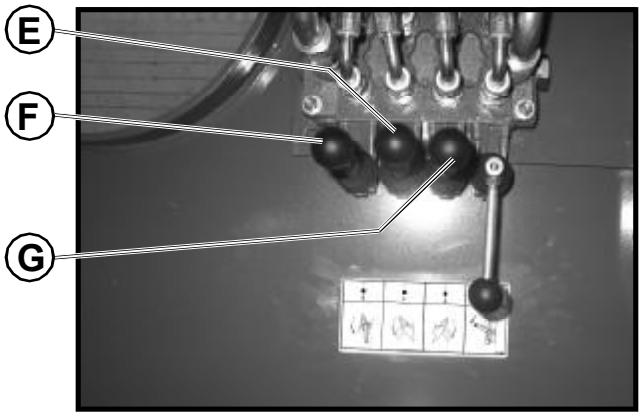
(iii) Place hopper extension *(Item A)* into the working position by operating the control *(Item E)*.

(iv) Place hopper extension *(Item B)* into the working position by operating the control *(Item F)*.

(v) Insert holding wedges *(Item H)* into position on Hopper rear Extension . Insert bolts to hold wedges in position.

(vi) Insert holding wedges *(Item I)* on side extension flaps to hold in position secure with Coach bolts. *(Wedges are stored underneath hopper during transport)*

Hopper flaps in Transport Position



Hopper Extensions Control bank

Hopper Extension Rear Wedges

Hopper Flaps in Working Position

3.8 PLACING DIRT CONVEYOR IN WORKING POSITION(Optional)

If your NFJ1175 is fitted with the optional Dirt Conveyor it will arrive with the Dirt Conveyor *(Item A)* folded into the transport position.Tounfold:

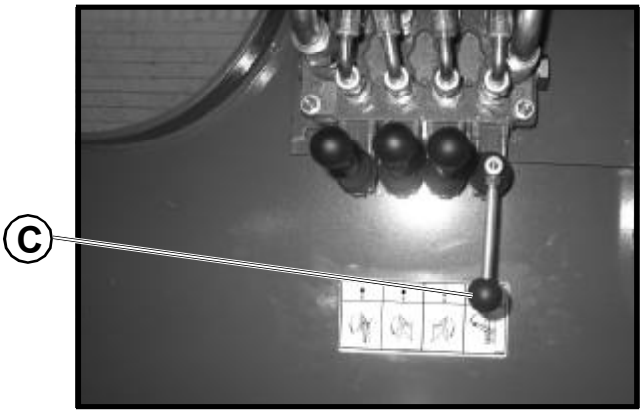
(i) Ensure the Hopper flaps have been placed in the working position prior to unfolding.

(ii) Remove the pin *(Item B)* from the transport support.

(iii) Activate the control *(Item C)* located underneath the hopper to unfold the conveyor into the working position *(Item. D).*

Dirt Conveyor in transport position

Control Bank Location



Dirt Conveyor control bank

Dirt Conveyor in working position

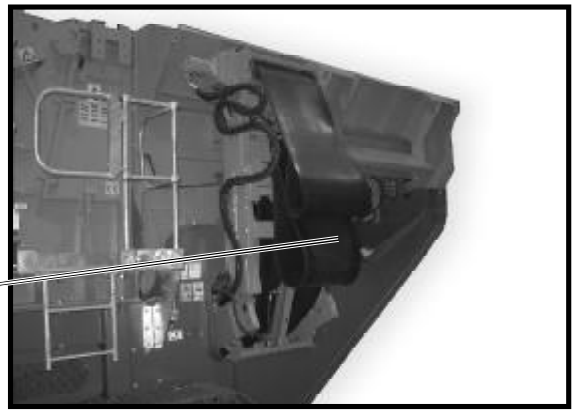
3.8.2 PLACING EXTENDED DIRT CONVEYOR IN WORKING POSITION(Optional)

If your NFJ1175 is fitted with the optional Ex-tended Dirt Conveyor it will arrive with the Dirt Conveyor *(Item A)* folded into the trans-port position.To unfold:

(i) Observe the Dirt conveyor to ensure all transpost restraints have been removed prior to unfolding.

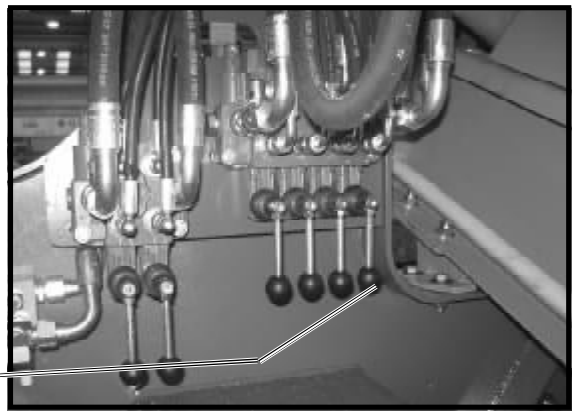
(ii) Remove the pin *(Item B)* from the trans-port support.

(iii) Activate the control *(Item C)* located un-derneath the hopper to unfold the con-veyor into the lowered position *(Item. D)*.



Dirt Conveyor in transport position

Control Bank Location



Dirt Conveyor control bank

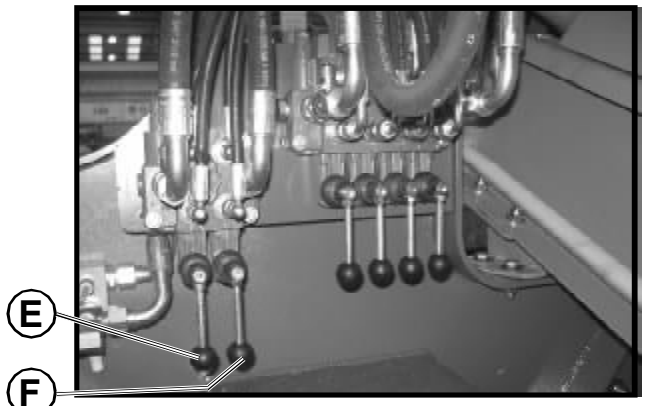
⚠ WARNING

HAZARDOUS NIP AND CRUSH POINTS IN THIS AREA

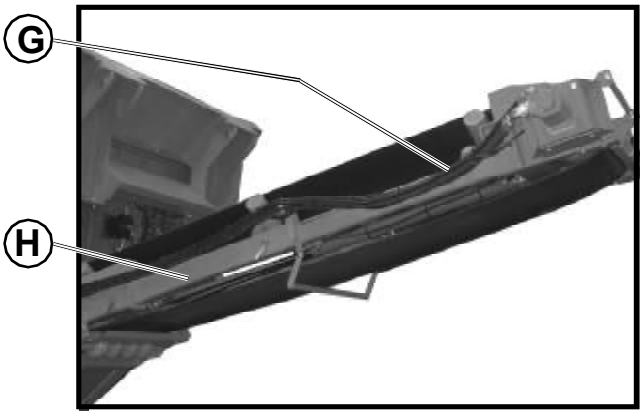
Dirt Conveyor lowered

(iv) Activate the control *(Item F)* located underneath the hopper to extend the lower section of the conveyor *(Item H)*. Ensure the belt is moving freely. Secure the conveyor in position by inserting the holding pins *(item I)* on each side of the conveyor.

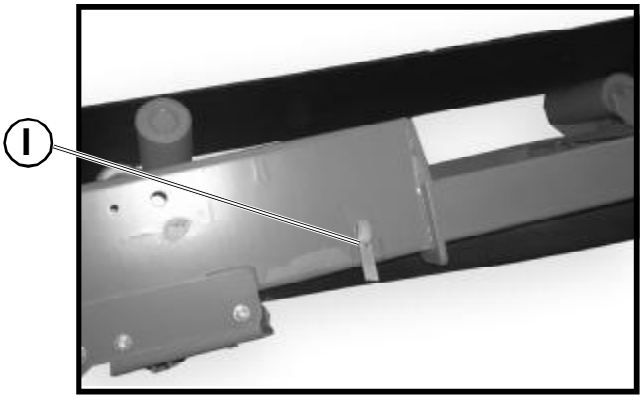
(v) Activate the control *(Item E)* located underneath the hopper to extend the upper section of the conveyor *(Item G)*. Ensure the belt is moving freely. Secure the conveyor in position by inserting the holding pins *(Item J)* on each side of the conveyor.



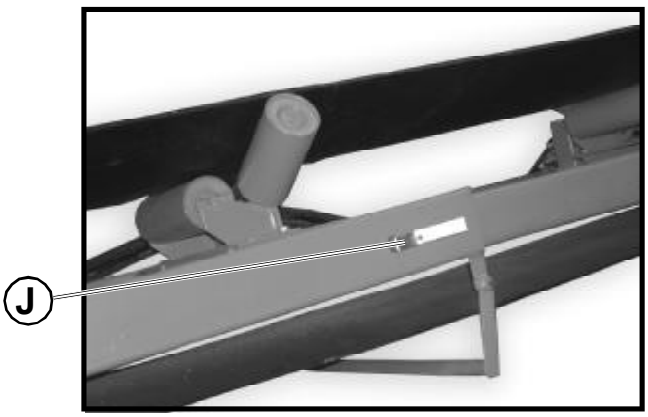
Dirt Conveyor Control bank



Dirt Conveyor in working position



Dirt Conveyor Lower Section Pin



Dirt Conveyor Upper Section Pin

3.9 UNFOLDING MAIN CONVEYOR

When you receive your Machine it will arrive with the Main Conveyors folded in the transport position *(Item A)*.

IMPORTANT

USE THE CONTROL LEVERS IN A SMOOTH AND CONTROLLED MANNER.
DO NOT RUSH OR JERK THE CONTROLS.
THIS WILL DAMAGE THE CONVEYOR

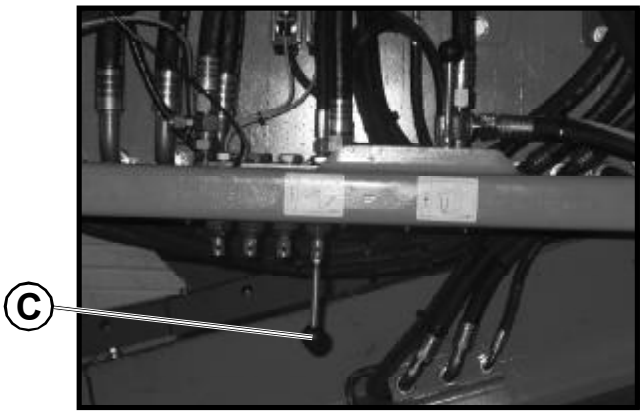
To Place the conveyor in the working position *(Item D)*.

(i) Locate the control Bank *(Item B)*. Operate the control *(Item C)* this will raise the main conveyor head section into the working position *(Item D)*.

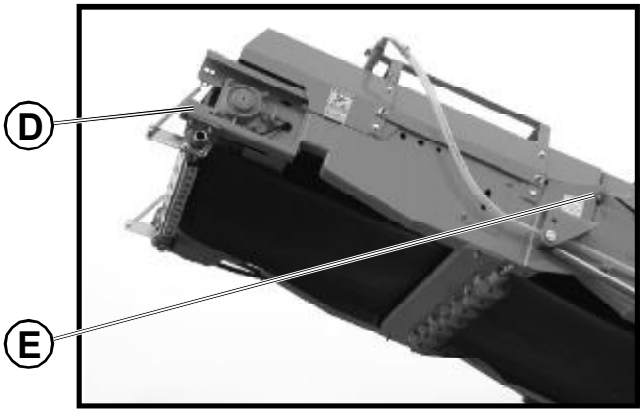
(ii) Secure the conveyor in the working position by inserting holding pins *(Item E)* and secure with safety clips to prevent pins falling out.Repeat for both sides of the conveyor.

Main Conveyor in Transport Position

Main Conveyor Fold Control Panel



Middle Grade Conveyor Control Panel



Main Conveyor in Working position

3.10 FITTING ACCESS LADDER

Your machine will arrive with the access ladder *(Item A)* folded up in the transport position. To place in the working position.

(i) Remove the bolts *(Item A)* securing the ladder in the transport position.

(ii) Lower the ladder into the working position and secure with the bolts *(Item B)*.

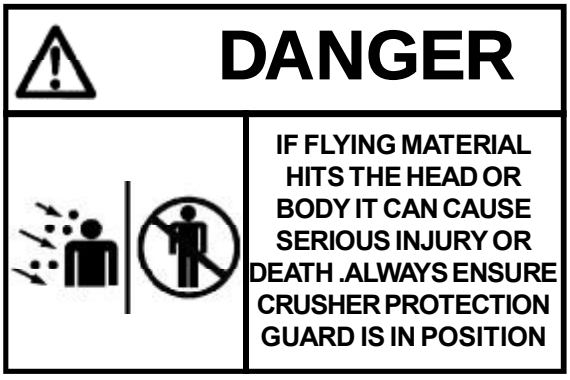
Ladder in Transport Position

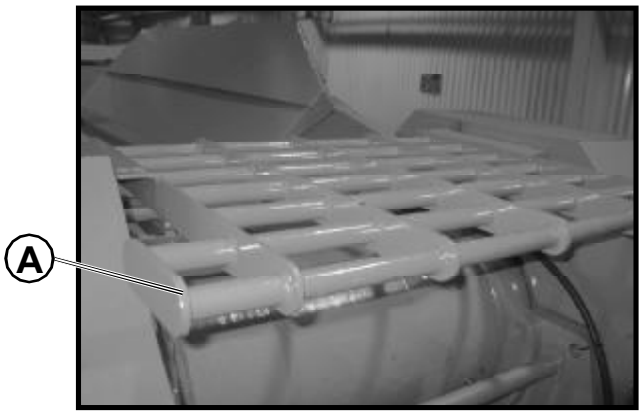
Ladder in Operating Position

3.11 CRUSHER PROTECTION GUARD

Your machine will arrive with the crusher guard in the transport position this is required to keep machine height to minimum for transport.

(i) Rotate the crusher guard *(Item A)* from the transport position . And secure into the Working position *(Item B).*





Crusher Guard in Transport Position

Crusher Guard in Working Position

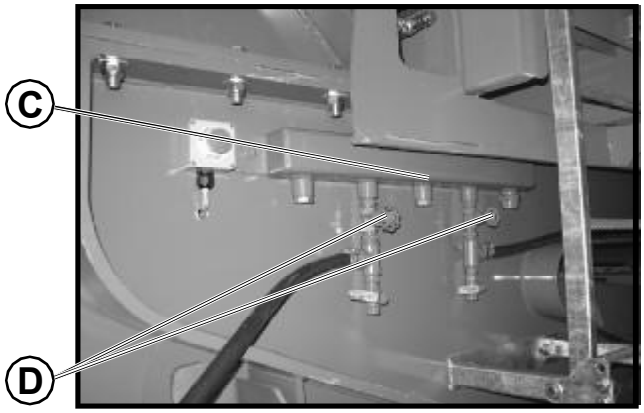
3.12 DUST SUPPRESSION

Dust Suppression is fitted on two locations, one on the feed in to the crusher and one on the main conveyor discharge.

To function:

(i) Connect a clean water supply to the Dust Suppression Manifold *(Item C)*.

(ii) The Gate valves *(Item D)* control the flow of water to the Dust suppression



Dust Suppression Manifold

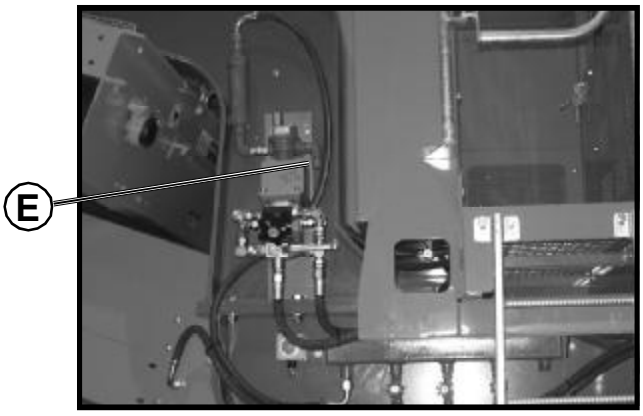
3.15.1 WATERPUMP(Option)

If a hydraulic waterpump is connected to the manfold. Connect the water supply to the inlet *(Item F)*

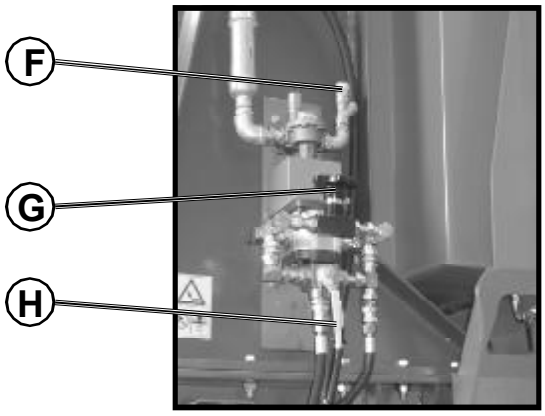
The oil Supply to the pump is controlled by the ball valve *(Item H).*
ON POSITION-Ball Valve at 6 o clock position *(Item H)*
OFF POSITION - Ball valve at 3 o clock position.

The water pump must not run without water. This will destroy the pump. Turn ball valve OFF if not in use.

The water pressure is contolled by the valve *(Item G).*



Waterpump Dust Suppression



Waterpump Dust Suppression

Contents

4.1 SAFETY INTRODUCTION ..... OP - 2

4.2 SAFETY BEFORE AND DURING OPERATION ..... OP - 11

4.3 PRE-OPERATION CHECK ..... OP - 12

4.4 INITIAL START-UP AND RUNNING IN ..... OP - 13

4.5 BLOCKED CRUSHER ..... OP - 14

SECTION 4 : OPERATION

4.1 SAFETY INTRODUCTION

1. Entanglement Hazard.
   Do not reach into an unguarded machine you can be pulled in.
2. In-running nip points can cause serious injury or even death.
   Your arm could be pulled in and amputated.
3. Switch off and lockout machine before opening or removing guard doors.

1. Beware of escaping Hydraulic oil. High pressure hydraulic oil can penetrate the skin causing serious injuries.
2. Always relieve pressure from the hydraulic system before attempting maintenance or adjustment. Use a piece of cardboard to check for oil leaks.
3. If oil is injected under the skin, it must be surgically removed or gangrene will result. Get medical help immediately.

DANGER

1. Electrical Hazard.
   Risk of serious injury or death.
2. Never work on the electrical system of any equipment unless you are thoroughly familiar with system details.
3. Work on the electrical system must only be carried out by a qualified electrician.

DANGER

1. Nip Hazard. Do not reach into an unguarded machine.
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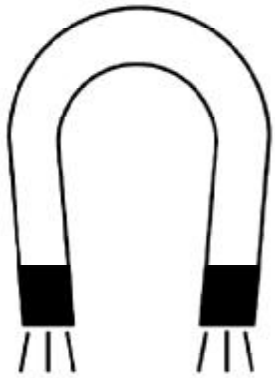


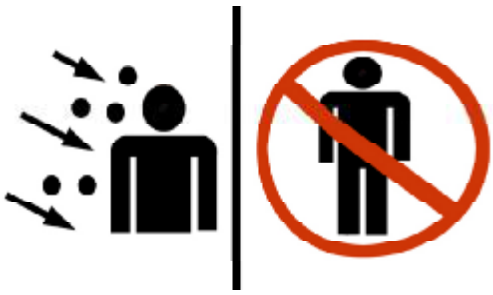
1. Electrical Hazard.
   Risk of serious injury or death.

2. Magnetic fields will affect personnel with pacemakers (within 3 metres), medical electronic implants or metallic prostheses.

3. Ferrous tools and materials could be attracted to the magnet with considerable force.

4. Do not place credit cards or watches near the magnet, they could be permanently damaged

DANGER

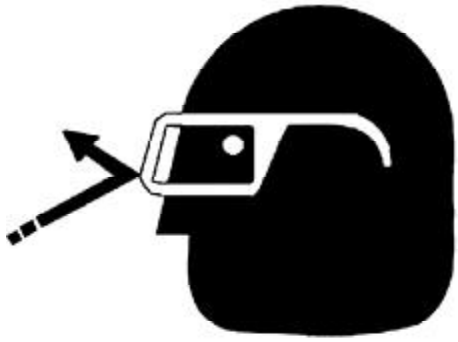


1. flying material Hazard.
   Risk of serious injury or death.

2. Platform is for Maintenance purposes only. Do not use platform when the plant is in operation

3. If Flying material hits the head or body it can cause serious injury or death .

DANGER

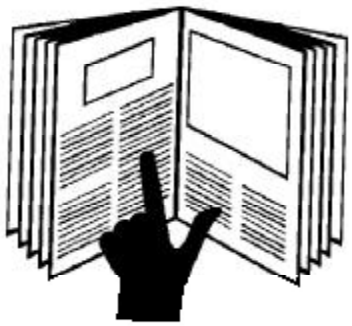


1. Flying material Hazard.
   Risk of serious injury or death.
2. Wear proper eye personalprotective equipment.
3. .Flying material can cause loss or degradation of eye sight.

DANGER

1. Crush Hazard.
   Risk of serious injury or death.
2. Keep clear of Track frames when the unit is in operation.
3. Death or serious injury can result from contact with the Track frames.

⚠ WARNING



1. Read and understand the operators manual before you work at this machine.
2. Study the different safety signs on this machine.
3. Only qualified and trained personnel should attempt to work at this machine.

⚠ WARNING

1. Loose or baggy clothing can get caught in running machinery.
2. Always wear correctly fitting (EN/ANSI approved) protective clothing.
3. Protective clothing includes Hard Hat, Safety Glasses, Ear Defenders, Overalls, Gloves, Steel Toed Boots and a High Visibilty Vest.

WARNING

1. Falling Rock Hazard.
   Do not walk near material discharge areas.
   You are at risk of serious injury or death.

2. Wear a hard hat at all times when working in the vicinity of the machine.

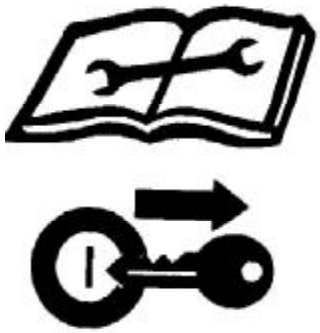
WARNING

1. Crush Hazard.
   Beware of dangerous Hand and Foot crush points.

2. Do not use any body parts e.g. hands or feet to check hole alignment or support machinery.

3. Always use pins and/or mechanical supports provided.

WARNING



1. When performing maintenance or adjustment to this machine always switch off engine and remove the ignition key, (if electric isolate power supply at source).
2. Keep ignition key on person during maintenance.
3. Never work alone.

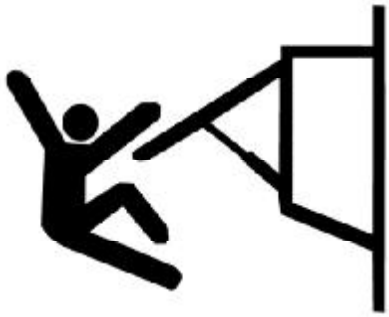
WARNING

1. Fall Hazard.
   Do not climb onto moving or working machinery. You could fall and be seriously injured.
2. Switch off and lockout machine before attempting maintenance or adjustment to this machine.
3. Always use a suitable lifting platform before attempting any maintenance work above 2 metres(6-6”).

WARNING

1. Burn or scald Hazard.
   Beware of hot surfaces at engine and exhaust areas. Be careful when draining hot coolants there is a risk of scalds.

WARNING



1. Impact Hazard.
   Beware of spring loaded doors opening unrestrained, facial injuries could result.

Lockout procedure.

WARNING

1. When performing maintenance or adjustment to this machine always isolate the machine and Lockout.
2. Keep lock on person during maintenance.
3. Never work alone.

The Isolator Switch *(Item A)* is located beside the battery box, the switch cuts the electric supply from the battery ensuring the machine cannot be operated. The machine must always be isolated prior to any maintenance work being carried out.

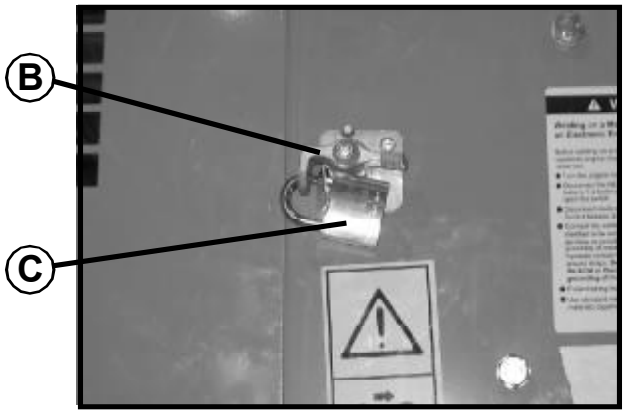
Isolator Switch "ON" Machine Operational

To Lockout:

(i) Rotate the switch *(item A)* to the "OFF" position *(Item B)*.

(ii) Attach a padlock *(Item C)* to secure in the lockout position.

(iii) During maintenance work always ensure the isolator is locked so the machine cannot be started accidently.



4.2 SAFETY BEFORE AND DURING OPERATION

Only qualified and trained personnel should attempt to operate this machinery.

1. Read and understand the Operators Manual before you start the machine.
Study the DANGER, WARNING, CAUTION and IMPORTANT safety signs on your machine.

2. Check your machine and have all systems in good operational condition.

- Check for broken, missing or damaged parts. Replace and make necessary repairs.
- Ensure all safety guards and emergency stops are fitted and in working order.
- See 'Daily Maintenance' for more detailed checklist before start-up.

3. Before starting, walk completely around the machine. Make sure no one is under it, on it or close to it. Let other workmen and bystanders know you are starting up and do not start until everyone is clear of the machine.

4. During operation, do not climb onto, over or under moving conveyor belts and rollers.
Always use ladders, steps and walkways when mounting and dismounting.

5. Hole alignment on mechanical supports must be checked and secured with pins provided and in accordance with safety signs.

6. Never check the tension of 'V' belts, drive chains and conveyors when machine is running.

7. Never work or stand beneath machinery or attachments as it is raised or lowered.

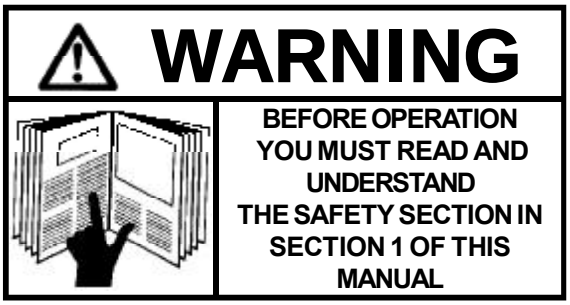
8. Never work or stand beneath machinery as they are being loaded with and/or discharging material.

9. Follow safe operating practices. Operate the machine controls smoothly. Avoid sudden stops, starts or changes in direction.

10. Only use emergency stop buttons or emergency stop lines *(if fitted)* in emergency situations or during safety drills.

11. After each day's operation, always run the machine dry; never leave material in the beltfeeder on conveyor belts or screenbox. Starting a machine with a full load will cause strain problems in your machine.

4.3 PRE-OPERATION CHECK



(i) Prior to operation ensure the machine is positioned on a level firm foundation with adequate space around the machine for operation, material stockpiles, maintenance and vehicular movement.

(ii) Ensure the material being feed into the crusher is below the size limitations recommended by the manufacturer.

(iii) Ensure all safety guards are in position.

(iv) Observe all safety instructions and ensure the correct protective clothing and equipment are used by operators.

(v) Check the Grizzly unit oil level and hydraulic oil level.

(vi) Chech the oil cooler fan and radiator for any build up of dust/dirt. Blow out dust/dirt if necessary.

4.4 INITIAL START-UP AND RUNNING IN

Introducing material to the Machine.

Some operators build a ramp at the side of the machine to give their shovel loader a better reach into the Hopper.

If you do build a ramp avoid contact with the side of the machine.

IMPORTANT

AVOID ANY CONTACT WITH THE SHOVEL, EXCAVATOR OR RAMP AGAINST THE SIDE OF THE MACHINE

Initial feed rate should be low gradually increasing up to normal rate over first days of operation.

During this period the belt pressure on each conveyor should be checked to ensure they are running at the correct pressure *(Max pressure is 210 Bar-3050 psi)(By service personnel only)*.

After initial period stop the machine and visually check belts, scrapers, skirting rubbers, rollers and nip guards for abnormal wear and obstructions.

Conveyor belts may behave different with material on them, retracking may be necessary *(see section 5)*.

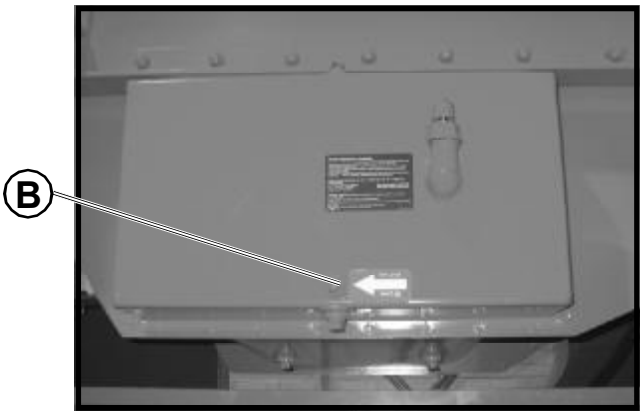
Check the quality of the crushed material.

Check the crusher vee belts*(Item A)* regularly for tension, particularly over initial few weeks of operation.

Frequently check the oil levels *(Item A)* in the vibrating unit.

Hopper Loading Area

Crusher Vee Belt Guard



Feeder Oil Level

DANGER

SWITCH OFF AND LOCKOUT BEFORE WORKING AT A MACHINE

DANGEROUS NIP POINTS EXIST

HIGH PRESSURE HYDRAULIC FLUID CAN PENETRATE THE SKIN CAUSING SERIOUS INJURIES

4.5 BLOCKED CRUSHER

It is the responsibility of the Owner, Machine Operator and safety officer of the *Terex® MPE* crusher to provide a safe means of unblocking a stalled crusher unit.

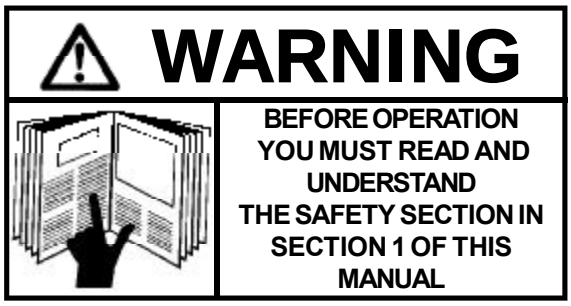
To comply with Health and safety law it is necessary to carry out adequate risk assessments of the procedures for unblocking a crusher.

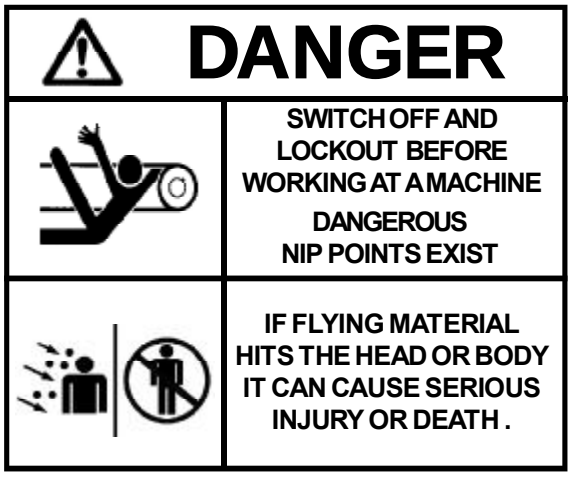
If the NFJ1175 Crusher begins to come under load the crusher hydraulic pressure in-creases and will cut the feeder off automati-cally after 3 seconds, this time delay allows the crusher more time to process the mate-rial. If after a further 7 seconds the crusher is still blocked and under pressure the crusher will come to a stop. The crusher has the facility to be run in reverse, to clear a jam. To reverse the crusher follow the procedure 4.10.1

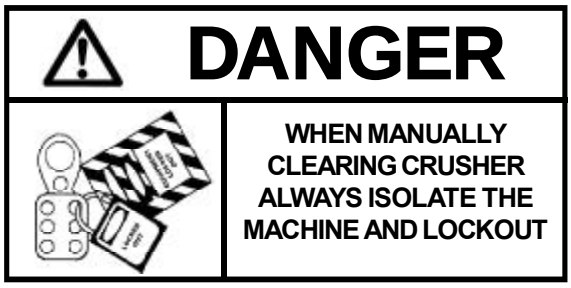
If the crusher stops during operation i.e Emergency stop activated or lack of fuel, the crusher may not be able to continue without clearing the crusher unit by reversing the crusher. *see 4.10.1*

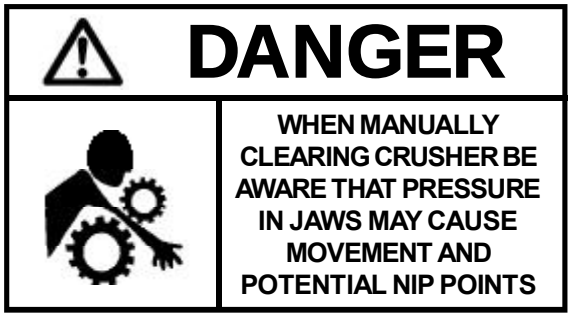
In a small number of instances the crusher reverse procedure may not be able to clear the crusher this is more likely if tramp metal is causing the blockage. In this situation the complete **machine must be shut down and locked out**. The blockage can then be cleaned manually.

When clearing the crusher by hand extreme caution must be taken, pressure in the jaws may cause the crusher to move after mate-rials have been remove, dangerous nip points exist.









4.5.2 Stalled Crusher Procedure

In the situation where an oversize or tramp piece of material causes the crusher to stall, and the toggle plate does not fail, it is possible that very high pressure will be sustained on the moving swing jaw (dependent on which position the stall occurred in the crushing cycle). This pressure may result in the object to be ejected upwards under high force and therefore will need to be released under controlled conditions to enable safe removal of the material that has created the stall condition.

⚠ WARNING!

CUTTING THROUGH THE TOGGLE PLATE WILL CAUSE THE RELEASE OF STORED ENERGY IN THE MOVING JAW WHICH MAY SWING VIOLENTLY BACK TOWARDS THE OPERATOR PERFORMING THE CUTTING OPERATION

BEFORE ANY ATTEMPT TO CARRY OUT THE PROCEDURES TO CLEAR THE STALLED CONDITION OF THE CRUSHER THE MACHINE MUST BE ISOLATED FROM ALL SOURCES OF SUPPLY AND THE SAFETY GRID AT THE MOUTH OF THE CRUSHER MUST BE SECURELY FASTENED IN PLACE. SUITABLE PERSONAL PROTECTION EQUIPMENT MUST BE WORN.

IMPORTANT

The procedure indicated below or any other procedure developed to remove tramp metals, or any other material that results in the stalling of the crusher should be incorporated into a ***permit to work*** procedure. This procedure should be maintained under the control of the component person appointed by the management.

The system should cover all factors of safety including notification to the manager, isolation of systems and locking out procedure, methods of removal, safe positions for persons and any other precautions as may be deemed necessary.

Procedure

1. Observe all safety warnings.
2. Isolate machine from all energy sources.
3. Ensure the safety grid is over the mouth of the crusher.
4. If the crusher is equipped with hydraulic drive, attempt to start the machine in reverse.

5. If the crusher is equipped with a hydraulic wedge adjust system, attempt to release the pressure on the jaw by pressing the jaw open button.

6. If none of the above operations are applicable, or successful, then the toggle plate must be cut. Before the cutting operation is commenced adequate and appropriate provision **MUST** be made to support the weight of the stored energy in order to prevent the violent swinging movement of the jawstock.

7. The toggle plate should heated across the centreline (i.e. between the holes), using a long-handled 'pepper pot' or similar, such that the operator can stand in a protected position, well clear of the back of the crusher frame. Heat the material to a dull red - starting at the mid-point, and working outwards either side in turn. Ideally, the toggle plate should be heated until it yields. The yielding of the toggle plate will reduce or eliminate the pressure on the trapped object. If the plate does not yield, then it is recommended that small cuts are made in either side of the toggle plate with an acetylene torch (say 50mm from each side) and the heating procedure repeated until yielding occurs. After the pressure is released due to the toggle plate yielding, the toggle plate should be cut in two and removed.

CONTENTS:

5.1 SAFETY INTRODUCTION ..... MA - 2
5.2 SAFETY BEFORE AND DURING MAINTENANCE ..... MA - 11
5.3 HYDRAULIC SAFETY ..... MA - 13
5.4 ELECTRICAL AND ENGINE SAFETY ..... MA - 14
5.5 HYDRAULIC SYSTEM ..... MA - 15
5.6 HYDRAULIC FLUIDS ..... MA - 16
5.7 FILTRATION ..... MA - 17
5.8 OILTANK, PIPES & FITTINGS ..... MA - 19
5.9 DIESEL ENGINE ..... MA - 20
5.10 CRUSHER ..... MA - 20
5.11 FEEDER ..... MA - 20
5.12 BELT TRACKING ..... MA - 21
5.13 BELT TRACKING MAIN CONVEYOR ..... MA - 22
5.14 BELT TENSIONING ON MAIN CONVEYOR ..... MA - 22
5.15 BELT TRACKING DIRT CONVEYOR (OPTIONAL) ..... MA - 23
5.16 BELT TENSIONING DIRT CONVEYOR (OPTIONAL) ..... MA - 23
5.16 CHANGING HYDRAULIC FILTERS & OIL ..... MA - 24
5.17 CHANGING ENGINE FILTERS & OIL ..... MA - 29
5.18 CHANGING FUEL FILTER ..... MA - 32
5.19 CHANGING GRIZZLY FEEDER GEARBOX OIL ..... MA - 33
5.20 CHANGING GEARBOX OIL IN TRACKS ..... MA - 34
5.21 TRACK ADJUSTMENT ..... MA - 35
5.22 CRUSHER MAINTENANCE ..... MA - 36
5.23 CHANGING FEEDER MESH ..... MA - 39
5.24 CHANGING MESH ON PRE-SCREEN (OPTION) ..... MA - 42
5.25 RECOMMENDED LUBRICANTS ..... MA - 43
5.26 GREASING SCHEDULE ..... MA - 45
5.28 SAFELY REMOVE & INSTALL BATTERY ..... MA - 47
5.29 MAINTENANCE SCHEDULES ..... MA - 50

SECTION 5 : MAINTENANCE

5.1 SAFETY INTRODUCTION

1. Entanglement Hazard.
   Do not reach into an unguarded machine you can be pulled in.
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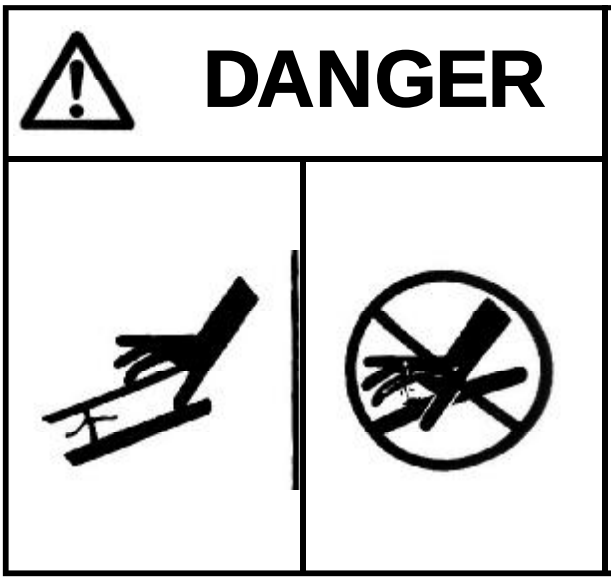
3. Switch off and lockout machine before opening or removing guard doors.



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2. Never work on the electrical system of any equipment unless you are thoroughly familiar with system details.
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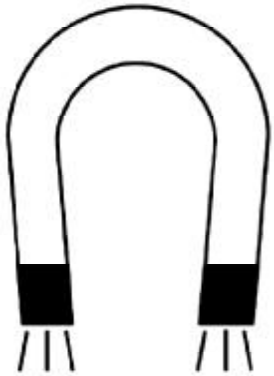

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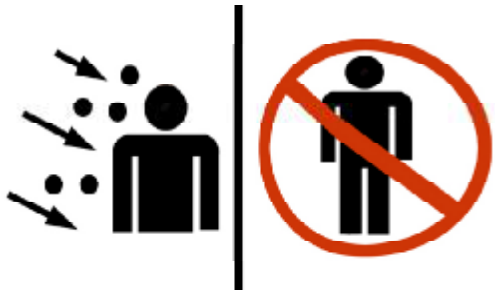

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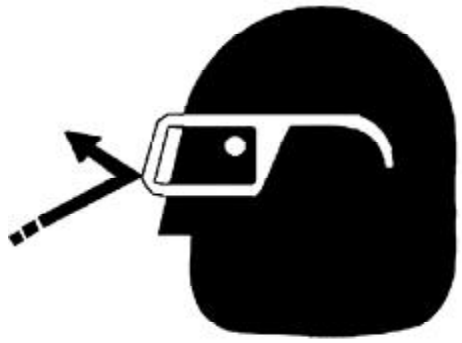

1. Flying material Hazard.
   Risk of serious injury or death.
2. Wear proper eye personalprotective equipment.
3. .Flying material can cause loss or degradation of eye sight.

DANGER

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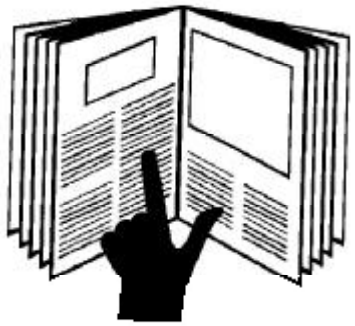
⚠ WARNING



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2. Study the different safety signs on this machine.
3. Only qualified and trained personnel should attempt to work at this machine.

⚠ WARNING

1. Loose or baggy clothing can get caught in running machinery.
2. Always wear correctly fitting (EN/ANSI approved) protective clothing.
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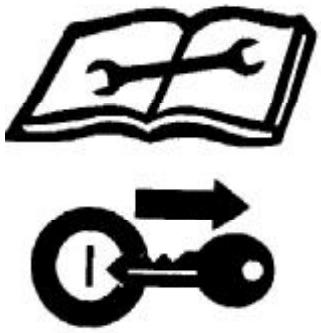

1. When performing maintenance or adjustment to this machine always switch off engine and remove the ignition key, (if electric isolate power supply at source).
2. Keep ignition key on person during maintenance.
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WARNING

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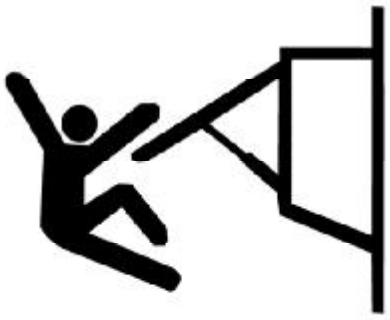
1. Burn or scald Hazard.
   Beware of hot surfaces at engine and exhaust areas. Be careful when draining hot coolants there is a risk of scalds.

WARNING



1. Impact Hazard.
   Beware of spring loaded doors opening unrestrained, facial injuries could result.

Lockout procedure.

WARNING	
	1. When performing maintenance or adjustment to this machine always isolate the machine and Lockout. **2.** Keep lock on person during maintenance. **3.** Never work alone.

The Isolator Switch *(Item A)* is located beside the battery box, the switch cuts the electric supply from the battery ensuring the machine cannot be operated. The machine must always be isolated prior to any maintenance work being carried out.

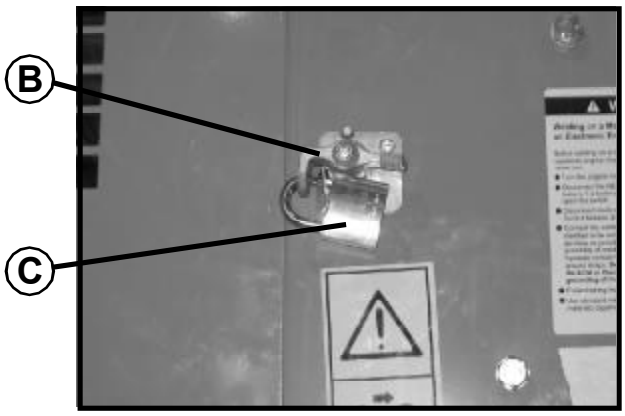
To Lockout:

(i) Rotate the switch *(item A)* to the "OFF" position *(Item B)*.

(ii) Attach a padlock *(Item C)* to secure in the lockout position.

(iii) During maintenance work always ensure the isolator is locked so the machine cannot be started accidently.

Isolator Switch "ON" Machine Operational



5.2 SAFETY BEFORE AND DURING MAINTENANCE

Maintenance should only be carried out by trained and qualified personnel.

1. Whenever maintenance or service is being carried out a minimum of two *(2)* persons should be present at all times. NEVER WORK ALONE.

2. Prepare yourself. Wear a hard hat, safety goggles, hearing protection and other protective equipment as required by job conditions. Do not wear loose clothing or jewellery that can catch on controls or moving parts. Long hair must be tied back.

3. Prepare the machine. Move the machine onto a level surface and apply parking brakes and/or wheel chocks. Shut off the engine and remove the key. Relieve all hydraulic pressure by returning controls to neutral. Secure all hydraulically operated attachments with pins provided.

4. Isolate all electrical supplies to the machine before starting any maintenance work.

5. Never attempt repairs or adjustments to the machine while it is running. Exempt to this rule: Belt tracking adjustments are only possible during working process *(see 5.11-5.12 Belt Tracking).*

6. Remove only guards or covers that provide access. Wipe away excess grease and oil.

7. Never leave guards off or access doors open when unattended. Keep bystanders away if access doors are open.

8. When working beneath raised equipment, always use blocks, jack-stands or other rigid and stable supports. Never work under unsupported equipment.

9. Performing work above 7ft *(2m)* always use an approved *(EN/ANSI)* safety harness. Always use walkways and platforms provided or a safe secure platform approved by the machine operators management. Do not use any unauthorised or unsafe platforms.

10. Never operate any type of engine without proper ventilation – EXHAUST FUMES CAN KILL. *(See Electrical and Engine Safety for more detailed checklist).*

11. Checking for hydraulic leaks. Beware hydraulic fluid under pressure can penetrate the skin or damage eyes. Fluid leaks under pressure may not be visible. Use a piece of cardboard to find leaks but do not use bare hand.

12. Wear safety goggles for eye protection. If fluid enters skin or eye, get immediate medical attention. *(See Hydraulic Safety for more detailed checklist).*

13. Clean or replace damaged, missing or painted over safety signs that cannot be read.

14. Rotating and moving parts must be inspected during maintenance and replaced if cracked or damaged. Excessively worn or damaged parts can fail and cause injury or death.

15. After maintenance, tighten all bolts, fittings and connections.
Install all guards, covers and shields. Replace or repair any damaged ones. Refill and recharge pressure systems with recommended fluids.
Start the engine and check for leaks. Operate all controls and make sure the machine is functioning properly. After testing, shut down, check the work you performed *(any missing cotter pins, washers, locknuts, etc.)*?
Recheck all fluid levels before releasing machine for operation.

16. NEVER make any modifications, additions or conversions which might affect safety without the supplier's approval.

5.3 HYDRAULIC SAFETY

1. Never disable or alter any hydraulic circuit or component without consulting with *NFLG* or your local dealer.

2. Any work carried out on a hydraulic system must be performed by an engineer with special knowledge and experience in hydraulics.

3. Relieve all pressure in hydraulic system by returning controls to the neutral position. Turn machine off and isolate power supply before any pipes, filter caps, filters or hydraulic fittings are disconnected or removed.

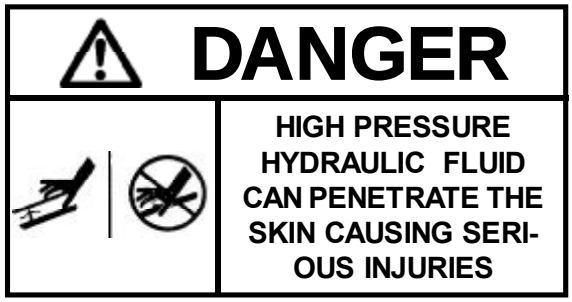
4. Ensure replacement hydraulic components are the same standard and quality as the original parts. Ensure they are fitted correctly.

5. Beware of hydraulic oil leaks. Hydraulic fluid under pressure can penetrate the skin or damage eyes. Fluid leaks under pressure may not be visible. Use a piece of cardboard to find leaks but do not use bare hand.

6. Wear safety goggles for eye protection. If fluid enters skin or eye, get immediate medical attention.

7. Replace any damaged hydraulic pipe. Leaks can cause fires. Always ensure a replacement pipe has the same specification as the original pipe, length, fittings, orientation and diameter. Ensure pipes are laid and fitted correctly.

8. Do not exceed safe limits. Never set a pressure relief valve to a pressure higher than that set at the factory, 172 Bar *(2500 PSI).* The Tracks are set at 245 Bar *(3550 PSI).*



5.4 ELECTRICAL AND ENGINE SAFETY

1. Never work on the electrical system of any equipment unless you are thoroughly familiar with system details.

2. Work on the electrical system must only be carried out by a qualified electrician.

3. If machine is electric-hydraulic or direct electric drive, the 'mains' electrical supply to the machine should always be isolated by unplugging/uncoupling the 'mains' power socket.

4. The electrical equipment of the machine should be inspected at regular intervals. Damaged cables or loose connections must be corrected immediately. Use only original fuses with the specified current rating.

5. It is recommended that an earth leakage safety switch be fitted in the supply line to the power point on site. Special care should be taken to ensure that earth wires are correctly connected.

6. Disconnect the battery ground leads whenever performing any electrical maintenance or servicing to your engine.

7. Battery posts, terminals and related accessories contain lead and lead compounds, chemicals known to the State of California to cause cancer and reproductive harm. Wash hands after handling.

8. Never short across the starter terminals of a battery as this can cause a fire and could also damage the electrical system.

9. Battery electrolyte contains acid and can cause serious injury. Avoid contact with skin and eyes. Use gloves and protective glasses.

10. Diesel engine exhaust emissions contain products of combustion which may be harmful to your health. Always operate the machine in a well ventilated area and if operating in an enclosed area, vent the emissions outside.

11. Do not touch any part of the engine or exhaust system. Allow the engine and exhaust to cool before performing any repair or maintenance.

12. Never fill the fuel tank with the engine running, while smoking or when near an open flame.

13. Never overfill the tank or spill fuel. If fuel is spilled, clean it up immediately.

5.5 HYDRAULIC SYSTEM

The Hydraulic system which is used in your machine was chosen for its effectiveness and resistance to climatic and operating conditions.

This must be kept topped up with the correct hydraulic fluid and regular checks must be made to ensure this.

The filter element requires changing when the indicator is in the red at operating temperature.

The system should be checked for leaks and the hydraulic oil analysed after 1000 hours and every 500 hours afterwards and replaced if necessary.

WARNING

BEFORE MAINTENANCE YOU MUST READ AND UNDERSTAND THE SAFETY SECTIONS IN THIS MANUAL

5.6 HYDRAULIC FLUIDS

Hydraulic fluids play an important part in any hydraulic system. They have two main functions.

> To transmit power
> To lubricate moving parts

As a power transmitting medium the fluid must flow easily and be as incompressible as possible.
In most hydraulic components, the fluid provides internal lubrication only. For long component life, fluids are available containing additives that have high anti-wear properties. The fluids are known as anti-wear type hydraulic oils, which are recommended for your machine.
In most cases the fluid is the only oil seal present. For example, there are no sealing rings between the spool and the body of the directional valve. As the sealing characteristics of the fluid depend on its retaining viscosity, it is important that the oil selected is capable of maintaining the minimum viscosity change over a wide range of operating temperatures.

Some adverse effects of incorrect viscosity are:

> Increased power consumption
> High oil temperatures
> Entrapment of air in oil,
> Increase pressure drop.
> Excessive wear or even seizure under heavy loads,
> poor pump performance.

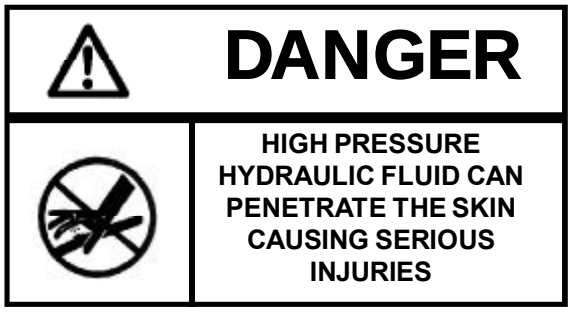
In addition to the qualities outlined earlier the selected fluid should also:

> Prevent rot
> Depress foaming
> Prevent formation of sludge, gum and varnish
> Retain its own stability, thereby reducing fluid replacement.
> Prevent corrosion and pitting.

For best operating results it is strongly recommended that the hydraulic fluid used is selected from the recommended range listed in *5.26 "Recommended Lubricants".*

In the service it is recommended that the hydraulic system is analysed every 500 hours and the oil changed if necessary. The hydraulic fluid should be changed after the machine has stopped work. The drain plug for the hydraulic system is mounted underneath the tank. In the event of pump or motor failure, both suction line and return line filters must be changed.

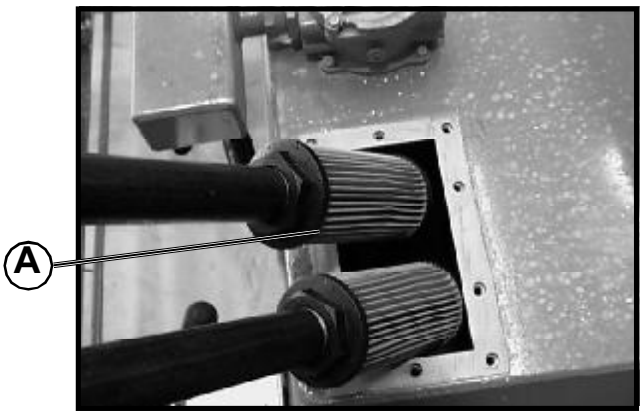
5.7 FILTRATION



In order to maintain the hydraulic oil at its peak performance, five different types of filter namely:

(ii) Suction line filters*(Item A)*
(iii) Return line filters*(Item B)*
(iv) Pressure Filter *(Item C)*
(v) Contamination Control Breather*(Item D)*

are fitted to the hydraulic system.



Suction Line Filters

Suction Line Filter:

5x 125 microns filter, which are matched to pump capacity, are fitted inside the hydraulic tank where they are readily accessible.
New filter element should be fitted at regular intervals (see routine maintenance schedule for frequency).

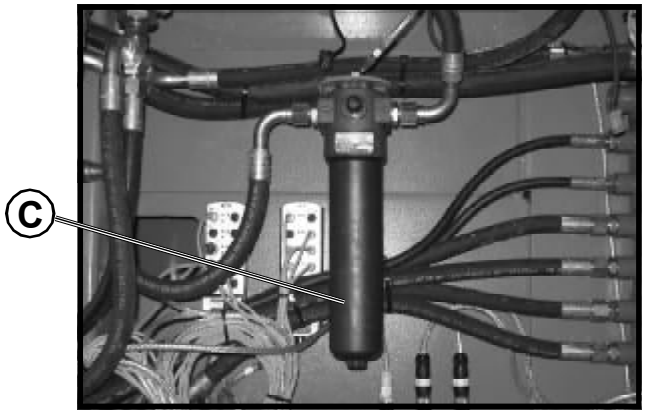
Return Line Filters

Return Line Filter:

The 2 x 10 micron filters, which are fitted on the tank, remove any tiny particles which can cause damage to the pumps and motors etc.
The filter element should be changed in accordance with the routine maintenance schedule.

Pressure Filter:

This 6 micron filter remove any tiny particles which can cause damage to the pumps and motors in the closed loop system.
The filter element should be changed in accordance with the routine maintenance schedule.



Pressure Filter

Condensation Control Breather

When the hydraulic oil in the tank is lowered, or pressure changes occur, air is drawn in through opens in the breather. First air passes through a fine, 2 micron solid particle filter. The air then passes through a diffuser to ensure maximum effectiveness within the silica gel chamber. Next water vapour is removed as the the air passes through a bed of silicon gel, the air then passes through a 2 micron particle filter and enters the tank clean and dry.

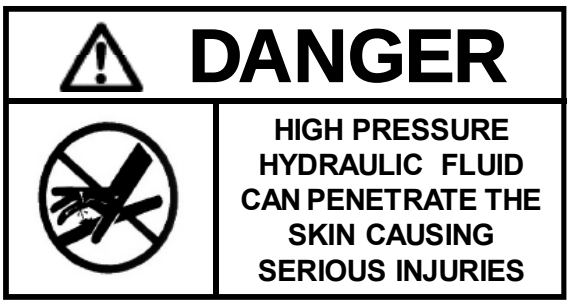
Condensation Control Breather

DANGER

HIGH PRESSURE HYDRAULIC FLUID CAN PENETRATE THE SKIN CAUSING SERIOUS INJURIES

5.8 OILTANK, PIPES & FITTINGS

In order to prevent contamination, the Oil Tank on your machine is of sealed construction. Access to the tank, if required, is via inspection covers which are mounted on the top of the tank, underneath the filler cap. The outside of the tank must be thoroughly cleaned before the removal of any of these covers.
It is very important that hydraulic pipes and hoses are checked regularly for leaks, scuffing and wear. Loose fittings should be securely tightened. Worn or leaking hoses should be replaced. Replacement hoses are available on request.



5.9 DIESEL ENGINE

The diesel engine is the heart of your machine and a great deal of care and thought has gone into its selection. The engine has a fuel capacity of 36 Litres per hour at 50% power.

In the interest of continued peak performance from your machine the engine must be maintained and cared for with meticulous attention in accordance with the manufacturers instructions as detailed in the "**Operators Handbook**", a copy of which is included with this manual.

Caterpillar C9 water cooled diesel engine

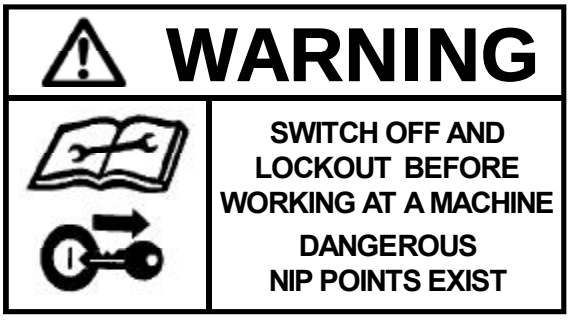
5.10 CRUSHER

The best way to keep the crusher in good operating condition is to adopt a planned program of inspection and preventative maintenance. It is recommended that a record be kept of the actual operating hours of the crusher, the inspection intervals, and the maintenance required. A new crusher should be examined after a relatively short period of operation. Detailed maintance instructions are outlinedin the 'Single Toggle Jaw crusher Instruction manual' a copy of which is included with this manual.

Jaw Crusher

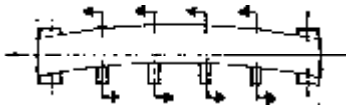
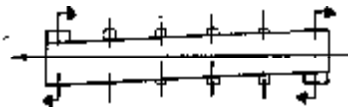
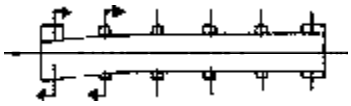
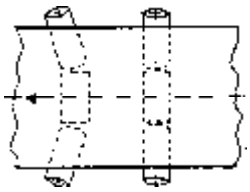
5.11 FEEDER

The greatest single factor extending the life of the vibrating Grizzly feeder is proper lubrication. Insufficent lubrication quickly damages the bearings. Also, excess oil in the vibrating unit increases oil agitation causing high operating temperature and higher horse power requirements. Cleaniness is vitally important. Dust and dirt will cause premature failure of the bearings. Clean all dirt from openings before removing caps. Use only clean oil to fill the vibrator housing.



5.12 BELT TRACKING

Oblique travel may cause fast deterioration of the belt. In order to assist you we have compiled some of the reasons for oblique travel with appropriate remedies.

Reasons for Oblique Travel	Remedies	Examples of Oblique Travel
1. Insufficient aligning of drums and Idlers	**a**. Align all drums, carrying idlers and return idlers at right angles to the travelling direction of the belt. **b**. Check that all drums and idlers are level to each other. If **a&b** are as above then carry out the following. Adjustments are made as indicated on **Figs.1,2&3** by turning drums and carrying idlers in direction of arrow until the belt is running straight and centred on the conveyor. Also return idlers may cause oblique belt travel and therefore they must be aligned at right angles to the travelling direction of the belt and after that adjustment can be made.	 **Fig.1.** Turning carrying idlers.  **Fig.2** Turn drums.  **Fig.3.** Turning both drum and carrying idlers.
2. Misalignment of frame.	Oblique travel can be reduced by turning forward outer carrying idlers 1-3 degrees in travelling direction of the belt. **See Fig.4:** if this does not correct the oblique travel then align the frame.	 **Fig.4** Turning forward carrying idlers.
3. Material built up on drums and idlers.	Clean idlers and drums. Check belt cleaners and replace if necessary. Possibly change to more efficient type.	

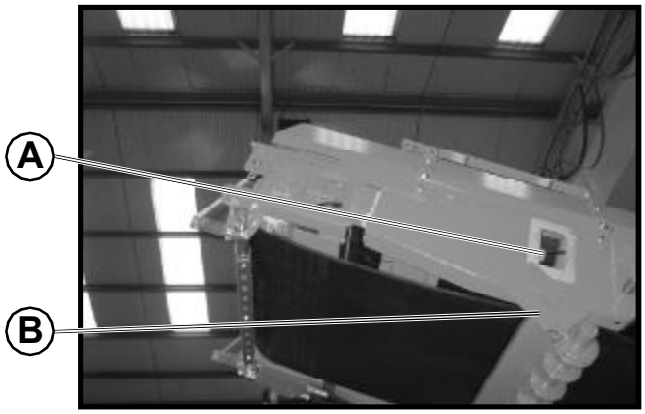
5.13 BELT TRACKING MAIN CONVEYOR

Belt tracking is performed at the Drive drum of the main conveyor. Both sides of the drive drum are adjustable.

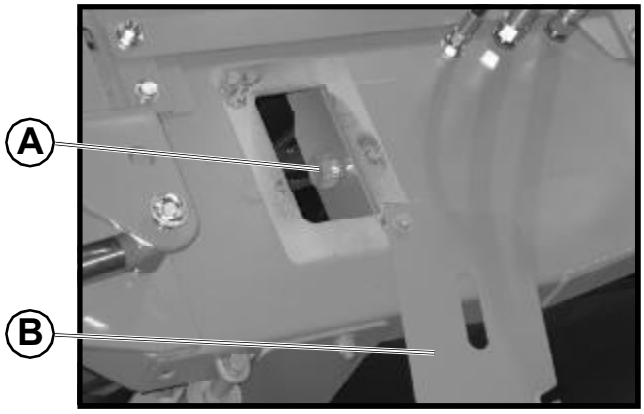
(i) Firstly determine which side requires adjusting.

(ii) To adjust the belt open the cover plate *(Item B)* rotate the nut *(Item A)* on the drum adjuster.
(Use the 7/8"(22mm)spanner-supplied in the toolbox)

A small movement in the adjuster i.e. less than 10mm should be sufficient. A greater movement will effect the tension of the belt.



Main Conveyor Drum Adjuster (Left)



Main Conveyor Drum Adjuster (Right)

IMPORTANT

BELT TRACKING AND TENSIONING ARE THE ONLY MAINTENANCE TASKS PERMITTED WHEN THE MACHINE IS RUNNING

5.14 BELT TENSIONING ON MAIN CONVEYOR

Belt tensioning is performed at the drive drum on main conveyor. Both sides of the drum are adjustable.
By rotating the nut *(Item A)* on both drum adjusters equally you can either tighten or loosen the belt.
(Use the 7/8"(22mm)spanner-supplied in the toolbox). The belt should be sufficiently tight to prevent slippage at the drive drum. If over tight the life of the belt will decrease.

DANGER

DO NOT REACH INTO A MOVING MACHINE DANGEROUS NIP POINTS EXIST

DO NOT CLIMB ONTO OR ACROSS MOVING BELTS FALL HAZARDS EXIST

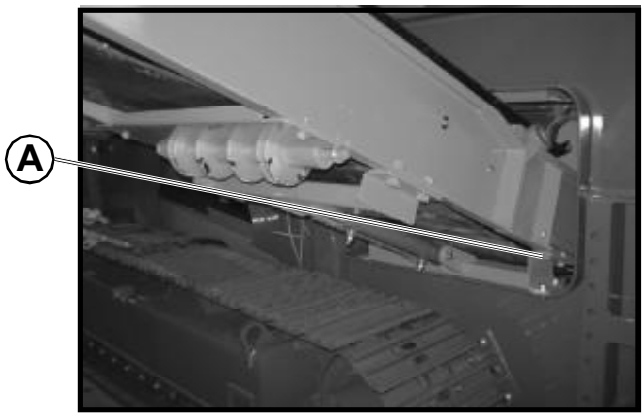
5.15 BELT TRACKING DIRT CONVEYOR (OPTIONAL)

Belt tracking is performed at the Idler drum of the Dirt conveyor. Both sides of the idler drum are adjustable.

(i) Firstly determine which side requires adjusting.

(ii) To adjust the belt rotate the nut *(Item A)* on the drum adjuster.

A small movement in the adjuster i.e. less than 10mm should be sufficient. A greater movement will effect the tension of the belt.



Dirt Conveyor Drum Adjuster (Right)

Dirt Conveyor Drum Adjuster (Left)

IMPORTANT

BELT TRACKING AND TENSIONING ARE THE ONLY MAINTENANCE TASKS PERMITTED WHEN THE MACHINE IS RUNNING

5.16 BELT TENSIONING DIRT CONVEYOR (OPTIONAL)

Belt tensioning is performed at the Idler drum on Dirt conveyor. Both sides of the drum are adjustable.
By rotating the nut *(Item A)* on both drum adjusters equally you can either tighten or loosen the belt. The belt should be sufficiently tight to prevent slippage at the drive drum.
If over tight the life of the belt will decrease.

DANGER

DO NOT REACH INTO A MOVING MACHINE DANGEROUS NIP POINTS EXIST

DO NOT CLIMB ONTO OR ACROSS MOVING BELTS FALL HAZARDS EXIST

5.16 CHANGING HYDRAULIC FILTERS & OIL

It is strongly recommended to change both Suction filter and Return filter elements when the Hydraulic oil is being changed.

5.16.1 Changing Return Filters

(i) Run the engine at full speed and ensure all the hydraulic equipment is operating.

(ii) Rotate open the hydraulic tank access covers *(Item A)* on the roof of the engine compartment at the left and right hand sides.

(iii) Turn engine off and return all control levers to the neutral position.

(iv) Open the Filler cap to release pressure inside the tank.

(v) Clean the area around the Return filters to prevent dirt contaminating the hydraulic oil when the filters are removed.

(vi) Loosen the bolts *(Item B)* on each filter *(Use a 13mm spanner).*

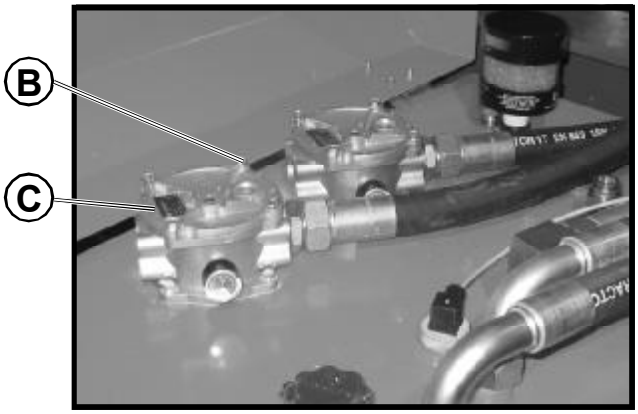
(vii) Rotate the Top plate *(Item C)* slightly to remove.

(viii) Remove the Filter element by pulling the removal handles. Dispose of the old filters safely.

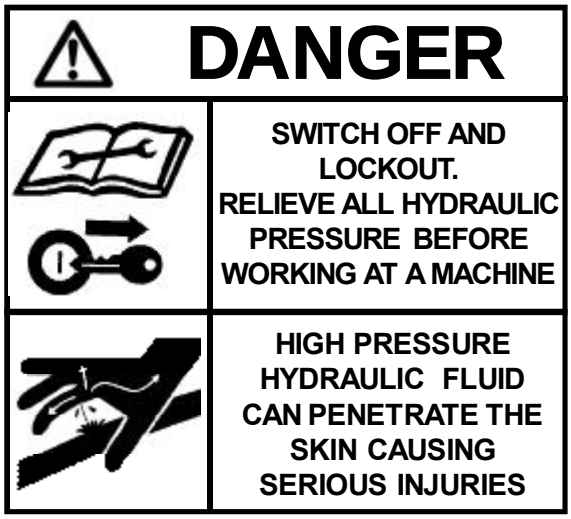
(ix) Insert new filter elements and replace Top Plate *(Item C)*.

(x) Tighten the filter retaining bolts.

Hydraulic Tank Protection Cover



Return Line Filters



5.16.2 Changing Suction Filters

(i) Turn engine off and return all control levers to the neutral position.

(ii) Open the Filler cap *(Item C)* to relieve any pressure inside the tank.

(iii) Remove the bolts *(Item A)(Use a 17mm spanner)* holding the inspection plate *(Item B)* down.

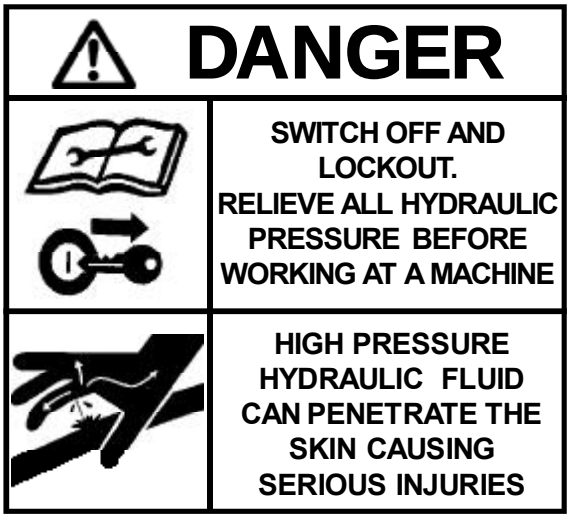
(iv) The Suction filters are fitted to the inside of the inspection cover. To Remove them lift off the inspection cover.

(v) Unscrew the Filters *(Item D)* from the internal Suction pipes *(Item E)* and replace with new filters. Hand tighten the filters.

(vi) Replace inspection covers and filler cap.

Hydraulic Tank Inspection Plate & Filler Cap

Internal Suction Filters (photograph for Illustration purposes not actual machine)



5.16.3 Changing Hydraulic Oil

It is recommended that the machines hydraulic oil is warm.This will enable the oil to flow more easily.

(i) Return all Hydraulic Cylinders to their closed position to return as much hydraulic oil back to the tank as possible.

(ii) Turn engine off and return all control levers to the neutral position.

(iii) Remove the drain plug *(Use a 27mm spanner)* at the base of the tank and drain to an external container large enough to hold all the oil. *(Hydraulic Tank Capacity:730 litres or 193 US gallons).*

(iv) Remove the inspection plate *(Item A)* on the hydraulic tank and flush the tank out with hydraulic oil to remove dirt or other particles. *(Use a 17mm spanner).*

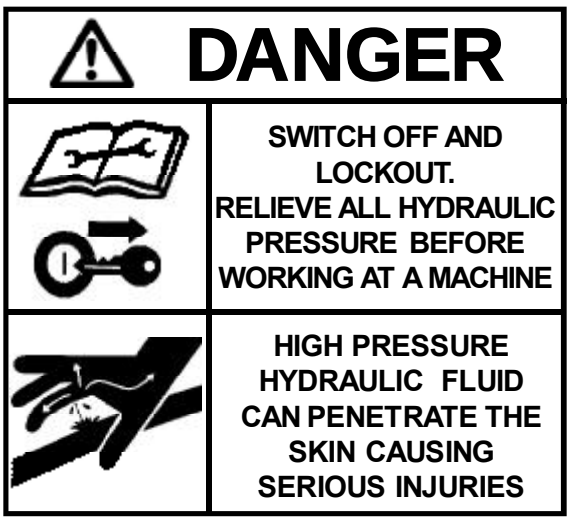
(v) Refill the tank until the oil is between the Min. and max. marks on the tank gauge *(Item B).*

(vi) Run the machine and operate hydraulic equipment to circulate oil.

(vii) Stop the machine and top up the hydraulic system.

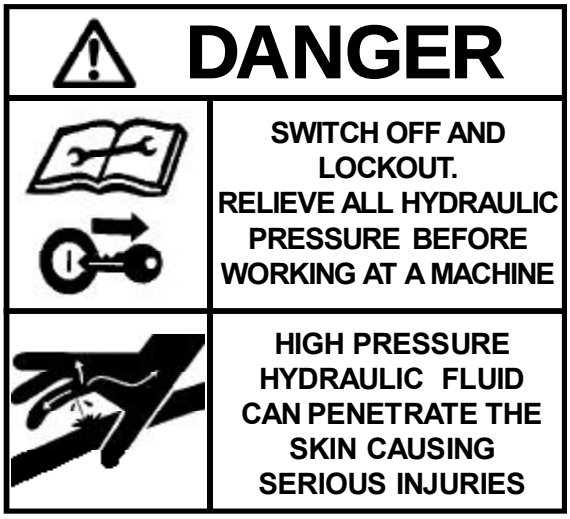
Hydraulic Tank Inspection Plate

Hydraulic Oil Level Guage



5.16.4 Changing Condensation Control Breather

The condensation control breather adsorbs moisure and therefore when maximum adsorption is reached, the silica gel *(Item A)* turns from gold to dark green to indicate that replacement of the breather is required.



5.16.5 Changing Pressure Filters

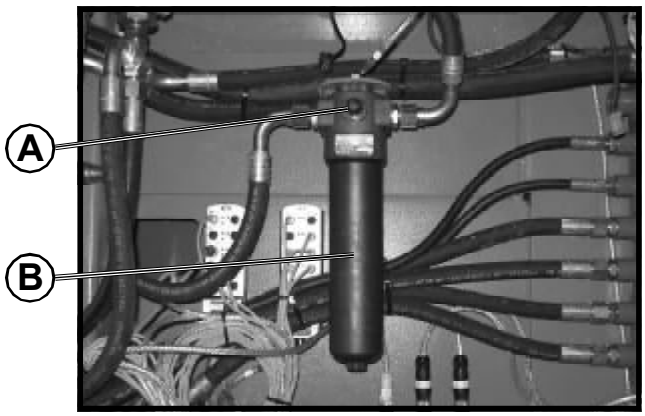
It is recommended that the Pressure Filters are changed every 250Hrs or when the indicator *(Item A)* is coloured red.To Change :

(i) Turn engine off and return all control levers to the neutral position.

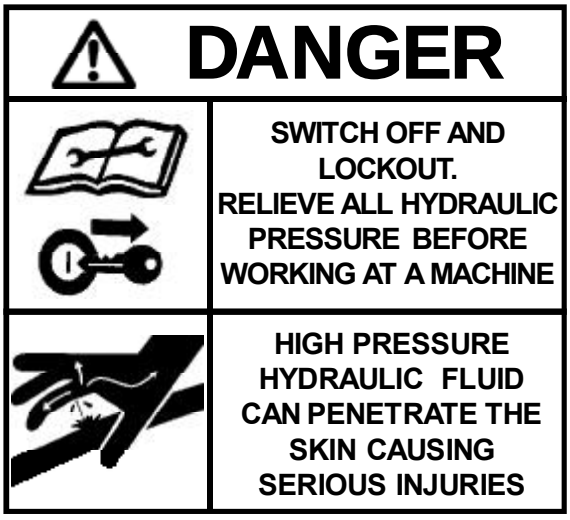
(ii) Screw off the filter housing *(Item B)*

(iii) Remove the filter element and replace with a new element.

(iv) Run the machine and operate hydraulic equipment to circulate oil.



Pressure Filter



5.17 CHANGING ENGINE FILTERS & OIL

(See the engine Operators Manual for more detailed information regarding the Engine)

5.17.1 Checking Engine Oil

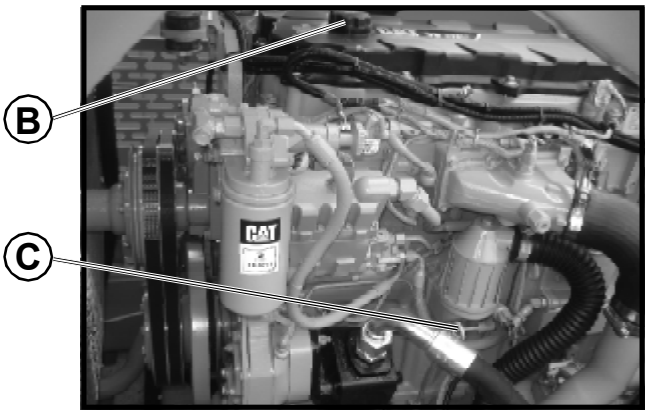
(i) Turn off the engine. Open the Engine Inspection Doors *(Item A).*

(ii) Remove dipstick *(Item C)* from the engine and clean. Re-insert the dipstick fully.

(iii) Remove the dipstick again and observe that the oil level is between the min. and max. levels. If the level is below the minimum mark oil must be added through the fillercap *(Item B)*.

(iv) It is important the oil level is not above the maximum level. If this happens some oil must be removed.

Engine Filters Inspection Door



Engine Oil Filler Cap

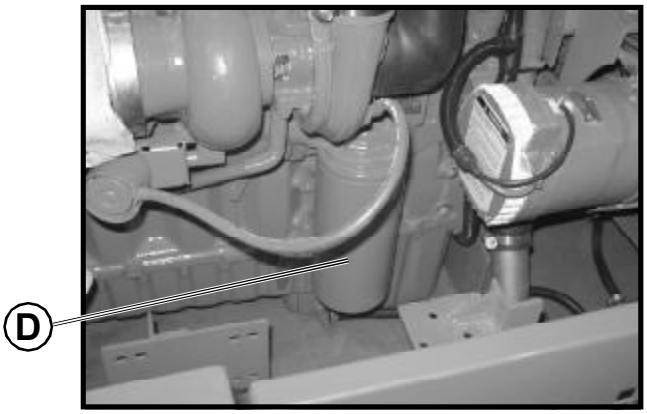
5.17.2 Changing Engine Oil

(i) Operate the engine to ensure the oil is warm, this enables the oil to flow more easily.

(ii) Stop the engine and open the fillercap.

(iii) Remove the drain plug located underneath the engine and drain the oil to an external container large enough to hold all the oil *(Approx Oil capacity: 31litres or 8.2 US gallons).*

(iv) When all the oil has drained off replace the drain plug and refill. Dip the engine. *(See 5.18.1)*

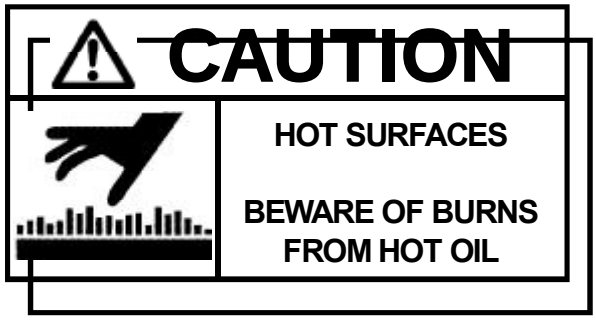


5.17.3 Changing Engine Oil Filter

It is recommended to change the Oil filter when changing the oil.

(i) Loosen the Filter *(Item D)* and dispose off properly.

(ii) Replace with new filter and hand tighten on.



5.17.4 Changing Air Filter Element

(For more details of Engine Air Filter Maintenance view the Engine Operators Manual)

The Air filter requires replacing if the service indicator*(Item A)* becomes red. When this happens the machine must be stopped and the Filter should be changed immediately.
Otherwise change at least every 250Hours for the Primary element and every 500 working hours for the secondary element.

(i) Turn off the engine**.** Unfasten the Filter Cover*(Item C)* by unclipping the holding clamps*(Item B).*

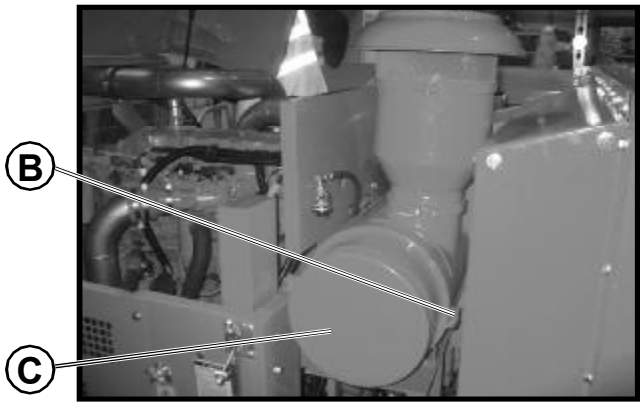
(ii) Inside the Air filter is two elements : **Primary air Cleaner Element** *(Item D)*. The primary Element should be changed every 250 working hours.
Secondary air cleaner element *(Item E)*. The Secondary element should be changed every 500 working hours.

(iv) Inspect the filter if there are any tears or damage to the filter element it must be replaced.

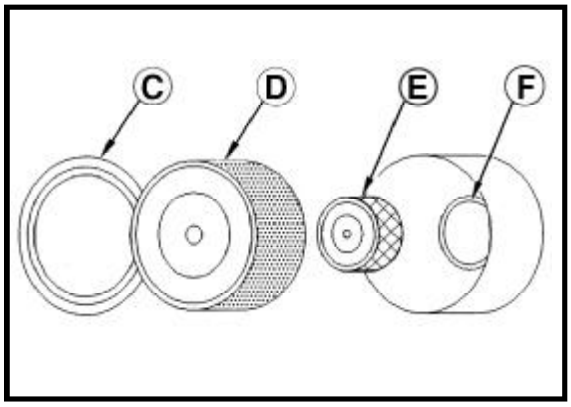
(v) Once Replacing completed re-attach the filter cover and reclip the holding clamps*(Item B)*.

(vi) Reset the Service indicator *(Item A)* by pressing reset button. The service indicator should also be replaced annually.

Service Indicator



Air Filter Housing



Air Filter Components

5.17.5 Checking Engine Coolant Level

(For more details of approved Engine Coolant and Coolant Level view the Engine Operators Manual)

(i) Check the engine coolant level when the engine is stopped and cool.

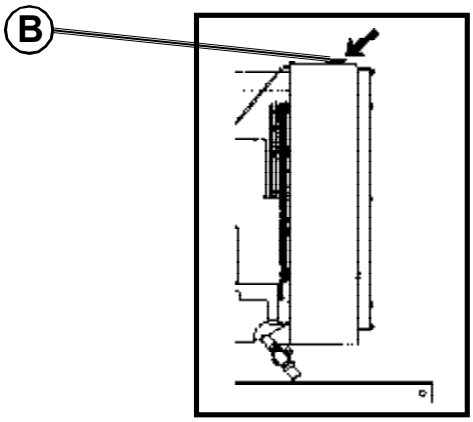
(ii) Open the Radiator Inspection guard *(Item A)*.

(iii) Remove the cooling system filler cap *(Item B)* slowly in order to relieve pressure.

(iv) View the Coolant Level through the top of the radiator.

(v) The coolant level should be maintained within 13mm (0.5 inches) of the bottom of the filler pipe.

Radiator Inspection Doors



WARNING

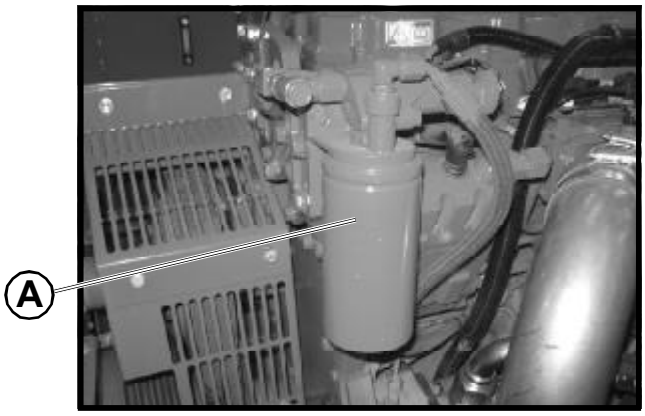
HOT COOLANT CAN CAUSE SERIOUS BURNS. TO OPEN THE COOLANT SYSTEM,STOP THE ENGINE AND WAIT UNTIL SYSTEM HAS COOLED. REMOVE FILLER CAP SLOWLY TO RELIEVE PRESSURE

5.18 CHANGING FUEL FILTER

(i) Unscrew the fuel filters *(Item A)* and carefully dispose off.

(ii) Fill the new filters with diesel and fit into position.

(iii) Hand tighten the new filters.



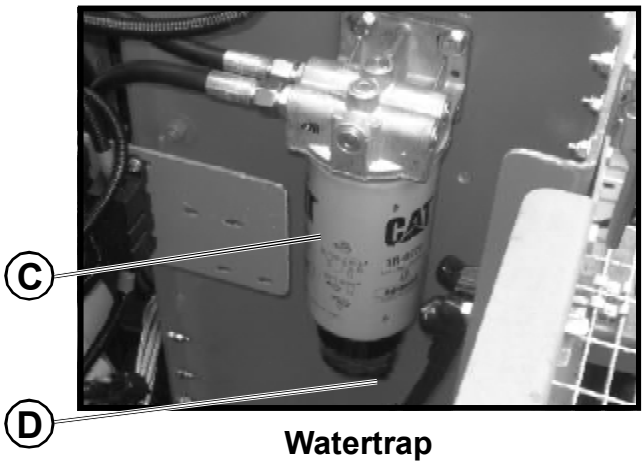
Engine Fuel Filter

5.19 WATERTRAP

5.19.1 Draining the Watertrap

(i) Loosen the draintap *(Item D)* on the base of the watertrap.

(ii) Allow the water to drain off. Tighten the draintap when all the excess water has been drained off.



Watertrap

5.19.2 Changing Watertrap Filter

To change the watertrap filter element *(Item C)* follow the procedure outlined.

(1) Stop the machine.
(2) Remove Element (Item C).
(3) Remove the watertrap bowl from the filter element.
(4) Oil the watertrap bowl as shown.
(5) Attach new element to watertrap bowl.
(6) Oil top of filter element.
(7) Attach the filter element.
(8) Use the priming pump (Item B) to fill the element and watertrap.
(9) Replace the bleed.

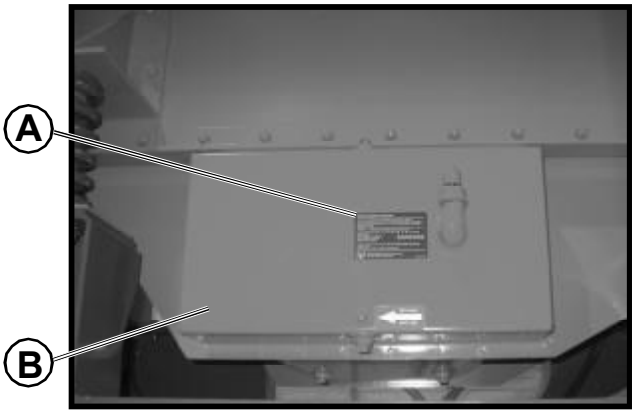
⚠ CAUTION

AVOID NAKED FLAMES WHEN WORKING ON THE FUEL SYSTEM. DO NOT SMOKE

5.19 CHANGING GRIZZLY FEEDER GEARBOX OIL

(see Maintenance Schedule for frequency)

This unit *(Item B)* is oil lubricated, and should be checked weekly , The oil also must be changed every 500Hrs. Use the proper lubricant type and quantity as shown on the lubrication plate *(Item A)* or Recommended Lubricants section.



Beltfeeder Gearbox Oil Unit

5.19.1 Checking oil Level

(i) Turn off the engine. Proper oil level can only be determined when unit is level and cold.

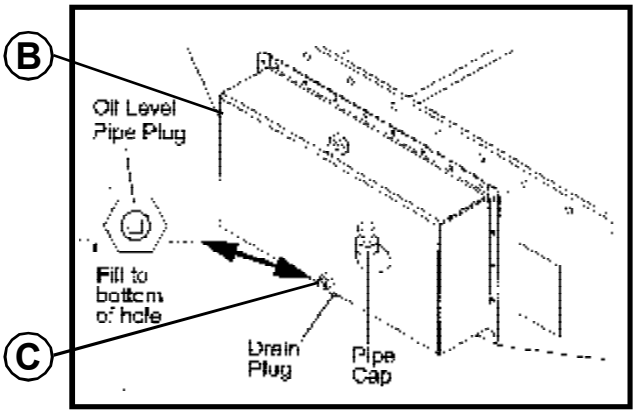
(ii) Remove oil level pipe plug*(see Illustration) (Item C)*

(iii) Oil Level should reach the bottom of the threaded oil level hole*(Item C)* when cold.

(iv) Add recommended lubricant as necessary with plug*(Item C)* re-placed. Allow time for oil to flow into centre housing before removing plug *(Item C)* to re-check level.

(v) Once oil is at proper level, re-install pipe plug.

DO NOT OVERFILL THE GEARBOX.



Gearbox Unit

Beltfeeder Gearbox Oil Uuit

5.19.2 Changing Oil

(i) Remove the drain plug and drain to a container capable of holding the oil. Dispose of the oil safely.Reinsert drain plug.

(ii) Follow the same procedure as 5.19.1 to fill the gearbox.

5.20 CHANGING GEARBOX OIL IN TRACKS

(see Maintenance Schedule for frequency)
More effective draining will take place if the oil is hot

(i) Turn off the engine.

(ii) To drain the gearbox. Clean the area around the gearbox plugs. Position the Plugs *(Items A&B)* as shown in the photogragh ensuring the plug *(Item B)* is at the bottom. Remove the plug *(Item A)* this will allow air into the gearbox to allow it to drain. *(Use a 8mm allen key to remove plug).*

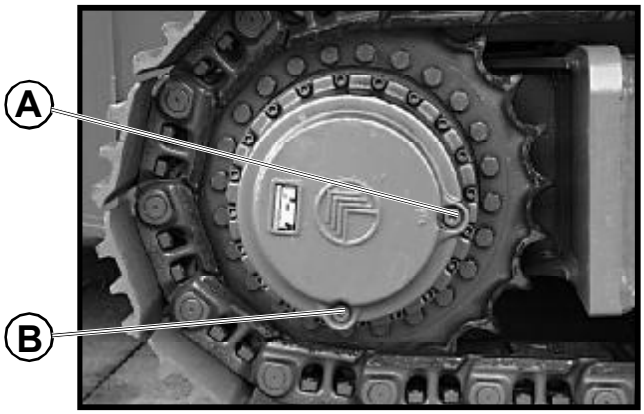
(iii) Remove the Drain plug *(Item B) (Use a 8mm allen key to remove plug)* and drain the oil to an external container large enough to hold all the oil.

(iv) Start the engine and move the tracks until the Gearbox is in the Fill Position with the Plugs *(Items C&D)* as shown in the photograph.

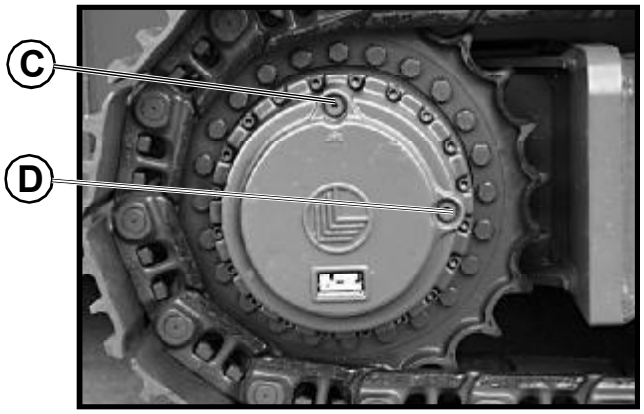
(v) Now refill the gearbox through the Filler plug *(Item C)* until the oil begins to flow from the level plug *(Item.D).*
DO NOT OVERFILL THE GEAR-BOX.*(Approximate oil quantity 2.5 Litres per Gearbox)*

(vi) Replace the plugs *(Item C&D).*

(vii) After a short period of operation visually check the oil level.



Tracks Gearbox in Drain Position



Tracks Gearbox in Fill Position

DANGER

	SWITCH OFF AND LOCKOUT BEFORE WORKING AT A MACHINE
	DANGEROUS NIP POINTS EXIST **CLOSE GUARDS AFTER MAINTENANCE**

IMPORTANT

THE GEARBOX OIL MUST BE CHANGED IN ACCORDANCE WITH THE MAINTENANCE SCHEDULES IN SECTION 5.26 OTHERWISE YOUR WARRANTY WILL BE EFFECTED

5.21 TRACK ADJUSTMENT

As the track wears, they become slack,and should be checked daily. To establish if the tracks require adjustment move the machine, a few metres, backwards and forwards on level ground, in a straight line. This enables the track to adopt its natural degree of tension. Measure the droop as shown in Fig 1
This measurement must be between 5mm- 20mm.

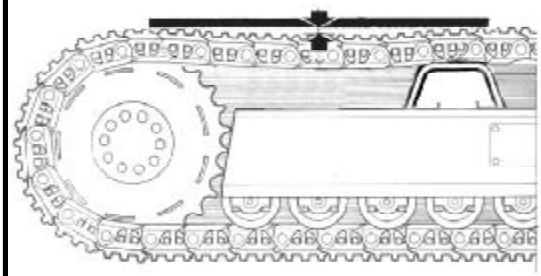


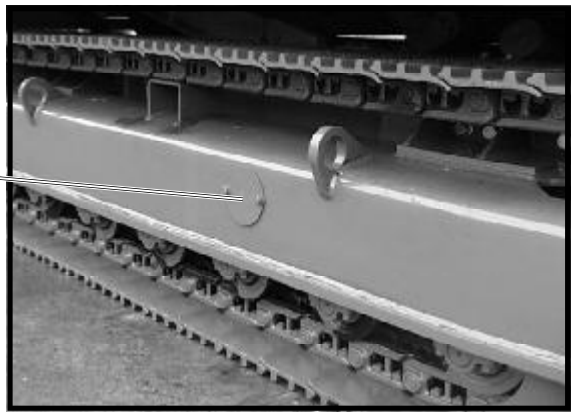
Fig 1

5.21.1 To Release Track Tension

(i) Remove the adjuster cover plate *(Item A)*. *(Use a 17mm Spanner).*

(ii) Loosen the Grease fitting *(Item B)* on the end of the track tensioning cylinder by one half turn only. *(Use a 22mm Spanner*). Allow grease to leave the tension cylinder therefore reducing the track tension.

(iii) Retighten the Grease fitting *(Item B)* and replace the cover plate *(Item A).*

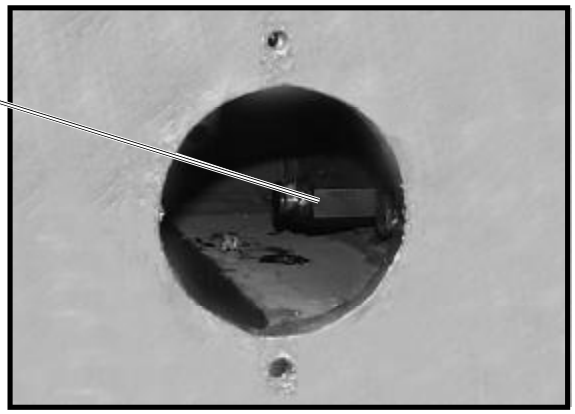


Track Adjustment Access Plate

5.21.2 To Increase Track Tension

(i) Remove the adjuster cover plate *(Item A)*. *(Use a 17mm Spanner).*

(ii) Using the special grease gun adaptor *(Item C)* attach to a standard grease gun and place over the grease fitting *(Item B)*.

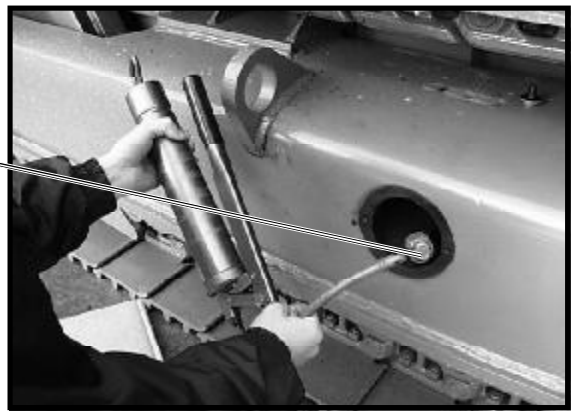


Track Adjustment Grease Fitting

(iii) Pump the track out to the correct adjustment.

(iv) When the track droop appears to be correct, move the machine backwards and forwards, and recheck the droop.

(v) Replace the cover plate *(Item A)*

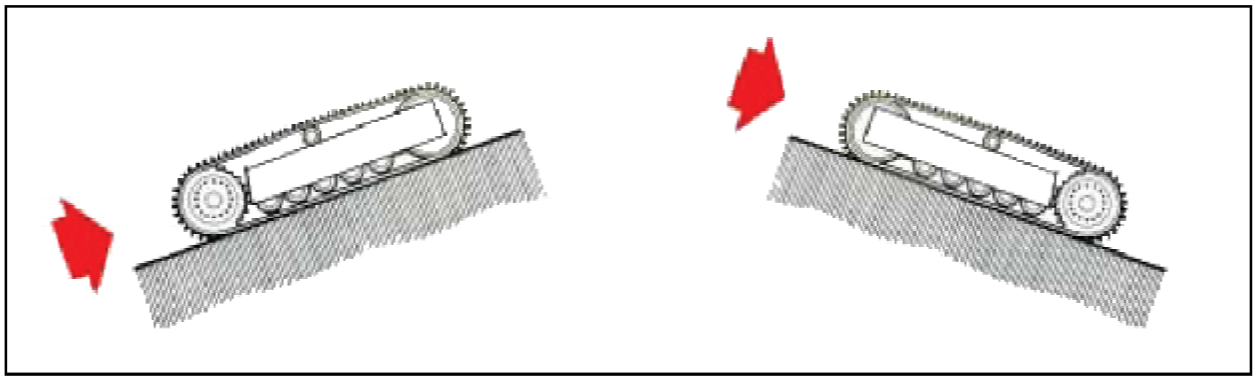


Track Adjustment Greasegun Fitting

5.21.3 TRACK MAINTENANCE

TRACKS - OPERATING PRECAUTIONS

When travelling up a gradient, the tracks should be driven forward (i.e. idlers first, drive sprocket to the rear). When travelling down a gradient, tracks should be driven sprocket first.



ALWAYS:-

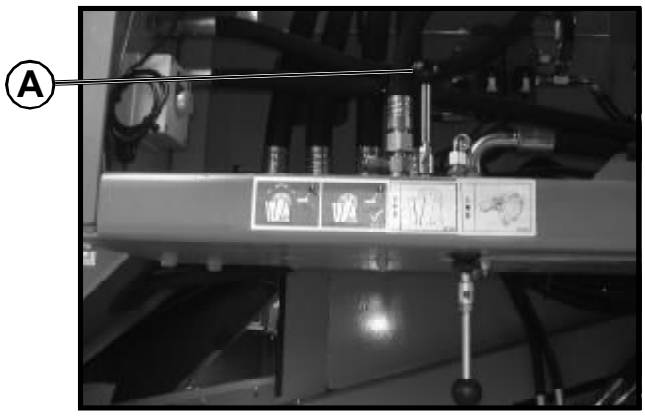
- Park the machine on flat, level ground. If it is necessary to park the machine on a gradient, the tracks should be solidly blocked.
- Ensure the terrain the machine is working on is firm enough to adequately support the machine.
- Ensure the machine is tracked at least 10m in either direction on a daily basis, to minimise risk of track chain seizure.
- Ensure the track systems are free from debris before moving the machine.
- Ensure the tracks are not frozen to the ground before moving the machine.

NEVER:-

- Attempt to track the machine if there is any build up of material around the tracks and drive sprockets.
- Attempt to track the machine if the tracks are frozen to the ground. Push or tow the machine when unable to free itself.

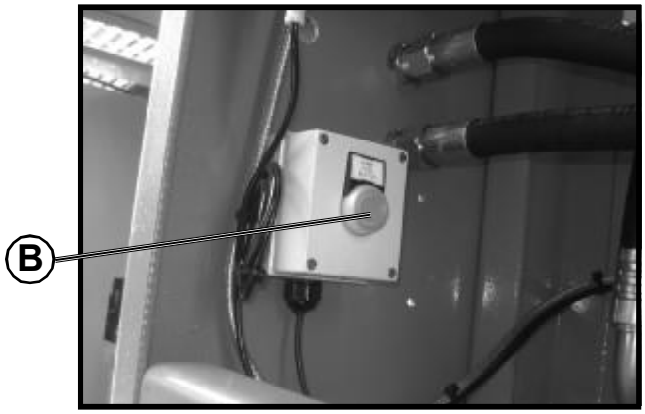
5.22 CRUSHER MAINTENANCE

All maintenance relating to the Crusher can be found in the Crusher Manual 'Single Toggle Jaw Crusher Instruction Manual'. This manual must be read and understood prior to operating or carrying out maintenance to the crusher unit.

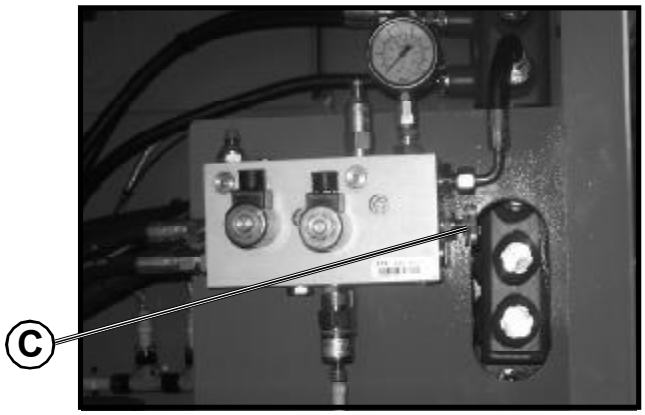


5.22.1 Adjusting Jaw Settings

In the Operation Section *(Section 3)* of the Crusher Manual is details of the Setting adjustment of the crusher. The adjustment uses a Hydraulic Cylinder between the two toggle wedges. The Crusher manual refers to the Hydraulic Cylinder being moved by a hydraulic pump. In the case of the *NFLG* NFJ1175 Crusher the hydraulic Cylinder is controlled by a control lever *(Item A)*.



The NFJ1175 is fitted with full hydraulic adjust therefore the tension/pullback rod spring compression is being controlled by a hydraulic cyclinder and circuit, the compression of the spring is controlled by the button *(Item B).*



The procedure to adjust is:

(i) Ensure the jaw is **not** running and the engine revs is around 1000rpm.

(ii) Screw out the manual restristor valve *(Item C)* fully.

(iii) Press and hold down the over-ride button *(Item B)*, allowing the tension to be relieved off the spring.

(iv) Keeping the button depressed pull the adjust lever *(Item A)* to obtain the desired setting. Release the button *(Item B)* to recharge pull rod.

WARNING

BEFORE MAINTENANCE YOU MUST READ AND UNDERSTAND THE SAFETY SECTIONS IN THE CRUSHER MANUAL

5.22.2 Vee Belt Tensioning

Periodically check the tension of the vee belt throught the inspection plate *(Item A).*(see mantainenance schedule for details). The deflection at Point D *(fig 1)* should be around 25mm(1") *(this assuming hand Force (point F) of 125 newtons).* Poorly tensioned vee belts cause damage to the belt and reduce RPM of the crusher, therefore productiion capacity.

To adjust the belt follow procedure below:

(i) Loosen the Bolt *(Item C)*.

(ii) Rotate the Nut *(Item B)* in the desired direction to either tension or loosen the Vee belt.

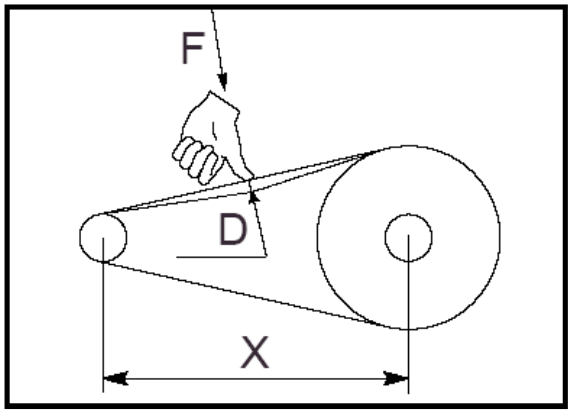
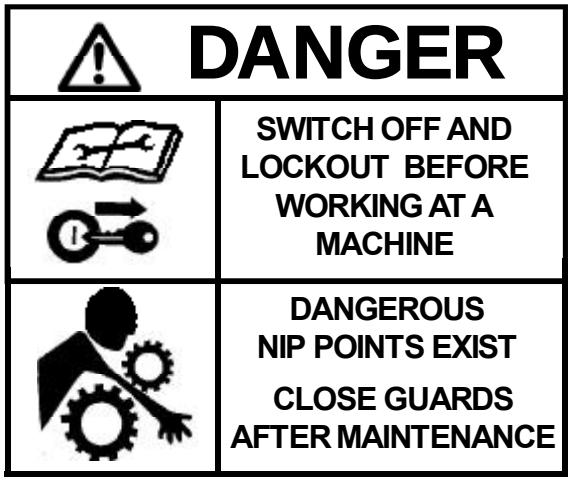


Fig.1

After replacing Vee Belts remember to check the tension after 4 hours run in time, check again after 24 hours.



5.22.2 Rebar Deflection Plate (Option)

If you machine is fitted with a Rebar deflection plate it is important to ensure the deflector plate is correctly set to ensure the belt is not damaged. Fig. 1 and fig 2 . The angle of the deflector plate must be sufficient to prevent rebar snagging on the belt. Remember that when the CSS settings of the crusher are readjusted the deflector plate also needs to be readjusted.

To adjust deflector plate:

(i) Loosen the nuts *(Item B)*. Repeat for the adjuster on the other side of the deflector plate.

(ii) Adjust the deflection to the required angle.

(iii) Tighten the nuts *(Item B)* to hold in position.

Note that if the angle of the deflector plate *(Item A)* is to large it will restrict the discharge of material from the crushing chamber.

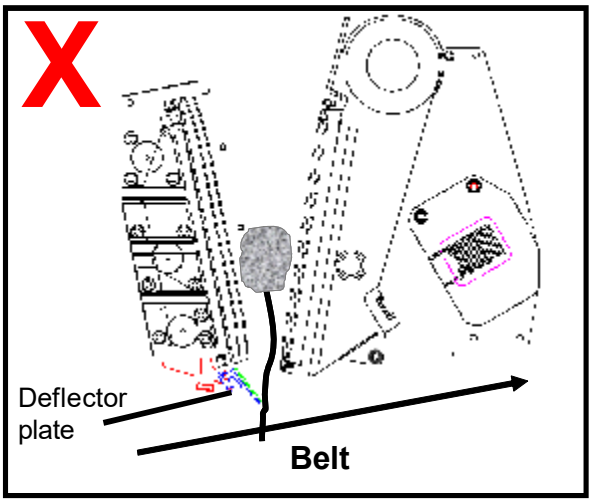


fig. 1 Incorrectly set Deflector Plate- rebar has missed the deflector plate

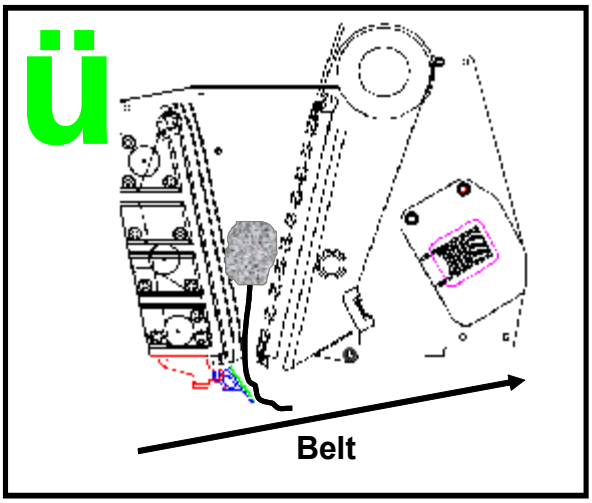
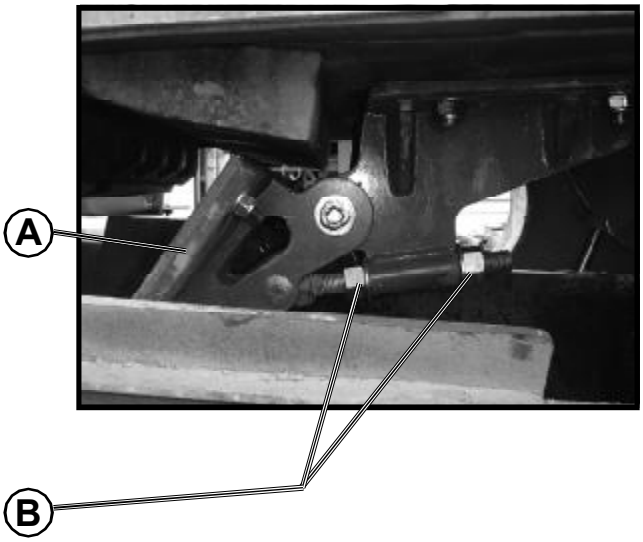


Fig. 2 Correctly set Deflector Plate- rebar deflected in belt direction.



5.23 CHANGING FEEDER MESH

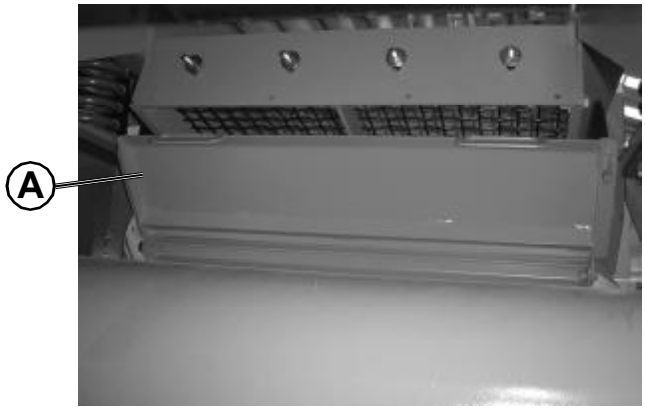
To change the feeder mesh:

(i) Lower the access Guard *(Item A)* to gain full access to the mesh frame *(Item B).*

(ii) Support the Mesh Frame *(Item B)* by placing a chain *(Item C).*

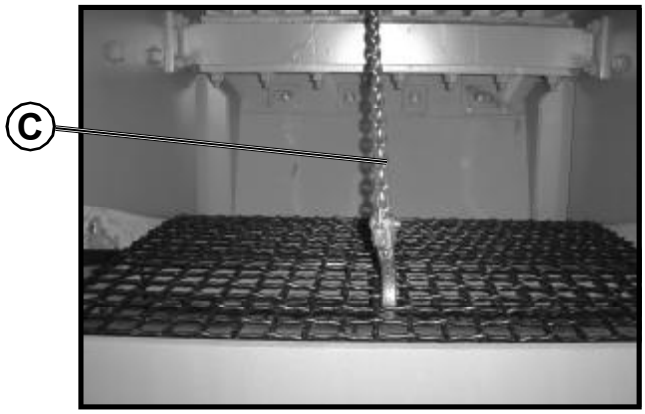
(iii) Remove all the Mesh frame securing bolts *(Item D)*

Procedure Continued on next page....

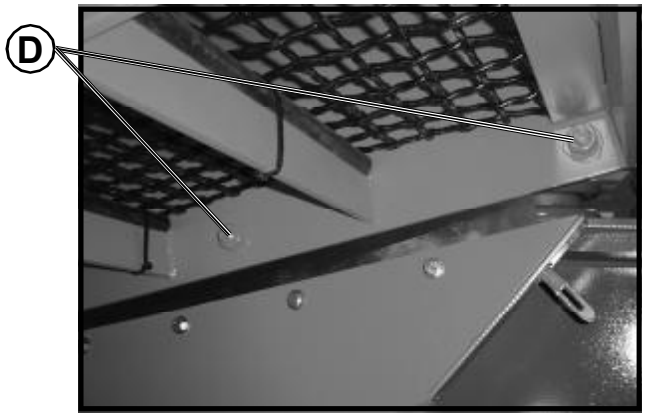


Feeder Mesh Access closed

Feeder Mess Access open



Mesh Frame support



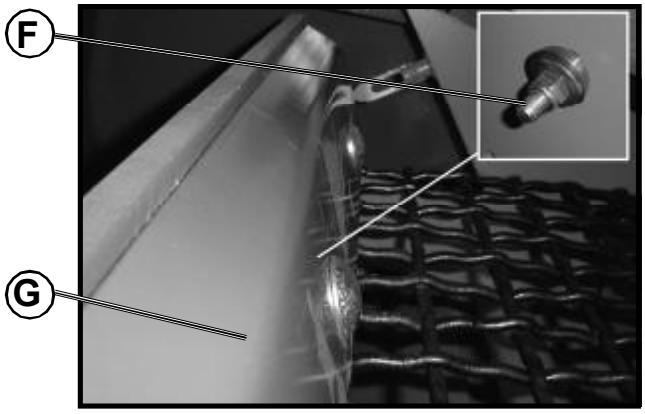
Mesh Frame Bolts

(iv) Lower the mesh frame *(Item E)* to the resting position.

(v) Remove the 4 securing bolts *(Item F)* and remove the Mesh clamps *(Item G).*

(vi) Remove and replace the Mesh. Replace clamps, ensure they are clean from material, retighten.

Mesh Frame Resting Position



Mesh Clamps

5.24 CHANGING MESH ON PRE-SCREEN (OPTION)

To change the Prescreen mesh:

(i) Access to the Pre-screen *(Item A)* can be gained underneath the Feeder.

(ii) Loosen the bolts *(Item C).* The Mesh will loosen and can be removed through the flap *(Item B)*

(iii) Clean material from around the mesh clamps, replace with new mesh.

Prescreen Option

Accessing Meshes

5.25 RECOMMENDED LUBRICANTS

Grease Lubrication

	SPECIAL SCREEN GREASE	SCREEN GREASE FOR HOT MATERIALS
B.P	ENERGREASE LS-EP2	---
TOTAL	MULTIS EP2	---
DUCKHAMS	ADAMAX LEP2	---
CASTROL	SPHEEROL EPL2	---
SHELL	ALVANIA EP2	LIPLEX EP2
ESSO	BECON EP2	UNIREX EP2
TEXACO	MULTIFAK EP2	---
CHEVRON	DURALITH EP2	---
GULF	GULFCROWN EP2	SUPERCROWN EP2
MOBIL	MOBILUX EP2	MOBILGREASE HP2
CENTURY	REGULUSA2EP	LUPLEX M2
F.A.G.	ARCANOL L135	---
ROCOL	SAPPHIRE	BG 151
OPTIMOL	LONGTIME PD	---
OMEGA	---	85

Above is a list of the recommended alternatives suitable for use with your machine. The above special screen greases are lithium base greases of Class 2 with extreme pressure additives.

Hydrauilc Oil Type : ISO VG 46 (Ambient Temperature 0ºC - 25ºC)
ISO VG 68 (Ambient Temperature 20ºC - 40ºC)

Antifreeze Type : BS 6580

RECOMMENDED LUBRICANTS FOR THE GEARBOX ON THE TRACKS

LUBRICANT	-20°C + 5°c IV 95 min	+5°C + 30°c IV 95 min	30°C + 50°c IV 95 min	30°C + 65°c IV 95 min
ESSO	Spartan EP 100	Spartan EP 150	Spartan EP 320	Compressor oil LG 150
AGIP	Blasia 100	Blasia 150	Blasia 320	Blasia S 220
ARAL	Degol BG 100	Degol BG 150	Degol BG 320	Degol BS 220
BP MACH	GR XP 100	GR XP 150	GR XP 320	SGR XP 200
CASTROL	Alpha SP100	Alpha SP150	Alpha SP 320	Alpha SN6
ELF	Reduetell SP 100	Reduetell SP 150	Reduetell SP 320	Oritis 125 MS Syntherma P30
CHEVRON	Non leaded Gear compound 100	Non leaded Gear compound 150	Non leaded Gear compound 320	
GULF		EP lubricant HD 150	EP lubricant HD 320	
I.P	Mellana 100	Mellana 150	Mellana 320	Telesia oil 150
MOBIL		Mobilgear 629	Mobilgear 632	Glycoyle 22/30 SHC 630
SHELL	Omala oil 100	Omala oil 150	Omala oil 320	Tivela oil SA
TOTAL	Carter EP 100N	Carter EP 150	Carter EP 320N	
KLUBER	Lamora 100	Lamora 150	Lamora 320	
ISO 3448	VG 100	VG 150	VG 320	VG 150-200

Machines are supplied with Mobilgear 629, NFLG Partcode 12.40.0629

Engine

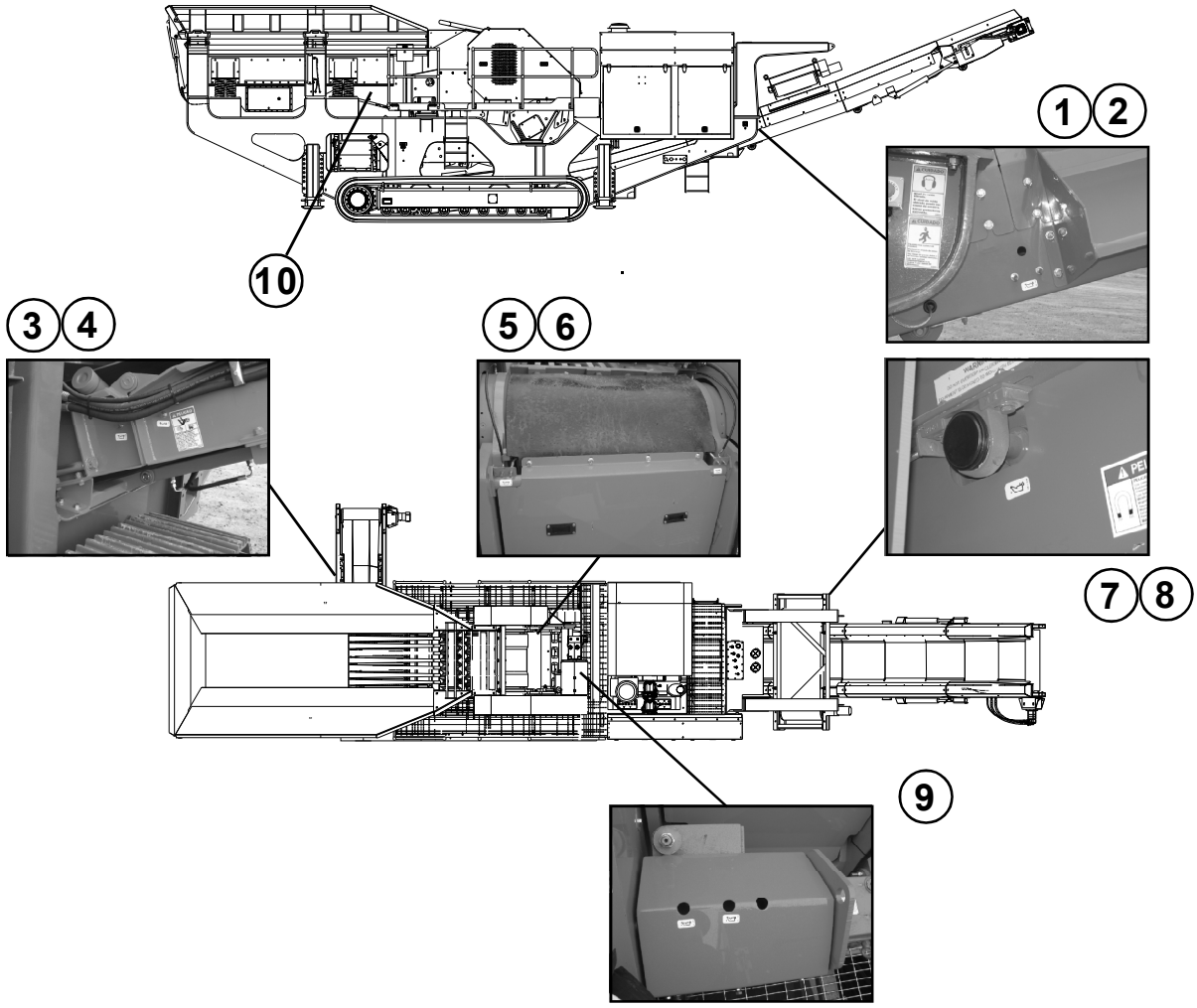
Refer to Engine Operation and Maintenance Manual.

Grizzly Feeder

	Ambient Temp. 40 to 95° F 4 to 35° C	**Ambient Temp.** 0 to 65° F -18 to 18° C
CHEVRON	Industrial Oil 220	Industrial Oil 150
EXXON	Teresstic 220	Teresstic 150
MOBIL	DTE Oil BB	DTE Oil Extra Heavy
SHELL	Molina 220	Molina 150
Texaco	Regal R & O 220	Regal R & O 150
ISO No.	ISO 220	ISO 150

5.26 GREASING SCHEDULE

The J -1175 Supertrak has low level greasing to all bearings.Grease pipes are returned from each bearing to ground level where maintenance can be performed with ease.
(See the maintenance schedules in Section 5.26 for a detailed checklist)



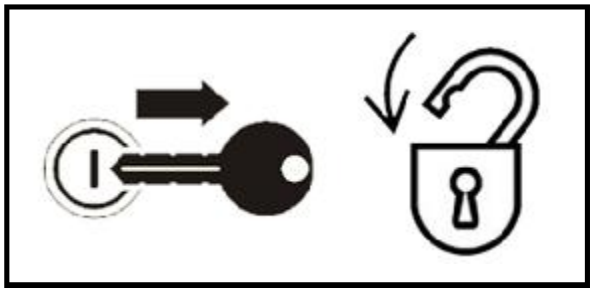
	LOCATION	QTY.	FREQUENCY	CAPACITY
1.	Main Conveyor Drive Drum	2	50Hrs	4 grams/2 pumps
2.	Main Conveyor Idler Drum	2	500Hrs	4 grams/2 pumps
3.	Dirt Conveyor Drive Drum*(Optional)*	2	50Hrs	4 grams/2 pumps
4.	Dirt Conveyor Idler Drum*(Optional)*	2	50Hrs	4 grams/2 pumps
5.	Jaw Swingstock Bearings	2	8Hrs	24 grams/12 pumps
6.	Jaw Main Frame Bearings	2	8Hrs	24 grams/12pumps
7.	Magnet Drive Side Bearings	2	50Hrs	4 grams/2 pumps
8.	Magnet Non-drive side Bearings	2	50Hrs	4 grams/2 pumps
9.	Jaw Layshaft Bearings	2	50Hrs	8 grams/4 pumps
10.	Prescreen Bearings *(Optional)*	*2*	50 Hrs	4 grams/2 pumps

5.28 SAFELY REMOVE AND INSTALL THE BATTERY

DO wear protective eyewear, clothing and gloves.

DO NOT expose the battery to flame, sparks, or incendiary devices, including cigarettes.

DO NOT lay tools across the top of the battery. A conductive metal might touch both terminals simultaneously causing a short, sparks, or a potential explosion.



(i) Turn off the Engine at Starter Panel and remove key.

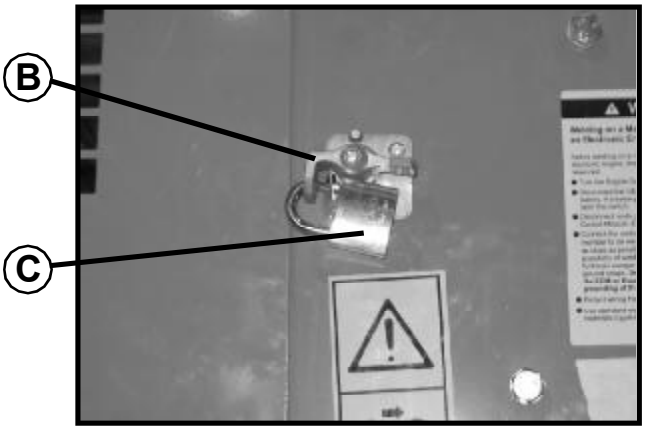
(ii) Rotate the switch *(Item A)* to the "OFF" position *(Item B).*

(iii) Attach a padlock *(Item C)* to secure in the lockout position.

(iv) Raise the engine inspection door *(Item D)*.

Procedure continued on following page. .

Isolator Switch "ON" Machine Operational



Isolator Switch "OFF" and Locked out Machine cannot operate

Engine Inspection Door

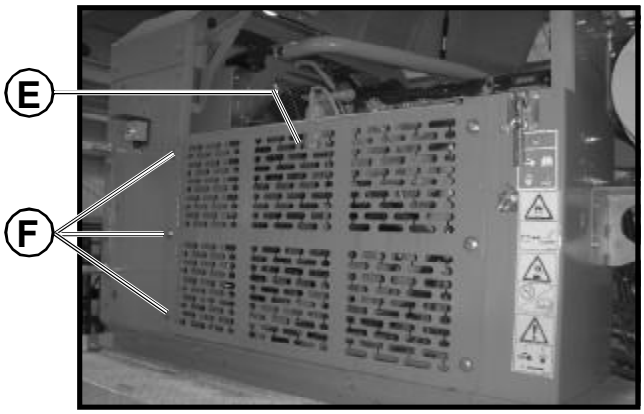
Procedure continued from previous page...

(v) Remove the side panel *(Item E)* by removing bolts *(Item F)*.

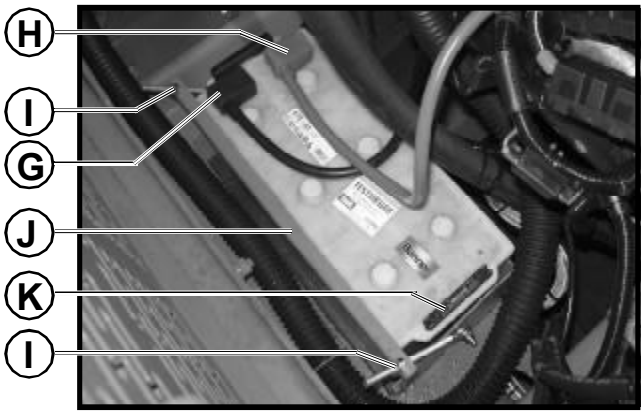
(vi) The battery terminals are labeled + (red/positive) and – (black/negative). Disconnect the negative terminal first *(Item G)* and flex the cable away from the battery. Next disconnect the positive terminal *(Item H)*.

(vii) Finally disconnect the top clamp that holds the battery down by removing bolts *(Item I)* and clamp *(Item J).*

(viii) Lift the battery straight up and out by holding it at opposite corners. As the battery weighs 101.4 pounds (46 kg) or more and should be lifted out with extreme care by 2 operators. Do not tip or drop the battery. Many newer batteries are made with a handy strap handles *(Item K)*.



Engine Compartment Side Panel



Fitted Battery

NOTE dumping the battery will be a hazard to the environment. Dispose of the battery by special disposal or at a recycling center.

IMPORTANT

DUMPING THE BATTERY WILL BE A HAZARD TO THE ENVIRONMENT. DISPOSE OF THE BATTERY BY SPECIAL DISPOSAL OR AT A RECYCLING CENTRE

(ix) Clean the battery cables, pan and clamp. If there is corrosion present (white powder), neutralize it with a mixture of baking soda and water using an old toothbrush.

NOTE when purchasing a new battery, confirm that the negative and positive terminal posts are on the same side of the battery as your old one.

Procedure continued on following page. .

Procedure continued from previous page...

(x) Carefully lower the new battery into place. Make sure it is sitting flat in the battery pan, and not on the lip.

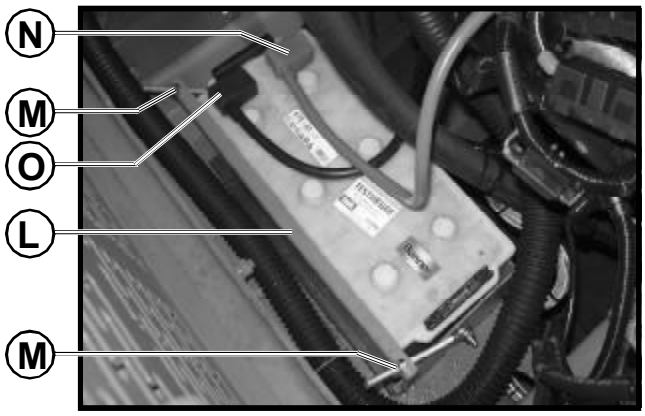
(xi) Position the top clamp *(Item L)* into place and secure the battery by tightening down the bolts *(Items M).*

(xii) Reconnect the positive terminal first *(Item N)*. Next reconnect the negative terminal *(Item O).*
The battery terminals are labeled
+ (red/positive) and
– (black/negative).

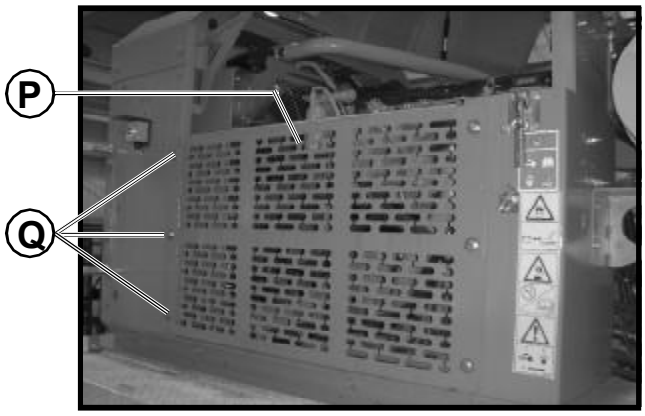
(xiii) Attach the side panel *(Item P)* by inserting bolts *(Items Q)*.

(xiv) Lower the engine inspection door *(Item R)*.

(xv) Rotate the switch *(Item S)* to the "ON" position.



Fitted Battery



Engine Compartment Side Panel

Engine Inspection Door

Isolator Switch "ON" Machine Operational

5.29 MAINTENANCE SCHEDULES

DATE_____

5.29.1 Daily Maintenance (10Hour) Schedule

General Maintenance

	Maintenance Task	Notes	Signature
Check For Material Bulid up	Check/Remove		
Check All Safety Guards are in Place	Check/Replace		
Check For Loose parts, missing nuts & bolts	Tighten/Replace		
Check Hydraulic hoses, Rams & pumps for leaks *(When tightening fittings use a Torque wrench and follow the torque setting on the following page)*	Tighten/Replace		
Check Hydraulic oil level in site gauge.	Check/Top up		
Check Diesel Level in fuel tank.	Check/Refill		
Check indicator on pressure inlet filter.	Check		

Conveyor Maintenance

	Maintenance Task	Notes	Signature
Check Conveyor belts for rips and tears	Check/Repare		
Check Belt Tension	Check/Tension		
Check Belt Alignment	Check/Align		
Check All rollers are free moving and free from obstruction	Check/Free		

Engine Maintenance *(See Engine operators manual for more details)*

	Maintenance Task	Notes	Signature
Check Engine Oil Level	Check/Top up		
Check Water Coolant Level	Check/top up		
Check Engine Air Cleaner Service Indicator	Check		
Check Colour in Contamination Control Breather on Hydraulic Tank.*(see 5.16.4)*	Check		

Track Maintenance

	Maintenance Task	Notes	Signature
Check Tension of Tracks	Check/Tension		
Check Tracks for oil leakage	Check/Repair		

Crusher Maintenance *(See Crusher operator's manual for more details)*

	Maintenance Task	Notes	Signature
Check for loose bolts and parts	Check		
Check the Vee belts for slippage	Check/Tension		
Check for unsual noise when operating	Check		

Feeder Maintenance

	Maintenance Task	Notes	Signature
Check oil level in vibrator unit.	Check/Top up		
Check Mesh Securing bolts are tight.	Check		
Check Mesh for wear.	Check		

5.29.2 Weekly Maintenance (50Hour) Schedule

DATE _____

General Maintenance

	Maintenace Task	Notes	Signature
Grease all Bearings *(See Greasing Schedule)*	Grease		
Drain the Watertrap	Drain		
After **first 50 Hrs** replace Return Filters	Replace		
After **First 50 Hrs**. change Gearbox oil in Feeder	Change	See 5.19	

Conveyor Maintenance

	Maintenance Task	Notes	Signature
Check Tension of Belt Scrapers	Check/Adjust/Replace		
Check Tension of Belt Skirting Rubbers - No spillage	Check/Adjust/Replace		
Check For Obstruction to Drums	Check/Remove obstruction		

Engine Maintenance *(See Engine operators manual for more details)*

	Maintenance Task	Notes	Signature
After **first 50Hrs** carry out full engine service	Service		
Air Filter pre cleaner	Clean up		
Battery Acid Level	Check/Top up		

Crusher Maintenance *(See Crusher operators manual for more details)*

Grease bearing while Crusher is warm and running.	Grease		
Check Jawface bolts are tight.	Check/Tighten		
Inspect Jawface wedges for wear and distortion.	Check/Replace		
Check that side liner bolts are secure.	Check/Tighten		
Check lower, middle, and Upper side liners for wear.	Check		
Check condition of tension rod.	Check		
Check condition of pullrod bushes.	Check/Rotate/Replace		

When tightening hydraulic fitting always use a torque wrench, as overtighting of fittings can break internal seals. Please follow the torque settings in the table below

Thread Size	Torque Ft/Lb.	Torque Kg/Mtr.	Torque N/Mtrs
¼" BSP	15	2.07	20.3
3/8" BSP	18	2.49	24.4
½" BSP	40	5.50	53.9
¾" BSP	60	8.30	81.3
1" BSP	85	11.75	115.2
1¼" BSP	110	15.20	149.0
1½" BSP	130	17.98	176.2

5.29.3 Monthly Maintenance

(250Hour) Schedule

DATE _____

General Maintenance

Check	Maintenace Task	Notes	Signature
Drain the Watertrap	Drain		
Change Hydraulic Pressure Filter	Remove/ Replace		

Engine Maintenance *(See Engine operators manual for more details)*

Check	Maintenance Task	Notes	Signature
Change Air Filter Primary (Outer) Element	Check/Replace	See 5.17.4	
Change Engine Oil	Drain & Refill		
Change Oil Filter & Fuel Filter	Remove & Replace		

Track Maintenance

	Maintenance Task	Notes	Signature
After **First 250 Hrs.** change Gearbox oil in tracks	Change		

Crusher Maintenance *(See Crusher Operators manual for more details)*

	Maintenance Task	Notes	Signature
Check end of toggle and toggle seats for wear or cracking.	Check/replace		
Inspect fixed and moving jawfaces for wear, distortion or growth.	Check		
Check the fixed Jawface to ensure that the growth of the jawfaces is not sufficent to allow them to contact the sides of crusher frame.	Check/readjust		
Check tightness of bearing housing bolts.	Check/Tighten		
Check fixed jawface wedges for distortion or wear preventing adequate retention of the Jawface.	Check/Replace		
Check moving Jawface retaining wedge for distortion or wear preventing adequate retention of the jawface.	Check/ Replace		

When tightening hydraulic fitting always use a torque wrench, as overtighting of fittings can break internal seals. Please follow the torque settings in the table below

Thread Size	Torque Ft/Lb.	Torque Kg/Mtr.	Torque N/Mtrs
¼" BSP	15	2.07	20.3
3/8" BSP	18	2.49	24.4
½" BSP	40	5.50	53.9
¾" BSP	60	8.30	81.3
1" BSP	85	11.75	115.2
1¼" BSP	110	15.20	149.0
1½" BSP	130	17.98	176.2

5.29.4 2 Monthly Maintenance (500Hour) Schedule

DATE _____

General Maintenance

	Maintenance Task	Notes	Signature
Adjust V Belt Tension	Check/Adjust		
Change Hydraulic Return Filters	Remove/ Replace		

Engine Maintenance

	Maintenance Task	Notes	Signature
Change the air filter secondary (Inner) element.	Change	See 5.17.4	

Feeder Maintenance

	Maintenance Task	Notes	Signature
Change the oil in the Feeder Unit.	Change		

5.29.5 4 Monthly Maintenance (1000Hour) Schedule

DATE _____

General Maintenance

Change	Maintenace Task	Notes	Signature
Hydraulic Oil	Analysis Oil and change if required **		
Suction Filters	Remove & Replace		

Crusher Maintenance *(See Crusher operators manual for more details)*

Change	Maintenance Task	Notes	Signature
Inspect the tension rod, checking for damage, wear, or cracks.	Check		

** Your local oil supplier will be able to analysis your hydraulic oil.

5.29.6 8 Monthly Maintenance (2000Hour) Schedule

DATE _____

Track Maintenance

	Maintenance Task	Notes	Signature
Change the Gearbox oil in the tracks	Change		

Engine Maintenance *(See Engine operators manual for more details)*

	Maintenance Task	Notes	Signature
Change Engine Coolant	Drain &refill		
Alternator and Starter Motor	Check		

CONTENTS:

6.1 SAFETY INTRODUCTION ..... TR - 2

6.2 SAFETY BEFORE AND DURING TRANSPORT ..... TR - 11

6.3 STOPPING THE MACHINE ..... TR - 13

6.4 PREPARING FOR TRANSPORT ..... TR - 13

6.5 CRUSHER PROTECTION GUARD ..... TR - 14

6.6 ACCESS LADDER ..... TR - 15

6.7 FOLDING MAIN CONVEYOR ..... TR - 16

6.8 PLACING DIRT CONVEYOR IN TRANSPORT POSITION (Optional) .... TR - 17

6.9 PLACING EXTENDED DIRT CONV IN TRANSPORT POSITION ..... TR - 18

6.10 HOPPER EXTENSION FLAPS ..... TR - 20

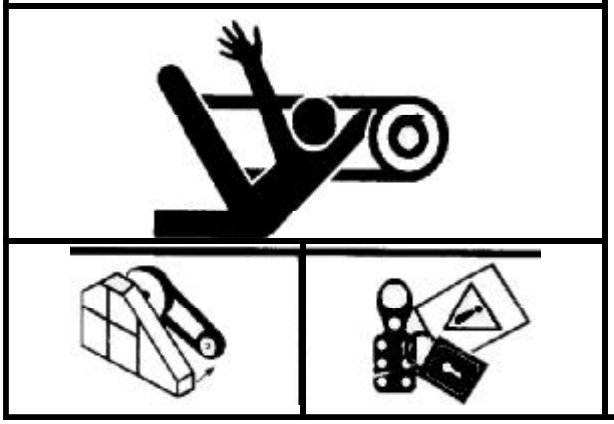
6.11 DETACHABLE WHEELED UNIT (OPTIONAL) ..... TR - 21

6.12 READY FOR TRANSPORT ..... TR - 23

6.13 PRIOR TO HAULAGE: ..... TR - 23

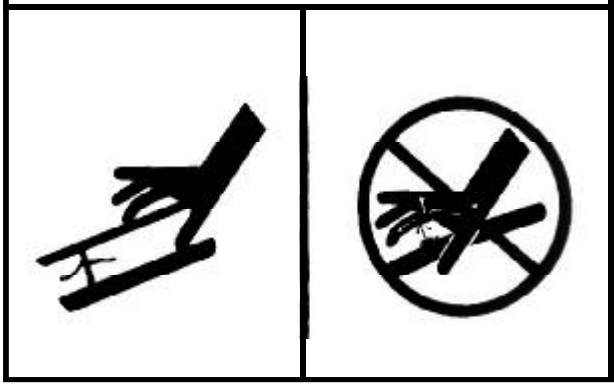
SECTION 6 : TRANSPORT
6.1 SAFETY INTRODUCTION

DANGER



1. Entanglement Hazard.
   Do not reach into an unguarded machine you can be pulled in.
2. In-running nip points can cause serious injury or even death.
   Your arm could be pulled in and amputated.
3. Switch off and lockout machine before opening or removing guard doors.

DANGER



1. Beware of escaping Hydraulic oil. High pressure hydraulic oil can penetrate the skin causing serious injuries.
2. Always relieve pressure from the hydraulic system before attempting maintenance or adjustment. Use a piece of cardboard to check for oil leaks.
3. If oil is injected under the skin, it must be surgically removed or gangrene will result. Get medical help immediately.

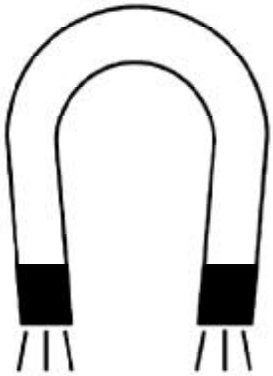
DANGER

1. Electrical Hazard.
   Risk of serious injury or death.
2. Never work on the electrical system of any equipment unless you are thoroughly familiar with system details.
3. Work on the electrical system must only be carried out by a qualified electrician.

DANGER

1. Nip Hazard. Do not reach into an unguarded machine.
2. In-running nip points can cause serious injury or even death. Your arm could be pulled in and amputated.
3. Switch off and lockout machine before opening or removing guard doors.

DANGER



1. Electrical Hazard.
   Risk of serious injury or death.

2. Magnetic fields will affect personnel with pacemakers (within 3 metres), medical electronic implants or metallic prostheses.

3. Ferrous tools and materials could be attracted to the magnet with considerable force.

4. Do not place credit cards or watches near the magnet, they could be permanently damaged

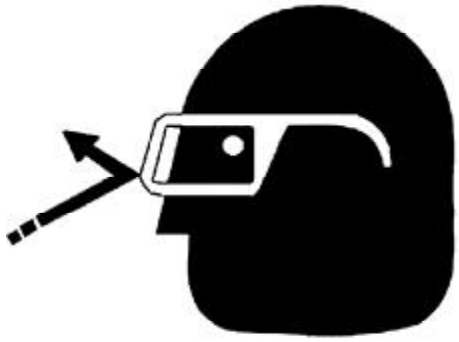
DANGER

1. flying material Hazard.
   Risk of serious injury or death.

2. Platform is for Maintenance purposes only. Do not use platform when the plant is in operation

3. If Flying material hits the head or body it can cause serious injury or death .

DANGER

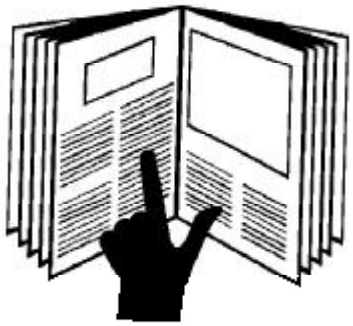


1. Flying material Hazard.
   Risk of serious injury or death.
2. Wear proper eye personalprotective equipment.
3. .Flying material can cause loss or degradation of eye sight.

DANGER

1. Crush Hazard.
   Risk of serious injury or death.
2. Keep clear of Track frames when the unit is in operation.
3. Death or serious injury can result from contact with the Track frames.

WARNING



1. Read and understand the operators manual before you work at this machine.
2. Study the different safety signs on this machine.
3. Only qualified and trained personnel should attempt to work at this machine.

WARNING

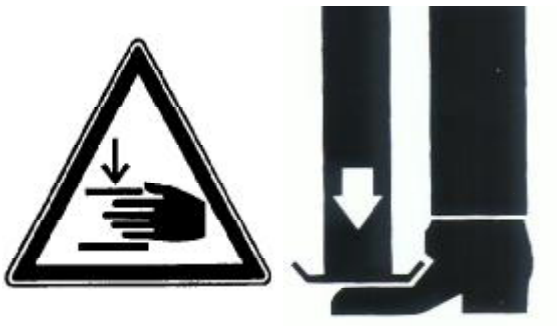
1. Loose or baggy clothing can get caught in running machinery.
2. Always wear correctly fitting (EN/ANSI approved) protective clothing.
3. Protective clothing includes Hard Hat, Safety Glasses, Ear Defenders, Overalls, Gloves, Steel Toed Boots and a High Visibilty Vest.

WARNING

1. Falling Rock Hazard.
   Do not walk near material discharge areas.
   You are at risk of serious injury or death.

2. Wear a hard hat at all times when working in the vicinity of the machine.

WARNING

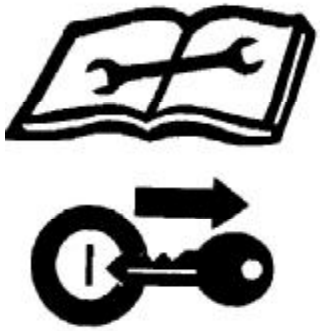


1. Crush Hazard.
   Beware of dangerous Hand and Foot crush points.

2. Do not use any body parts e.g. hands or feet to check hole alignment or support machinery.

3. Always use pins and/or mechanical supports provided.

WARNING



1. When performing maintenance or adjustment to this machine always switch off engine and remove the ignition key, (if electric isolate power supply at source).
2. Keep ignition key on person during maintenance.
3. Never work alone.

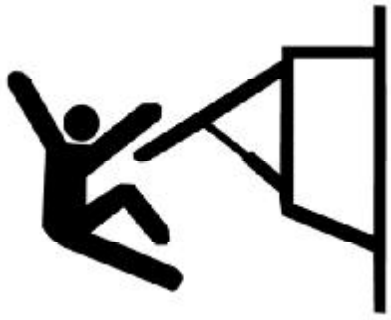
WARNING

1. Fall Hazard.
   Do not climb onto moving or working machinery. You could fall and be seriously injured.
2. Switch off and lockout machine before attempting maintenance or adjustment to this machine.
3. Always use a suitable lifting platform before attempting any maintenance work above 2 metres(6-6”).

WARNING

1. Burn or scald Hazard.
   Beware of hot surfaces at engine and exhaust areas. Be careful when draining hot coolants there is a risk of scalds.

WARNING



1. Impact Hazard.
   Beware of spring loaded doors opening unrestrained, facial injuries could result.

Lockout procedure.

WARNING

1. When performing maintenance or adjustment to this machine always isolate the machine and Lockout.
2. Keep lock on person during maintenance.
3. Never work alone.

The Isolator Switch *(Item A)* is located beside the battery box, the switch cuts the electric supply from the battery ensuring the machine cannot be operated. The machine must always be isolated prior to any maintenance work being carried out.

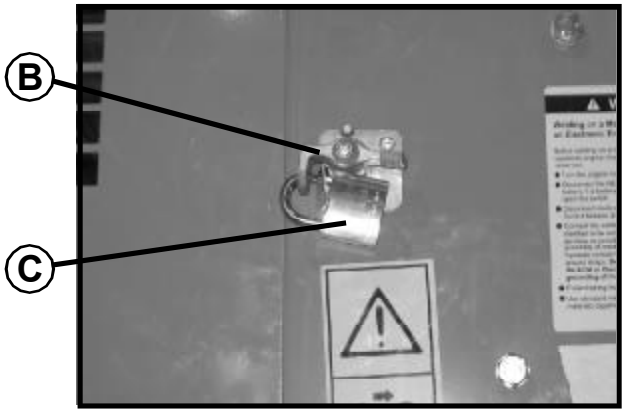
To Lockout:

(i) Rotate the switch *(item A)* to the "OFF" position *(Item B).*

(ii) Attach a padlock *(Item C)* to secure in the lockout position.

(iii) During maintenance work always ensure the isolator is locked so the machine cannot be started accidently.

Isolator Switch "ON" Machine Operational



Isolator Switch "OFF" and Locked out Machine cannot operate

6.2 SAFETY BEFORE AND DURING TRANSPORT

1. The preparations to move equipment by articulated lorry should be supervised by a minimum of two (2) persons.

2. Ensure persons transporting the machine adhere to all Safety Signs and procedures.

3. Before transportation on public roads, ensure the machine has been properly secured with no loose material left in or on the machine.

4. Ensure wheel nuts are torqued between 500 to 550 ft.lb (69 to 76 kg.m) prior to transport. (Wheeled machines only)

5. Check your tyres for:-
- correct pressure
- cuts or bulges
- nails or spikes
- uneven or excessive wear
- missing valve caps

6. Check your wheels for:-
- damaged rims
- missing or loose wheel nuts or bolts
- obvious misalignment

7. Have cuts or punctures repaired by authorised personnel before adding air. Beware an over-inflated tyre can explode and cause serious injury or death.

8. Extreme caution is required when transporting machinery on site. Soft or uneven ground may cause accidents.

9. When transporting a machine on a truck or trailer, know the overall height to avoid contacting overhead obstructions such as bridges, power lines etc.

6.3 STOPPING THE MACHINE

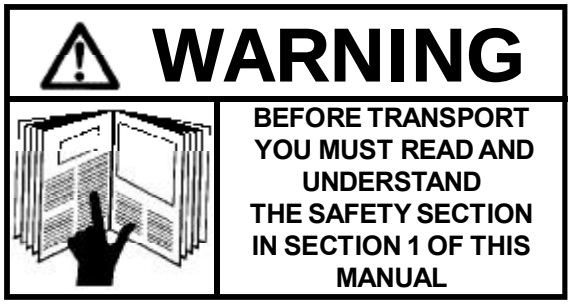
Prior to stopping the machine ensure that all maerial has been processed through the machine and that there is no material on the conveyors.

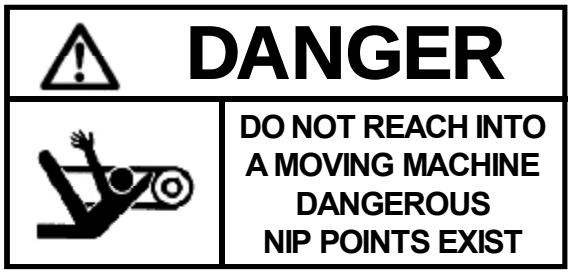
To stop the machine follow the sequence outlined in the Operation section *(4.6).*

6.4 PREPARING FOR TRANSPORT

Inspect the the Feeder, crusher and all the Conveyors to ensure that the machine is clear of all material.The machine should be washed to remove all debris.

It is advisable to have an operator on the opposite side of the machine to ensure areas around machine are clear of personnel when folding conveyors, as serious crush points exist with folding conveyors.



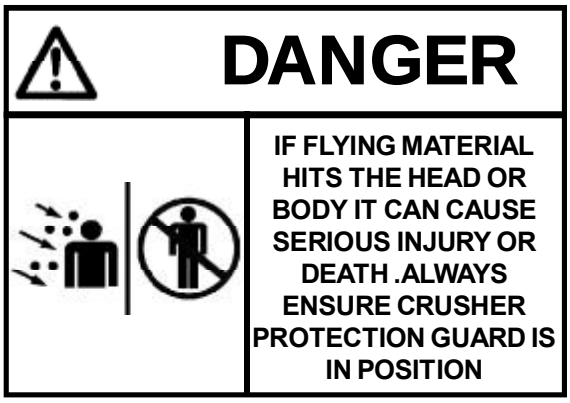


6.5 CRUSHER PROTECTION GUARD

Prior to transporting the crusher the guard *(Item A)* must be placed in transport position, this is required to keep machine height to minimum for transport.

(i) Rotate the crusher guard *(Item A)* from the working position. And secure into the Transport position *(Item B).*

Crusher Guard in Working Position

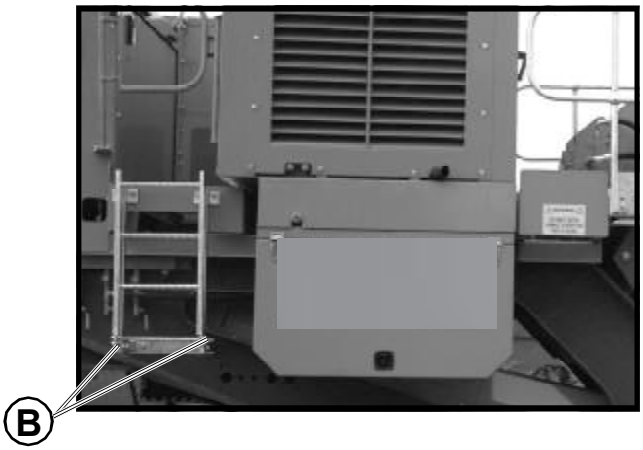


Crusher Guard in Transport Position

6.6 ACCESS LADDER

Prior to transport fold the access ladder from the working position *(Item A)* to the transport position *(Item B)* and secure in position.

Ladder in Operating Position



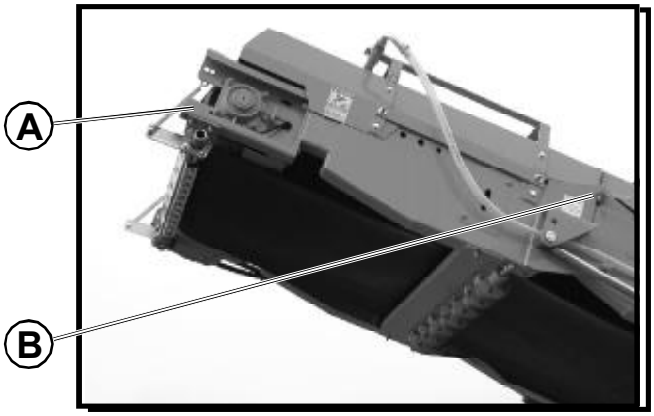
Ladder in Transport Position

6.7 FOLDING MAIN CONVEYOR

When loading onto a low loader have the conveyor in the working position to avoid damage. The conveyor should only be lowered once the machine is loaded. To change the conveyor from the working position *(Item A)* to the transport position *(Item E)* :

(i) Remove the pin *(Item B)* on each side of the conveyor.

(ii) Locate the control Bank *(Item C)*. Operate the control *(Item D)* this will fold the main conveyor head section into the transport position *(Item E)*.

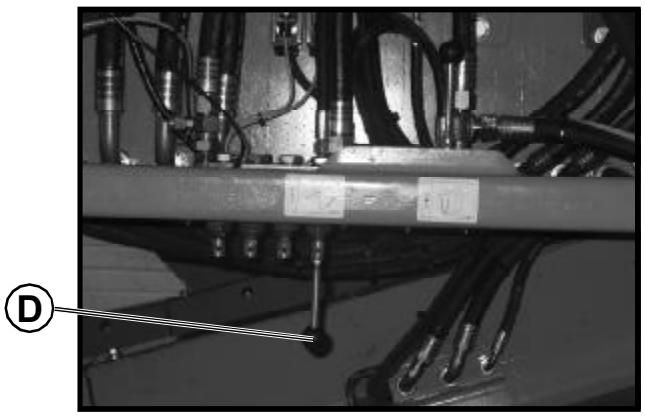


Main Conveyor in Working position

Main Conveyor Fold Control Panel

IMPORTANT

USE THE CONTROL LEVERS IN A SMOOTH AND CONTROLLED MANNER. DO NOT RUSH OR JERK THE CONTROLS. THIS WILL DAMAGE THE CONVEYOR



Middle Grade Conveyor Control Panel

IMPORTANT

DO NOT LOAD CRUSHER ONTO LOWLOADER WHEN THE CONVEYOR IS FOLDED.

Main Conveyor in Transport Position

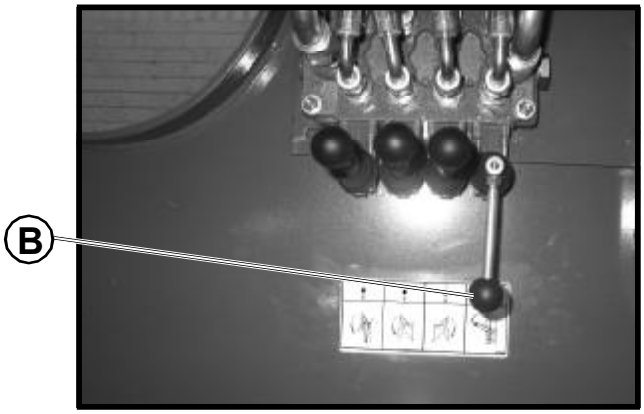
6.8 PLACING DIRT CONVEYOR IN TRANSPORT POSITION (Optional)

If your NFJ1175 is fitted with the optional Dirt Conveyor *(Item A)* it will need to be folded into the transport position*(ItemD)*.To fold:

(i) Operate the control *(Item B)* located underneath the hopper to fold the conveyor into transport position.

(ii) Place a pin *(Item B)* into the transport support to hold in position.

Dirt Conveyor in working position



Dirt Conveyor control bank

WARNING

HAZARDOUS NIP AND CRUSH POINTS IN THIS AREA

Control Bank Location

Dirt Conveyor in transport position

6.9 PLACING EXTENDED DIRT CONVEYOR IN TRANSPORT POSITION (Optional)

If your NFJ1175 is fitted with the optional Extended Dirt Conveyor *(Item A)* it will need to be folded into the transport position.Tofold:

(i) Remove the Pins *(Item C)* from the Conveyor Upper section.

(ii) Remove the Pins *(Item D)* from the conveyor lower section.

(iii) Activate the control *(Item E)* located underneath the hopper to retract the upper section of the conveyor *(Item A)*. Ensure the belt is moving freely.

(iv) Activate the control *(Item F)* located underneath the hopper to retract the lower section of the conveyor *(Item B)*. Ensure the belt is moving freely.

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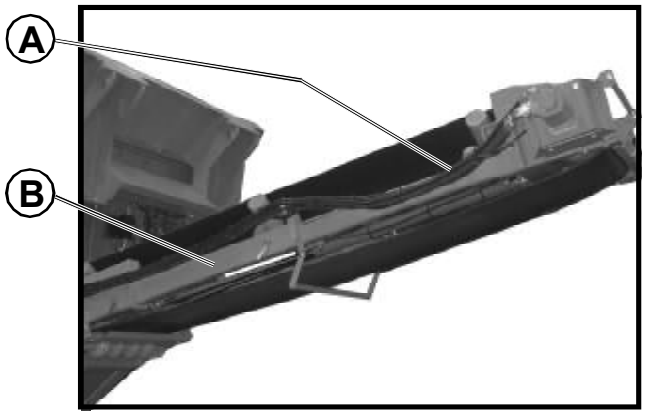
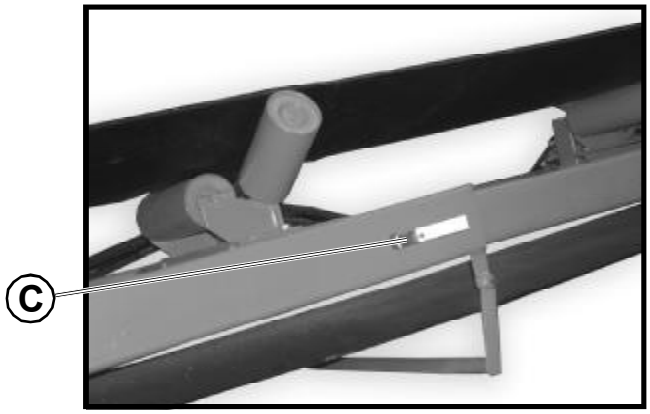
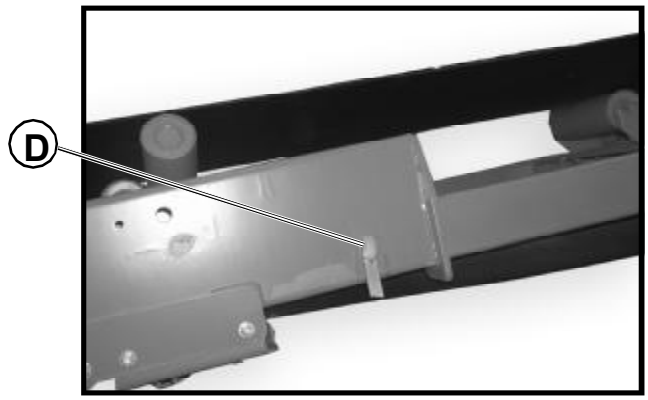


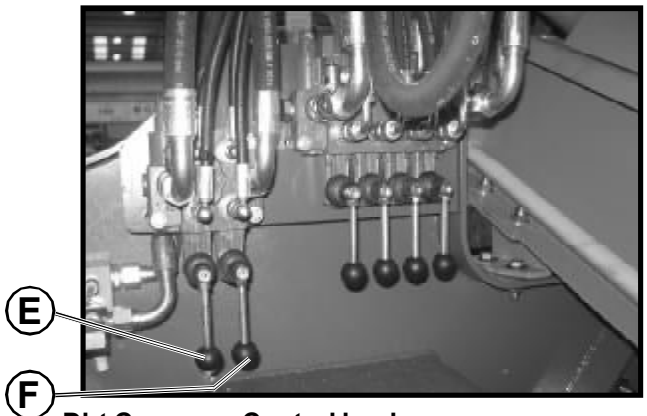
Dirt Conveyor in working position



Dirt Conveyor Upper Section Pin



Dirt Conveyor Lower Section Pin



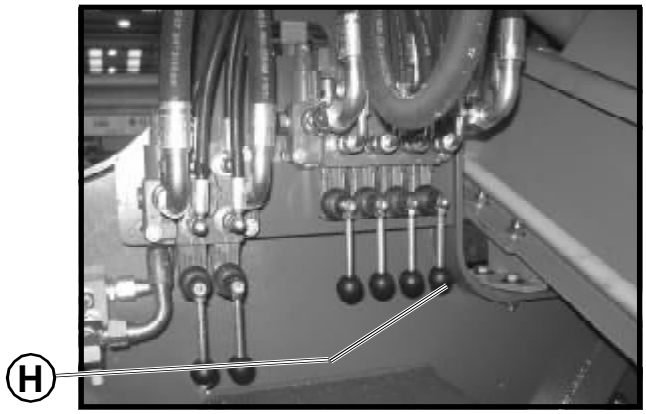
Dirt Conveyor Control bank

(iv) Activate the control *(Item H)* located underneath the hopper to fold the conveyor into the transport position *(Item J)*.

(v) Secure in the transport position by inserting a securing pin *(Item I)*

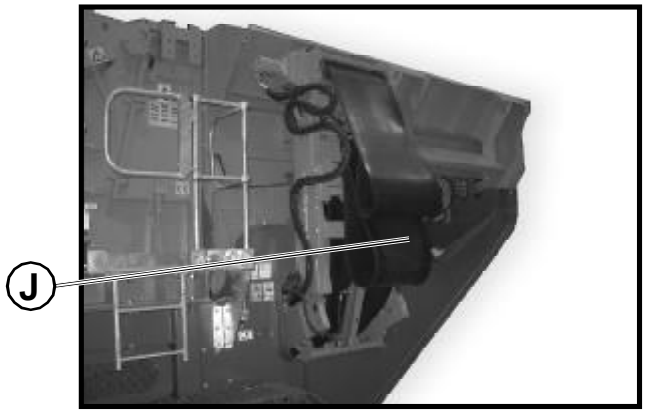
(vi) Ensure the belt *(Item J)* has been tied to prevent excessive movement during transport.

Dirt Conveyor lowered



Dirt Conveyor control bank

Control Bank Location



Dirt Conveyor in transport position

6.10 HOPPER EXTENSION FLAPS

To meet height regulations the Extension flaps*(Item A & B)* must be folded into the transport position *(Items H, I & J)*. To place into the transport position:

(i) Remove the Supports *(Item C)* from each of the holding positions and store in the toolbox for transport.

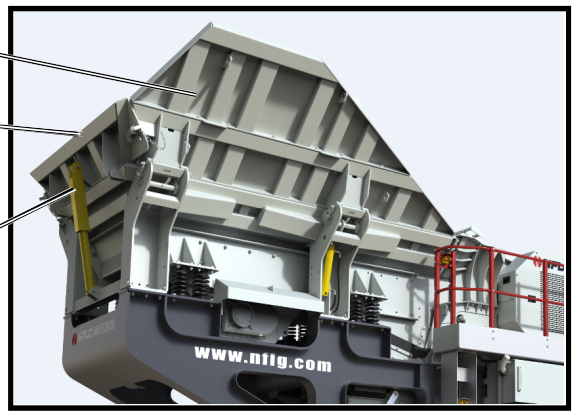
(ii) Remove the holding wedges *(Item D)* from the Rear extension flap.

(iii) Locate the control Bank *(Item K)* .

(iv) Place hopper extension *(Item I)* into the transport position by operating the control *(Item F)*.

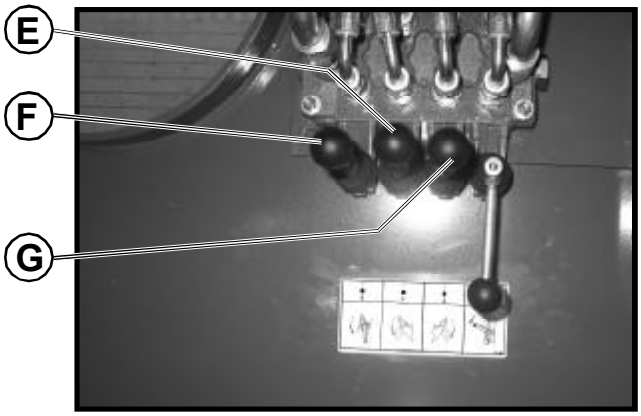
(v) Place hopper extension *(Item H)* into the transport position by operating the control *(Item G)*.

(vi) Place hopper extension *(Item J)* into the transport position by operating the control *(Item E)*.



Hopper Flaps in Working Position

Hopper Extension Rear Wedges



Hopper Extensions Control bank

Hopper flaps in Transport Position

DANGER

DO NOT CLIMB INTO WORKING OR MOVING MACHINERY FALL HAZARDS EXIST

WARNING

HAZARDOUS NIP AND CRUSH POINTS IN THIS AREA

6.11 DETACHABLE WHEELED UNIT (OPTIONAL)

To reattach the wheeled unit:

(i) Lower the front Jack legs *(Item A)* by operating the control levers *(Items D&E)*

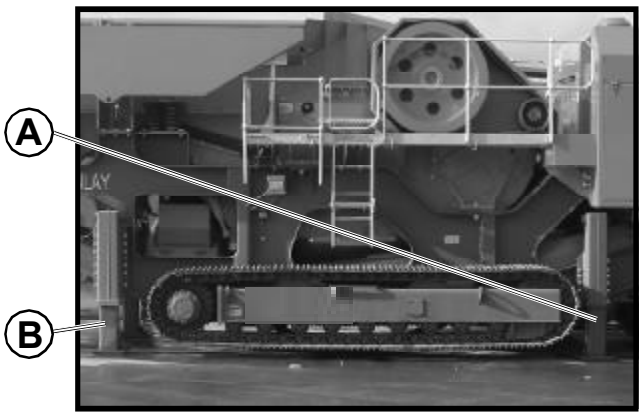
(ii) Raise the front Jack legs *(Item B)* by operating the control lever *(Item C).*

(iii) Push the wheeled unit *(Item G)* into position until the locator *(Item F)* has located in the correct place.

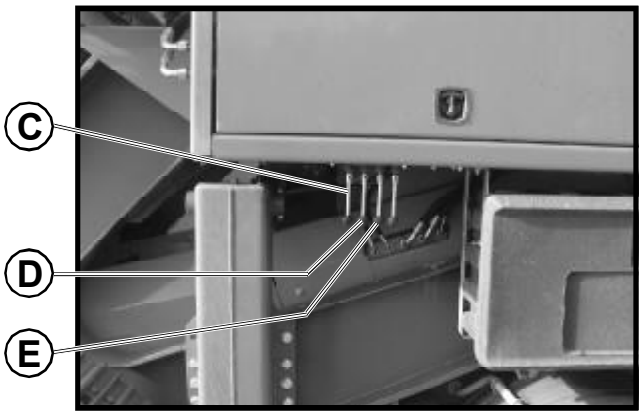
(iv) Lower the Front jacklegs *(Item A)* until the locator *(Item H)* is in position on both sides of the machine.

(v) To lock in position push the lever *(Item I)* up and rotate, repeat for both sides of the machine.

(vi) Tighten the locknut *(Item J)* to secure.

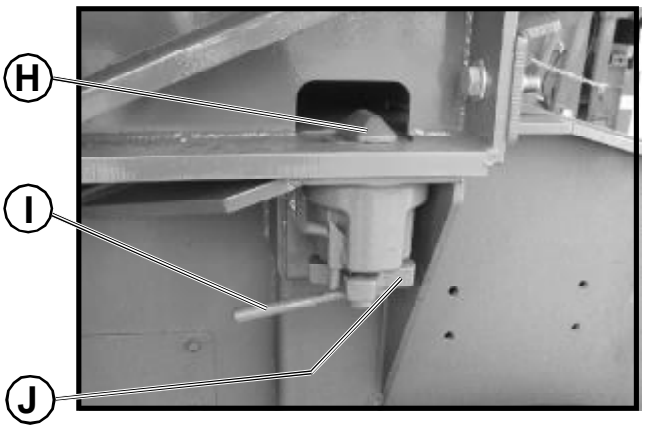


Jack Legs



Jack Leg Controls

Weeled Unit in Position



(vii) Remove the pin *(Item A)* and push the Towbar *(Item D)* back into the transport position.

(viii) Remove the pins*(Item E)* securing the lightboard in the stored position*(Item C),* pull the lightboard out and secure in the transport position.

(ix) Ensure the lightboard connector *(Item D)* is connected.

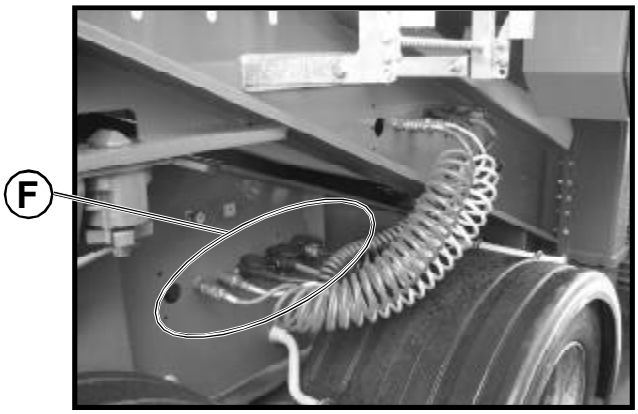
(x) Connect the lightboard and Air brake connections *(Item F)*

WARNING

HAZARDOUS NIP AND CRUSH POINTS IN THIS AREA

Wheeled Unit Towbar

Wheeled Unit Lightboard



Lightboard & Air Brake Connections

6.12 READY FOR TRANSPORT

The machine should now be ready for transport. Using the Radio control handset or Wander lead handset load the machine onto the trailer.

Secure machine with Chain shackles to the Lowloader.

Use securing eyes *(Item A)* fitted on the machine undercarriage on either side.

IMPORTANT

DO NOT TRANSPORT THIS MACHINE WITHOUT IT BEING PROPERLY SECURED ONTO THE LOW LOADER

6.13 PRIOR TO HAULAGE:

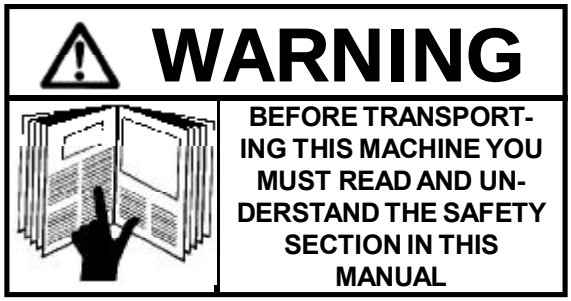
(i) Recheck for loose parts or debris on the machine

(ii) Ensure all pins are secured with 'R' pins and Split pins

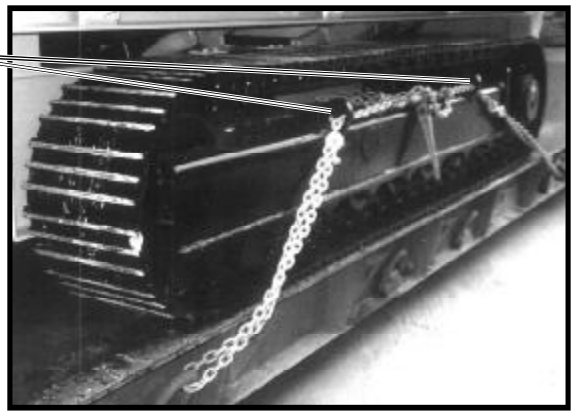
(iii) Raise the support legs fully.

(iv) Ensure wheel nuts are torqued between 500 to 550 Ft.lb *(69 to 76 kg.m)* prior to transport *(Bogie Option)*

(v) Ensure persons transporting the machine adhere to all safety signs and procedures.



Machine on Lowloader



Tracks Secured on Lowloader

福建南方路面机械有限公司

Fujian South Highway Machinery Co.Ltd

NFJ1175 Tracked Jaw Crusher

Control system

Electrical Control System Manual
NFJ1175

This is the commissioning and operation manual of the electrical system of the NFLG NFJ1175 tracked jaw crusher.

File number: NFLGSMS-DQ-NFJ1175-GV02

Version number: GV.02

Revision record: 2nd Writing

Release date: June 15, 2022

Catalog

Catalog **2**
1 User notice **3**
2 Precautions for safe use **4**
2.1 Safety level 4
2.2 Electrical and engine safety 4
2.3 Equipment operation and maintenance precautions 5
3 System introduction **6**
3.1 System composition 6
3.2 Main feature 6
4 Electrical installation specifications **7**
5 System debugging **8**
5.1 Preparation before commissioning operation 8
5.2 Debugging of main components 8
6 Display operating instructions **16**
6.1 Main menu interface 16
6.2 Function information interface 17
6.3 Parameter setting interface 18
6.4 Node information interface 29
6.5 Tracked walking mode interface 31
6.6 Automatic production mode interface 34
6.7 Manual production mode interface 37
6.8 Blockage clearing mode interface 38
6.9 Folding mode interface 40
6.10 Fault information interface 41
6.11 Fault record interface 42
6.12 Engine information interface 43
6.13 Engine failure history interface 44
6.14 Application selection interface 45
7 Fault tracing **46**
User feedback form **52**

1 User notice

The copyright of the NFJ1175 tracked jaw crusher control system belongs to Fujian South Highway Machinery Co., Ltd. (hereinafter referred to as NFLG). NFLG reserves the right of final interpretation and modification of the function, structure and appearance of the control system (including control software, supporting electrical equipment, operation manual and other related facilities).

The copyright of the NFJ1175 tracked jaw crusher control system software (including the program parts of the on-board controller, man-machine interface, etc.) belongs to the NFLG copyright system and is protected by copyright law and international treaties, No organization or individual may reverse engineer, decompile or illegally disassemble this software or any part of it! If the software (or any part thereof) is copied, disseminated, modified or destroyed without authorization, NFLG will reserve the right to pursue its legal responsibility, and require it to bear the necessary responsibility for the losses caused by all parties!

NFLG reserves the right of final interpretation and modification of the specification of this control system under the condition that the patent and commercial ownership have been registered. Without the official permission of the company, any organization or individual shall not copy, reproduce, photograph, modify or republish this manual! NFLG will reserve the right to pursue legal responsibility and financial loss for any violation of copyright regulations.

This manual contains information necessary to use this equipment. Please read this manual carefully and fully understand it. Before you use this equipment for production activities, our company believes that you have read this manual carefully and can operate and maintain it in accordance with the requirements of this manual. During the use of the equipment, the company cannot guarantee to avoid damage to the equipment caused by man-made sabotage or natural disasters.

The parameters in this manual are only used to describe the production equipment of NFJ1175 tracked jaw crusher. Before publishing, the documents have been carefully checked, but there are unavoidable omissions or inappropriateness. In order to provide better service and technical support, we will sincerely accept any suggestions and comments related to this manual, conduct regular audits, and provide better service and technical support. Make the necessary changes in the next version of the document to meet your needs to the greatest extent possible. We sincerely thank you for your support of our work!

Please treat this manual as a permanent part of the device, always keep it with the device and keep it safe.

2 Precautions for safe use

Before equipment installation, operation, maintenance and overhaul, please be sure to read and understand the relevant chapters in this manual and the mechanical operation manual attached to the equipment, and be familiar with the meanings of all labels, nameplates, safety information and precautions on the equipment before using this equipment, if strictly followed, will be of great help to your security.

2.1 Safety level

In this instruction manual, the severity of safety precautions is divided into three categories: Danger, Warning and Caution to warn you of different potential dangers or remind you to pay attention to relevant information.

Danger:

This represents an imminently hazardous situation and it is assumed that when equipment is incorrectly installed, operated and maintained, it will, if not avoided, result in death or serious injury.

Warning:

This indicates a potentially hazardous situation, which is envisaged when equipment is incorrectly installed, operated and maintained, which, if not avoided, could result in death or serious injury.

Caution:

This indicates that a potentially hazardous situation exists, and it is envisaged that when equipment is incorrectly installed, operated and maintained, it may result in minor to moderate injury, if not avoided, with the possibility of serious consequences depending on the situation.

For the location of all safety signs on the equipment and their meanings, please refer to the mechanical operating manual that came with the equipment.

Note: This equipment is only allowed to be operated and maintained by personnel who have been fully trained by our company's professionals and are qualified to operate. NFLG is not responsible for any consequences arising from failure to follow the provisions of this manual!

2.2 Electrical and engine safety

☆Do not operate the electrical system of the equipment unless you are familiar with the ins and outs of the system and must only be operated by a qualified electrical technician.

☆The electrical system of this equipment must be checked regularly. Damaged cables or loose connectors must be repaired immediately. Only use genuine fuses that meet the specified amperage ratings.

☆Always disconnect the battery ground lead before performing any maintenance work on the electrical system or servicing the engine. When welding work on the equipment, it must be ensured that the battery pack is isolated.

☆Never fill the fuel tank with the engine running, while smoking, or near an open flame.

☆Never short-circuit the positive and negative poles of the battery, as this may cause a fire and damage the electrical system.

☆Batteries contain sulfuric acid, an electrolyte that can cause severe burns and generate explosive gases. Wear personal protective equipment and avoid contact with skin, eyes or clothing during maintenance.

2.3 Equipment operation and maintenance precautions

☆Staff should wear protective equipment necessary for safe production, such as safety helmets, protective gloves, insulating rubber shoes, etc.

☆It is strictly forbidden for untrained and unqualified personnel to start the machine, operate it and take shifts without authorization. Staff with different responsibilities perform their own duties, and it is strictly forbidden to mix and change positions.

☆Before starting the device or before moving the device, check around the device. Make sure that no one is under, on, or near the device. Inform other operators and those present that you are starting or moving equipment and do not activate or move equipment until everyone is evacuated.

☆During the production process of the equipment, all action mechanisms are in operation, and no one is allowed to enter the production site without permission.

☆If there is no need for overhaul and maintenance, ensure that all control cabinet doors are completely closed to prevent dust from entering the cabinet.

☆The equipment site staff should always put safety production in the first place and maintain a sufficient safety distance from various action mechanisms on the equipment.

☆The emergency stop switches at various positions on the equipment should be very familiar, so that when an emergency occurs, the emergency stop operation can be performed as soon as possible.

☆ When overhauling, maintaining and maintaining the equipment, the battery isolation switch must be turned off, and the battery isolation switch must be locked at the same time. Before starting work, check that the disconnected parts are not live, grounded, or shorted.

☆Smoking and teasing are strictly prohibited within the scope of the equipment. Non-conductive fire extinguishing equipment (such as dry powder fire extinguisher) necessary for safe fire protection should be equipped around the

equipment. Do not use foam to extinguish the fire to avoid electrical short circuit and electric shock accident when using fire extinguishing equipment.

☆According to the actual situation on site, set up equipment safety operation procedures. All personnel involved in the operation and maintenance of the machine must be familiar with the various clauses listed in the safety operation regulations, and strictly abide by them, and resolutely prevent personal injury and equipment accidents caused by misoperation.

☆After the shift work is completed, the staff should remove the ignition switch key, do the final equipment inspection and cleaning, fill in the work log, and cut off the main power switch and battery isolation switch before leaving.

For more safety information, see the mechanical operating manual that came with the equipment.

3 System introduction

3.1 System composition

NFJ1175 tracked jaw crusher control system, the main hardware is composed of high-performance vehicle controller, display screen, node module, internationally renowned low-voltage electrical appliances, complete sets of wiring harnesses and control cabinets and other hardware, combined with the company's self-developed professional control software system to achieve remote control of tracked jaw crusher and automatic production control.

The control system is powered by 24V DC, which is composed of two DC 12V batteries connected in series. The battery group connected in series is charged in real time by the small generator that comes with the engine. The control system takes the vehicle controller as the main station, the display screen and multiple node modules as the slave stations to form the controller area network (CAN for short), the signal acquisition of each sensor, the control of each solenoid valve, etc. are all through the channel of the node module to accomplish. In addition, the high-performance on-board controller also supports the J1939 protocol, which can facilitate communication with the ECU of the engine, and realize the speed control of the engine and the reading of effective information. The matching engine-specific wiring harness and casting-integrated equipment wiring harness provide a stable and reliable line basis for the control system.

3.2 Main feature

☆The main low-voltage electrical appliances adopt world-renowned brands such as ABB, Weidmüller, Rockwell, and Siemens.

☆Adopt IFM vehicle controller imported from Germany: protection level IP67, anti-vibration, external circuit self-diagnosis and other protection functions.

☆The 7-inch IFM color display screen imported from Germany is adopted: the high-resolution, high-performance, all-Chinese control interface brings users an intuitive and responsive operation experience.
☆The complete and comprehensive fault alarm system adopts the form of pictures and texts, and can save the latest 200 fault history records for easy viewing.
☆The system field bus adopts the most advanced CAN communication technology in the world, which is stable and reliable, with strong anti-interference and simple wiring.
☆The equipment wiring adopts heavy-duty connectors\casting integrated connectors to realize signal docking, and the connection is firm and beautiful.
☆Sound and light alarms are given before the equipment is turned on or during walking to remind people around the equipment to pay attention to safety.
☆When the equipment fails, an audible and visual alarm is automatically issued, and there are corresponding protection measures to ensure the safety of the equipment and personal safety.
☆The parameter setting flexibility is high to meet the application requirements of different working conditions, and all parameters can be restored to factory settings with one key.
☆Using the advanced J1939 communication technology, the fault of the EFI engine can be diagnosed on-site.
☆The control of the crushing machine introduces a compensation algorithm, which can ensure that the efficiency of the hydraulic system remains constant in the case of sudden load changes, so that the performance of the main machine is not affected.

4 Electrical installation specifications

(1) The whole machine must have good grounding protection measures, and the grounding resistance is less than or equal to 4Ω.
(2) The wiring should be neatly and reliably fixed to avoid the appearance of coils. The openings reserved for the wire routing in the mechanical part should be covered with broken plastic hoses around the holes to protect the wires and prevent the harness from being worn and scratched which result insulation performance declines. If the damaged wire needs to be replaced, the specification and model of the wire should meet the requirements of the drawing.
(3) Multiple wires should be bundled with cable ties or wrapping tapes, and there should be enough bending length to keep the wire harness free from tension during the movement.
(4) Each connector must be locked/fastened in place. When connecting the connector of the cabinet and the connector of the wiring harness, there should be a one-to-one correspondence. Only after the label is aligned, the connector can be connected.

(5) When wiring, pay special attention to the color of the wire used for the ground wire, and both ends of the wire should have a number.
(6) Before the cabinet is hoisted, the lifting lug bolts should be carefully checked to ensure their firmness. The cabinet must be fixed and locked with a special shock-absorbing pad.
(7) The wiring should make the bend radius as large as possible, avoid sharp corners, and avoid places of friction and high temperature.
(8)All other wiring harness connections must be made before connecting the battery wiring harness to the engine and equipment electrical systems.
(9)The engine wiring harness should be secured to the engine case or engine mounting bracket in a suitable position using cable ties.
(10) The bolt terminals of the engine should be firmly locked, and must be covered with a suitable pipe end sheath for insulation protection.

5 System debugging

5.1 Preparation before commissioning operation

(1) Before debugging the control system, please read and understand the instruction manual and electrical schematic diagram in detail.
(2) Check whether the wiring of the components in the cabinet is loose, and check whether the connectors of the equipment wiring harness and the engine wiring harness are connected reliably. Before the new machine is powered on or before the engine is started, the wiring terminals of each device and the locked parts of the snap/thread locks of each connector should be Re-tighten. Due to the large vibration of the equipment during operation, the wiring terminals of each device in the cabinet and the connectors of the equipment wiring harness should be regularly reinforced and locked during the daily maintenance process. If the wire is found to be deformed by heat, change color, emit peculiar smell or damage the insulation, it should be replaced immediately. At the same time, it is recommended to clean the dust on the electrical components in the electrical cabinet every month.
(3)Before closing the isolation switch, check the voltage of the engine battery. If the voltage is lower than 24V, the battery should be charged.

5.2 Debugging of main components

(1) Debug instructions:

After the commissioning preparations are ready, the electrical staff and the mechanical commissioning personnel must be present at the same time before the equipment can be powered on and tested. Unless otherwise specified, the debugging of each component is carried out under its own parameter setting interface, and the values are all system default values.

Note: In addition to the debugging of the engine itself, the first advance in the debugging of other components is that the engine must be started.

(2) Control power debugging:

① The reliability of the emergency stop switch must be ensured before power is applied.

② Before debugging the control power supply, check whether the wiring of the DC 24V power supply of each branch is correct according to the drawing, and ensure that the power line number of each branch should be exactly the same as the drawing.

③ After each circuit breaker is closed, check whether the voltage of each power supply output terminal meets the requirements of the actual application standard.

④ After the key switch is powered on, after the warning is over, observe the display screen. If a fault message pops up, please check first, and then perform other operations after the fault is eliminated. For specific fault information and troubleshooting guide, please refer to Chapter 7.

(3)Engine debugging:

① This model uses an EFI engine, and it must be ensured that the communication between the control system and the engine ECU is normal.

② After the key switch is powered on, you should first observe the three indicator lights on the operation cabinet panel (Figure 6.2.3) - from left to right are waiting for running, warning, and shutdown. At this time, the engine ECU will conduct a self-check. The three indicator lights are on at the same time under normal condition, and then go out at the same time after 2-3 seconds. If any indicator light is not off or the three indicator lights are not on at the same time, do not start the engine.

③ The meanings of the three indicators on the control cabinet panel are, from left to right, waiting for operation, warning, and shutdown. If the waiting operation indicator light does not go out, it means that the temperature of the engine intake manifold is too low, and the intake air heater is working. After the heating is completed, the indicator light will automatically go out. If the warning or shutdown indicator light does not go out, it means that the engine is faulty, and an “engine fault” alarm will pop up on the display screen. At this time, you should enter the “engine fault information” interface to check the specific SPN code, FMI code and text information, and then troubleshoot the problem. Except that the spontaneous failure of the engine ECU will limit the start of the engine, other related interlocking conditions include: emergency stop circuit, hydraulic oil level, hydraulic oil temperature and CANBUS bus status. Before the engine is started, it should be ensured that there is no related alarm.

④ Turn the key switch to the ignition gear and hold it. After the engine starts, release the key switch. If the engine fails to start, you should first find out the cause and solve the ignition operation again.

Note: The time of the key switch in the ignition gear should not exceed 10 seconds, and the interval between starting the engine each time should not be less than 5 minutes, otherwise it will lead to the possibility of the starter motor or starter relay being burned out.

Figure 5.2.3 Engine self-test indicator light

⑤ After the engine is started normally, the state of the engine and hydraulic pressure should be carefully observed. If there is any abnormality, the engine should be turned off immediately and checked. After finding out the cause and solving it, the engine can be restarted.

⑥ After the engine runs normally, let the engine run at idle speed for 3 to 5 minutes, then enter the "System Control Parameters" interface, manually adjust the engine speed, let the speed slowly and steadily increase to the maximum, and run continuously for 20 to 30 minutes.

⑦ If you need to know more about the status information of the engine during the starting or running of the engine, you can enter the "Engine Information" interface to view it at any time or start the engine on this interface.

(4) Auxiliary channel debugging:

① Enter the "System Control Parameters" interface (see Chapter 6.3 for details), and press the [Auxiliary 1 Start and Stop] and [Auxiliary 2 Start and Stop] buttons in sequence. Auxiliary output 1 or auxiliary output 2 status on the display changes from "off" to "on".

② Referring to the drawing, observe the channel output indicator of the corresponding node module, which should be solid yellow.

③ The auxiliary channel is only used as the spare output of the control system. In the standard model configuration, these two channels are not used. If you really need to use it, and it is not used as a separate start-stop interface, you must choose an appropriate start-up method.

(5)Cooling fan debugging:

Enter the "Hydraulic oil temperature control parameters" interface, and press the cooling fan start and stop button. The status of the hydraulic oil cooling fan on the display changes from "off" to "on". Observe whether the hydraulic oil cooling fan operates normally.

Unlike other components, the commissioning of the cooling fan does not require starting the engine first. Because the cooling fan is a high-power component, it is not suitable to run for a long time when the engine is not turned on, generally not more than one minute. If it really needs a long debugging time, please start the engine first, and then carry out the debugging.

In the actual operation of the equipment, once the temperature of the hydraulic oil is greater than or equal to 40° C, after a certain delay, the cooling fan will automatically turn on. On the contrary, once the hydraulic oil temperature is less than 40 ℃, after a certain delay, the cooling fan will automatically turn off. (The temperature value can be tested by the signal simulator output 4-20mA)

(6)Main conveyor belt debugging:

① Before debugging, the tail of the main conveyor belt must be lowered to the lowest level (the folding interface should be entered to perform related operations), otherwise the main conveyor cannot be started and a prompt will appear on the screen. Do not arbitrarily disable the tail proximity sensor function, as doing so may cause unpredictable damage to the main conveyor.

② Manually adjust the engine speed to 1200rpm or above. This is critical to avoid false alarms of "low main conveyor belt speed".

③ At the beginning of commissioning, you must first observe the direction of rotation. If the belt reverses, stop the main conveyor belt immediately, turn off the power supply, and adjust the position of the main conveyor belt hydraulic motor in and out of the oil pipe.

④ Adjust the deviation of the belt accordingly. If there is any abnormality, the main conveyor belt should be stopped immediately.

⑤ After the belt runs normally, observe the speed of the belt conveyor on the display screen, the speed should be ≥65m/min, if the speed value is too low or there is no speed, the screen will pop up an alarm and stop the main conveyor. Please refer to Chapter 7 for fault details. If the main conveyor belt speed wheel sensor function is disabled, the speed value has nothing to do with the running state of the main conveyor belt.

(7)Debugging of iron remover:

Note: The iron remover is optional and may not be installed on the machine.

① Enter the "Iron Remover Control Parameters" interface, and press the [Iron Remover Start and Stop] button. The status of the iron remover on the display changes from "off" to "on".

② If the iron remover is not installed on the machine, refer to the drawings and observe the channel output indicator lights and solenoid valve connector indicators of the corresponding node modules, which should be solid yellow. If the iron remover is installed on the machine table, you should first observe its direction of rotation. If the iron remover is reversed, stop the iron remover,

turn off the power supply, and adjust the position of the inlet and outlet pipes of the iron remover hydraulic motor.

(8)Side belt conveyor debugging:

Note: The side belt conveyor is optional and may not be installed on the machine.

① Enter the "Side Belt Conveyor Control Parameters" interface, and press the [Side Belt Conveyor Start and Stop] button. The side belt conveyor status on the display changes from "off" to "on".

② If the side belt conveyor is not installed on the machine table, refer to the drawings to observe the channel output indicator lights and solenoid valve connector indicators of the corresponding node modules, which should be solid yellow. If the side belt conveyor is installed on the machine table, you should first observe its turning direction. If the side belt conveyor is reversed, stop the side belt conveyor, turn off the power and adjust the position of the hydraulic motor of the side belt conveyor.

(9)Host pull rod debugging:

① If the tie rod pressure is less than the lower limit of the charging pressure (default is 46Bar), the hydraulic system will automatically charge the tie rod with liquid until the tie rod pressure is greater than the upper limit (50Bar by default) and automatically stop charging. The tie rod pressure signal is read in real time, as long as the engine is running, the tie rod filling process will be completed automatically.

Note: If the pressure of the tie rod reaches the upper limit, the moment when the liquid-filled solenoid valve loses power, the pressure of the tie rod rises sharply, indicating that there may be problems in the hydraulic or mechanical parts, and the relevant hydraulic or mechanical parts should be checked.

② In the "folding mode" interface, "automatic mode" interface, "manual mode" interface, "blocking mode" interface or "crushing host control parameter" interface, if the tie rod filling time exceeds 15 seconds, the filling of the tie rod has not been completed, the display screen will pop up an alarm of "filling time is too long" and interrupt the automatic filling process. At this time, an investigation should be carried out to find out the reason why the tie rod cannot be filled with liquid normally.

③ Adjust the jaw gap when the crusher is not running. A pull rod pressure relief button is installed on the side of the hydraulic cabinet. Press and hold the button with one hand and observe that when the pressure on the pull rod pressure gauge drops to close to 0Bar, the other hand operates the corresponding hydraulic handle to adjust the jaw.

Note: If the pressure relief button cannot be normally relieved or the pressure relief is incomplete after operating the pressure relief button according to the

above steps, please do not adjust the gap between the jaws. It should be checked first and resolved before proceeding.

⑩Crusher debugging:

①Before commissioning, check whether there is foreign matter in the crushing chamber, check whether the speed proximity switch is installed correctly; ensure that the pressure of the tie rod is normal, and the abnormal pressure of the tie rod will prohibit the start of the crushing machine; adjust the gap of the discharge port of the machine to the specified range.

②Manually adjust the engine speed to 1400rpm or above. This is critical to avoid false alarms of "low speed of the main conveyor belt" or "low speed of the crusher".

③Only after the main conveyor belt is started, the jaw crusher is allowed to start. The operation of the jaw crusher can be either forward or reverse. After the jaw crusher is started, you should first observe the steering of the inertia wheel. Clockwise is forward rotation, and counterclockwise is reverse rotation. The steering should match the direction of the console startup icon pressed on the display. If it is not consistent, stop the jaw crusher, turn off the power and adjust the positions of the two sets of solenoid valve connectors on the swash plate pump (first ensure that the channel wiring on the corresponding node module is correct), and restart the host.

④During the start-up process of the jaw crusher, observe the current values of the swashplate pump 1 and swashplate pump 2 on the display screen. If the current feedback value is not synchronized with the given value, the jaw crusher does not act and the feedback value is less than 200mA, the power should be turned off and check that the wiring is properly and securely connected. Under normal circumstances, the given mA value of the proportional valve is not much different from the feedback mA value.

⑤The debugging speed of the jaw crusher is 50% by default regardless of whether it is forward or reverse. It can be adjusted arbitrarily according to the needs of debugging. It is not recommended to set the speed below 50% for debugging. After the jaw crusher starts normally, if the alarm of "low crusher speed" appears, under the condition of ensuring the correct installation of the proximity switch, it may be caused by the too high number of action pulses set by the proximity switch. Just lower the number of action pulses of the proximity switch, please contact relevant professionals for specific operations.

⑥After the jaw crusher starts normally, no matter whether it is forward or reverse, it needs to be idling for 20-30 minutes continuously to observe whether the operation is stable, whether the vibration is within the normal range, and whether there is any abnormal sound.

⑪ Vibration feeder debugging:

①Unfold the upper part of the vibrating feeder and fix it in place to ensure that there is nothing on the vibrating feeder.

②At the beginning of debugging, the speed value of the feeding belt conveyor should be adjusted to 50%. After starting, if the vibration of the vibrating feeder is abnormal, it should be stopped immediately, the main power supply should be disconnected, and an investigation should be done.

③During the start-up process of the vibrating feeder, observe the current value of the feeder speed on the display screen. If the current feedback value is not synchronized with the given value, the feeder does not move and the feedback value is less than 200mA, the power should be turned off and check that the wiring is properly and securely connected. Under normal circumstances, the given mA value of the proportional valve is not much different from the feedback mA value.

④After the feeder runs normally, adjust the speed of the feeder, the upper limit is 100%, and the lower limit is 10%, observe whether the speed change of the feeder is normal, and measure the amplitude of the feeder with relevant instruments if necessary. Speed 100% corresponds to the parameter value of "maximum speed set point of feeder" under the interface of "Feeder Control Parameters", speed 0% corresponds to the parameter value of "minimum speed set point of feeder belt" under the interface of "Feeder Control Parameters". See chapter 6.3 for details. These two parameter values can be preset according to the actual application.

(11)Tracked walking debugging:

Note: Be sure to raise the tail of the main conveyor belt before the tracked travel operation!

①Enter the "Tracked Walking Mode" interface and select the control mode (wired or wireless). For the specific remote control operation instructions, please refer to chapter 6.6. It is stipulated that the vibrating feeder is the tail and the main conveyor belt is the head. The operator's standing direction should look from the tail to the head. The left hand side is the left forward or backward direction of the tracked, and the right hand side is the right forward or backward direction of the tracked.

②Carry out the track steering test: operate the direction keys on the remote control one by one, and observe whether the movements of the left and right tracks are consistent with the direction keys on the remote control.

③Carry out the crawler walking test: press the left and right forward or left and right backward direction keys at the same time, and observe whether there is any deviation in the process of tracked forward or backward. If a slight deviation is found, it can be corrected by adjusting the given value of the tracked proportional valve. The maximum difference between the left and right direction adjustment should not exceed 50mA. If a serious deviation is found, first check whether the hydraulic or mechanical aspects exist question.

④The wireless remote control device of this model is standard. After the wireless remote control function is fully activated, the green alarm light should be on. At this time, other buttons of the wireless remote control can be used (except the activation button on the remote control).

⑤When the tracked is running, there are two modes for the control of the engine speed—manual and automatic, which can be changed under the "Control System Parameters" interface. The default is to manually control the engine speed. In the "Tracked Walking Mode" interface, the speed of the engine can be increased or decreased through the high and low speed buttons, thereby controlling the speed of the tracked steering or walking. If the automatic engine control mode is selected, the engine speed will automatically rise to the speed value set by the "Tracked High-speed Travel Engine Speed" parameter after the running warning for 7s ends. Due to the limitations of the workshop site, it is recommended to use the manual mode to change the engine speed for track steering or walking tests to ensure safety.

6 Display operating instructions

6.1 Main menu interface

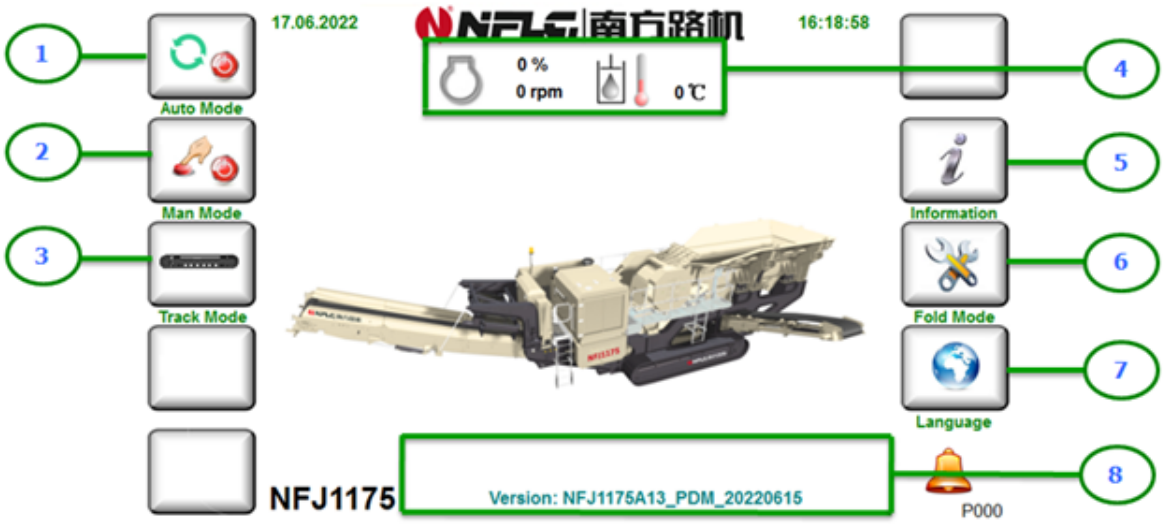


Figure 6.1.1 "Main Menu" interface

1—Enter the "Automatic Production Mode" interface (see Chapter 6.6 for details)

2—Enter the "Manual Production Mode" interface (see Chapter 6.7 for details)

3—Enter the "Tracked Walking Mode" interface (see Chapter 6.5 for details)

4—Display of engine load percentage, engine speed and hydraulic oil temperature.

5—Enter the "Function Information" interface (see Chapter 6.2 for details)

6—Enter the "Folding Mode" interface (see Chapter 6.9 for details)

7—Enter the "Language Selection" interface (see Chapter 6.10 for details)

8—PDM program version number and specific machine number.

The "Main Menu" interface appears automatically after the system is powered on. During the start-up warning process, all buttons on the screen are not allowed to operate. If the equipment has no fault alarm, after the warning is over, the user can enter the corresponding interface to perform the next operation as needed. If the equipment has a fault alarm, after the warning is over, the interface will automatically jump to the "Alarm Information" interface, listing the current fault alarm information.

6.2 Function information interface

Figure 6.2.1 "Function Information" interface

1—Enter the "Parameter Setting" interface 2—Enter the "Password Input" interface

3—Enter the "Date Settings" interface 4—Enter the "Engine Information" interface

5—Enter the "Node Information" interface 6—Enter the "Alarm Information" interface

7—Enter the "Screen Brightness Adjustment" interface 8—[Exit] button

This interface provides the entry of various important function information of the system.

6.3 Parameter setting interface

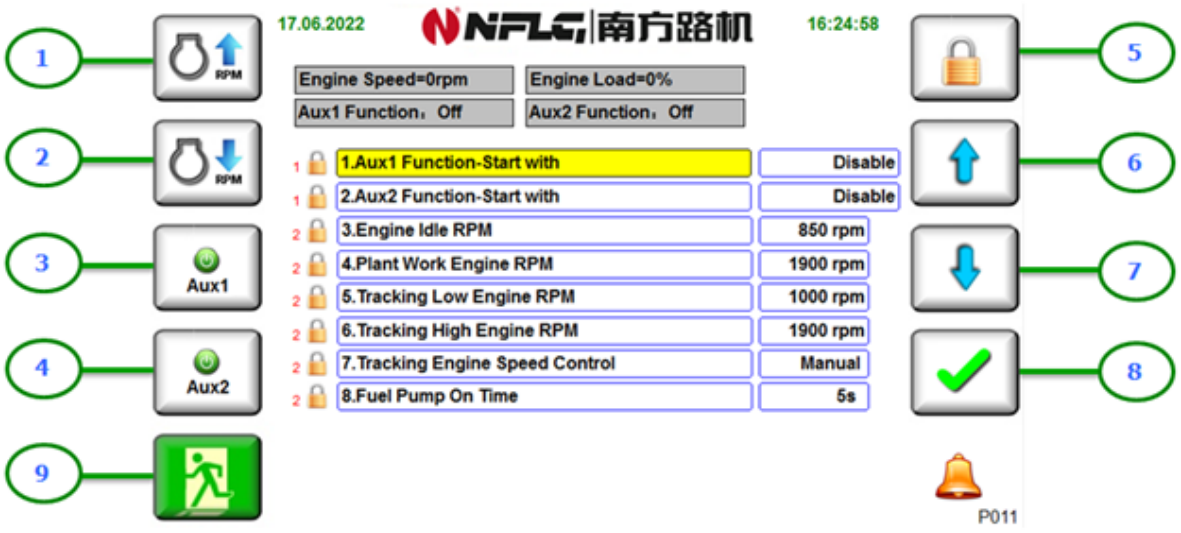
In the "Main Menu" interface, select the [Function Information] button (Figure 6.1.1-5), and it will enter the "Function Information" interface, and then select the [Parameter Setting] button to enter the "Parameter Setting" interface, as shown in Figure 6.3.1 shown.

Figure 6.3.1 "Parameter Setting" interface

This interface integrates the setting functions of all parameters of the system, and users can make reasonable changes according to different working conditions.

Note: The following content only introduces some parameter interfaces for common operations. Before modifying any parameter, please read the manual carefully, understand and master it thoroughly before proceeding.

(1) System control parameter interface



1—[Engine Speed Increase] button, press and hold to increase the engine speed, the function of this button will not be described below.

2—[Engine Speed Reduction] button, press and hold to reduce the engine speed, the function of this button will not be described below.

3—[Auxiliary 1 Start/Stop] button can start or stop the output of auxiliary channel 1.

4—[Auxiliary 2 Start/Stop] button can start or stop the output of auxiliary channel 2.

5—Enter the "Password Input" interface. If the password is valid, the icon will display the unlocked state. If the password is invalid, the icon will display the locked state. The function of this button will not be described later.

6-The function of the [Parameter Item up Selection] button is equivalent to rotating the knob counterclockwise. The function of this button will not be described below.

7-The function of the [Parameter Item down Selection] button is equivalent to rotating the knob clockwise. The function of this button will not be described below.

8-[Confirm Selection] button, the function is equivalent to pressing the center of the knob, the function of this button will not be described below.

Main parameter setting description:

①Start mode selection - auxiliary output 1 - this function parameter is to select the start mode of auxiliary channel 1. You can not use it, or you can choose to follow other components to start at the same time. The startup mode selection of auxiliary output 2 is also the same, and will not be introduced here.

②Engine idle speed, working mode engine speed, tracked low-speed running engine speed, crawler high-speed running engine speed can be reasonably set according to the actual working conditions, and the default values can be used in general. The engine idle speed setting should not exceed 900rpm.

③Tracked travel engine control mode - there are two modes of manual and automatic. When the equipment is running on the "Tracked Walking Mode" interface, if manual is selected, the engine speed can only be adjusted through the [Engine Speed Increase/Decrease] button; If Auto is selected, the engine speed will be automatically adjusted. The engine speed corresponding to the left/right forward and left/right reverse of the unilateral track is the fifth parameter, and the engine speed corresponding to the simultaneous forward and reverse of the bilateral track is the sixth parameter.

④The duration of the engine fuel pump action - the default is 5s. The working time depends on the location of the diesel tank and the length of the oil pipe.

(2) Hydraulic oil temperature control parameter interface

1—[Cooling Fan Start and Stop] button can start or stop the output of the cooling fan, which is only valid in this interface.

Main parameter setting description:

①Hydraulic oil temperature sensor function selection - release or maintain the interlocking relationship between hydraulic oil temperature and hydraulic oil temperature sensor status to the control system by disabling or activating, the default value is "active".

Note: Changing the value of this parameter to "Disabled" may have serious consequences, please use it after careful consideration. This disable function is only applicable when the temperature sensor fails. In order to ensure the normal operation of the equipment and emergency use, please replace the temperature sensor of the same type as soon as possible, and the parameter value must be changed to "active" again.

②Hydraulic oil high temperature warning value - If the hydraulic oil temperature is higher than or equal to this set value for more than the delay time of the third item, the system will pop up an alarm to prompt the operator. This function is valid when the "hydraulic oil temperature sensor function selection" parameter is set to "active".

③Hydraulic oil high temperature shutdown value - if the hydraulic oil temperature is higher than or equal to this set value for longer than the delay time of the sixth item, the system will pop up an alarm and stop the production process immediately. This function is valid when the "hydraulic oil temperature sensor function selection" parameter is set to "active".

⑤Hydraulic oil high temperature stop feeder function selection - disable or activate to release or maintain the interlock relationship to the vibrating feeder

when the hydraulic oil temperature exceeds the value set by "Hydraulic oil high temperature warning value", the default value is "active" ".

(3) Main conveyor belt control parameter interface

1—[Main conveyor belt start and stop] button can start or stop the output of the main conveyor belt, which is only valid in this interface.

Parameter setting description:

①Start delay time - the waiting time after the main conveyor belt receives the start signal until the solenoid valve is energized. This delay time applies in any operating mode.

②Stop delay time - this delay time is applicable to the sequential stop process under the interface of "automatic production mode" or "manual production mode". After the crusher stops, the main conveyor belt stops the delay countdown.

Under normal circumstances, the default value of the start delay time and stop delay time of the component can be used. When it is really necessary to adjust the parameters to better meet the production requirements, the user can make appropriate changes.

③Speed wheel sensor function selection - release or maintain the interlocking relationship of the speed wheel sensor input signal to the control system by disabling or activating, the default value is "active".

④Tail proximity sensor function selection - disable or activate to release or maintain the interlocking relationship between the input signal of the rear proximity sensor of the main conveyor belt on the tracked running or the main conveyor belt running, the default value is "active".

Note: If the parameter value of the above ③④ is changed to "disabled", it may lead to serious consequences, please use it after careful consideration. The disable function is only applicable to the case of sensor failure. In order to

ensure the normal operation of the device and emergency use, please replace the sensor of the same type as soon as possible, and the parameter value must be changed to "active" again.

⑤Main conveyor belt speed low alarm value - if the "speed wheel sensor function selection" is "active", this parameter is valid. When the main conveyor belt is started, if the actual speed is lower than the set value, the system will pop up an alarm and stop the main conveyor.

(4) Crusher control parameter interface

1—Information bar. It should be noted here that, if the host pressure = 0bar (0Bar), the value displayed in the brackets is the given value of the program, and the value displayed outside the brackets is the feedback value of the sensor. Similarly, the tie rod pressure, swash plate pump 1 and swash plate pump 2 numerical display also follow the same pattern.

2—Enter the interface of "More Parameters of the Host", see Figure 6.3.4.

Parameter setting description:

① Start delay time - the waiting time after the crusher receives the start signal to start the given current output signal. This delay time applies in any operating mode.

②Stop delay time - this delay time is applicable to the sequential stop process in the "automatic production mode" or "manual production mode" interface. After the feeder stops, the crusher will start to stop the delay countdown.

③Crusher speed sensor function selection - by disabling or activating to release or maintain the chain relationship between the speed proximity switch input signal and the crusher, the default value is "active".

Note: Changing the value of this parameter to "Disabled" may have serious consequences, please use it after careful consideration. This disable function

is only applicable when the speed proximity switch fails. In order to ensure the normal operation of the equipment and emergency use, please replace the speed proximity switch of the same model as soon as possible, and the parameter value must be changed to "active" again.

④The upper limit of the tie rod filling pressure, the lower limit of the tie rod filling pressure and the pressure value of the tie rod for stopping the crusher are the key parameters for the system to control the tie rod of the crusher. The default values can be used, and it is not recommended for users to change them without authorization.

⑤Delay time for stopping the crushing machine - if the system detects that the pressure of the tie rod is lower than the value set in the sixth item, it will stop the operation of the crushing machine after delaying the time set by this parameter value.

⑥Initial fill duration - The maximum time allowed for the automatic filling process of the drawbar. If the charging of the rod is not completed beyond this time, the system will pop up an alarm and stop the automatic charging.

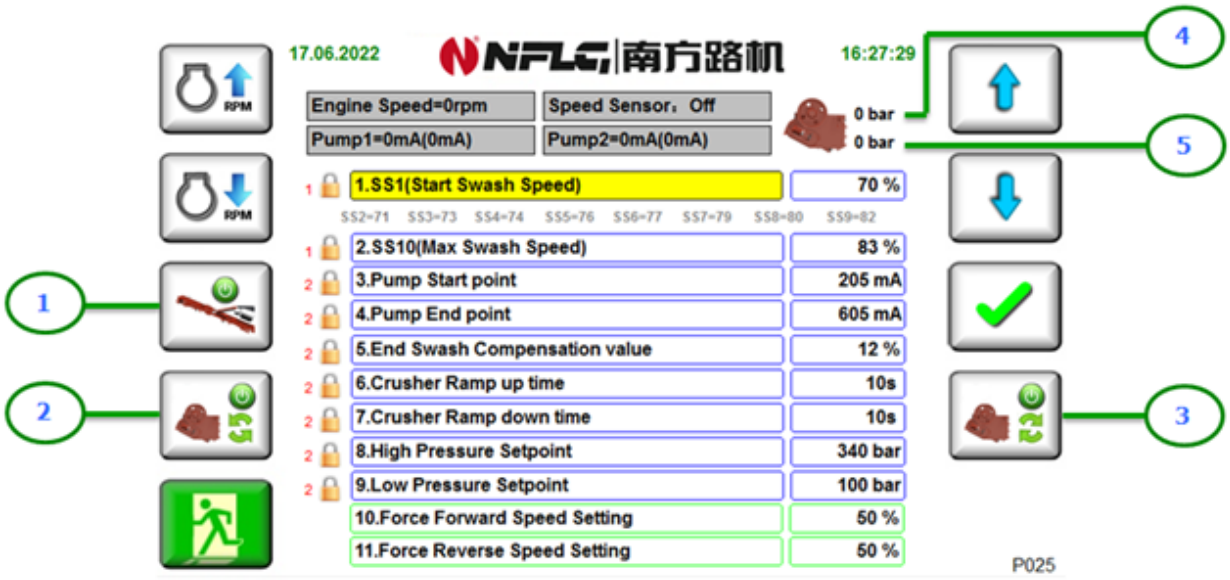


Figure 6.3.4 "crusher more parameters" interface

1—[Main Conveyor Belt Start and Stop] button can start or stop the output of the main conveyor belt, which is only valid in this interface.

2—[Start and Stop of Crusher Reverse Rotation] button can start or stop the crusher to make it run in reverse, which is only valid in this interface.

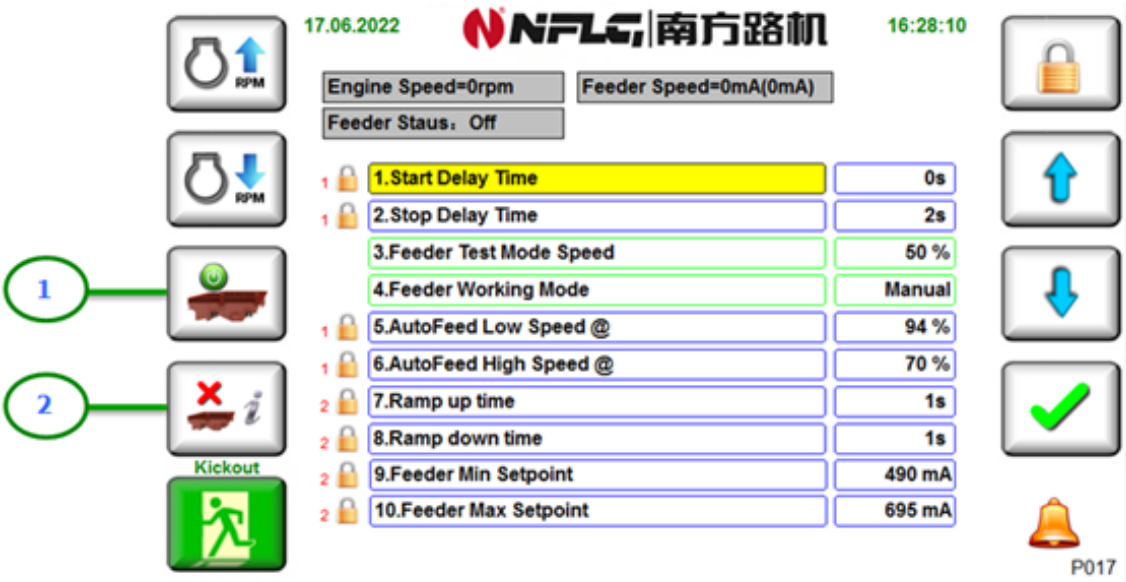
3—[Start and Stop of Forward Rotation of Crusher] button can start or stop the crusher to make it run forward, which is only valid in this interface.

Parameter setting description:

Note: The parameters of this parameter interface are not allowed to be changed by the user without authorization!

① SS1 and SS10 correspond to the minimum and maximum speeds among the 10 adjustable speeds of the crusher in the "Automatic Production Mode" and "Manual Production Mode" interface, and the other speeds SS2 to SS9 are automatically calculated by the system.
②The initial value of the swash plate pump - corresponding to 0% of the swash plate speed - when the crusher starts, the given mA value of the swash plate opening starts from the current value, not from 0 mA. This parameter value is based on the electrical characteristics of the proportional valve.
③The initial value of the swashplate pump - corresponding to 100% swashplate speed - the maximum given mA value to limit the swashplate opening.
④The combination of the terminal compensation value of the swash plate pump, the high setting value of the swash plate opening compensation pressure and the low setting value of the compensation pressure of the swash plate pump can realize the compensation calculation of the efficiency loss of the swash plate pump. The crusher keeps running at a constant speed at all times during the production process to avoid the phenomenon of falling speed.
⑤Forced crusher forward and reverse speed setting values - these two parameter values are only used for the debugging of the crusher. They are valid under this interface. During the debugging of the crusher, reasonable adjustments can be made according to the needs.
⑥Ramp-up time of swash plate pump—when the crusher starts, the time required for the proportional valve that controls the opening of the swash plate pump to rise from 0mA to a given mA value, so as to realize the soft start of the crusher.
⑦Ramp-down time of swash plate pump—when the crusher starts, the time required for the proportional valve that controls the opening of the swash plate pump to drop from a given mA value to 0mA, so as to realize the soft stop of the crusher.

(5) Feeder control parameter interface



1—[Start and stop of feeder] button can start or stop the output of the feeder, which is only valid in this interface.

2—Enter the interface of "Pause Parameters of Feeder", see Figure 6.3.5.

Parameter setting description:

①Start delay time - the waiting time for the feeder to start the given current output signal after receiving the start signal. This delay time applies in any operating mode.

②Stop delay time - this delay time is applicable to the sequential stop process in the "automatic production mode" or "manual production mode" interface. After the automatic stop button is pressed, the feeder stop delay countdown starts.

③Feeder test mode speed - This parameter value is only used for the debugging of the feeder, and it is valid under this interface. During the process of debugging the feeder, it can be adjusted reasonably according to the needs.

④Feeder working mode selection - choose whether the speed of the feeder is manually given or automatically given, if it is automatically given, the speed of the feeder will be automatically adjusted within a certain speed range according to the pressure of the crusher. The general principle is as follows:

When the pressure percentage of the crusher is greater than the parameter value set in the fifth item, the feeder speed will run at the mA value set in the ninth item. On the contrary, when the crusher pressure percentage is less than the parameter value set in the sixth item , the feeder speed will run at the mA value set in item 10. Then, when the crushing host pressure percentage is between the fifth parameter and the sixth parameter, the change of the feeder speed is inversely proportional to the change of the crushing host pressure percentage and changes linearly. In general, the default values of these parameters can be used.

⑤Feeder Speed Ramp Time - The time it takes for the proportional valve to ramp from 0 mA to a given mA value when the feeder starts.

⑥ Feeder Speed Ramp Down Time - The time it takes for the proportional valve to ramp down from a given mA value to 0 mA when the feeder starts.

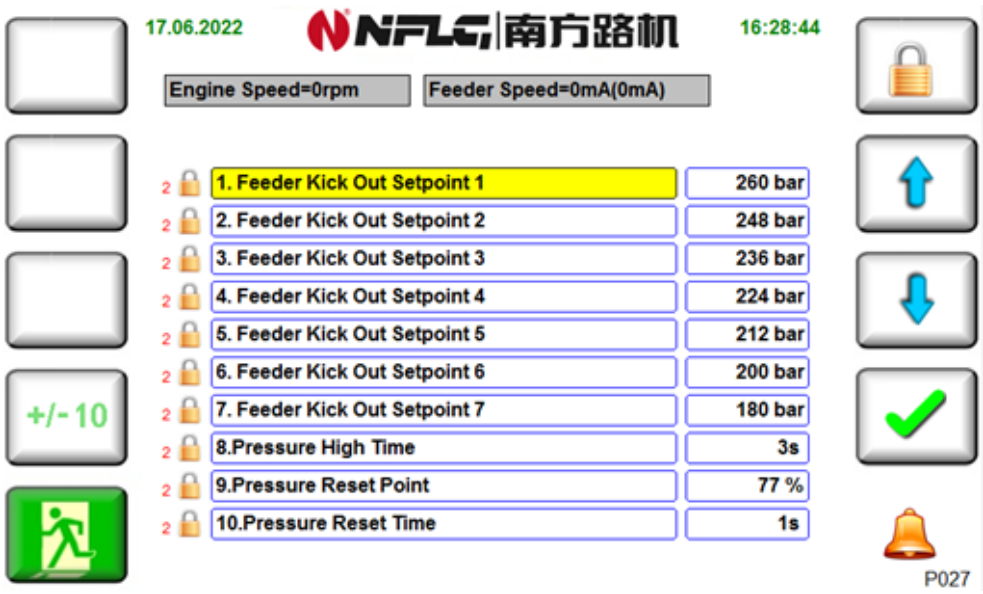


Figure 6.3.5 "Feeder Pause Parameters" interface

Parameter setting description:

The parameters of this interface are the functions of automatic pause and automatic restart of the feeder. According to the characteristics of different stone materials, the system has seven applications to choose from. Each application corresponds to different crusher pressure setting values. Once the crusher pressure exceeds the set value and lasts for a certain time (the eighth parameter), the feeder will automatically pause, the feeder will restart automatically only when the pressure of the crusher returns to the allowable range of the pressure set value of the crusher (parameter 9) and lasts for a certain period of time (parameter 10).

The pause function of the feeder will work regardless of whether the feeder's working mode is manual or automatic.

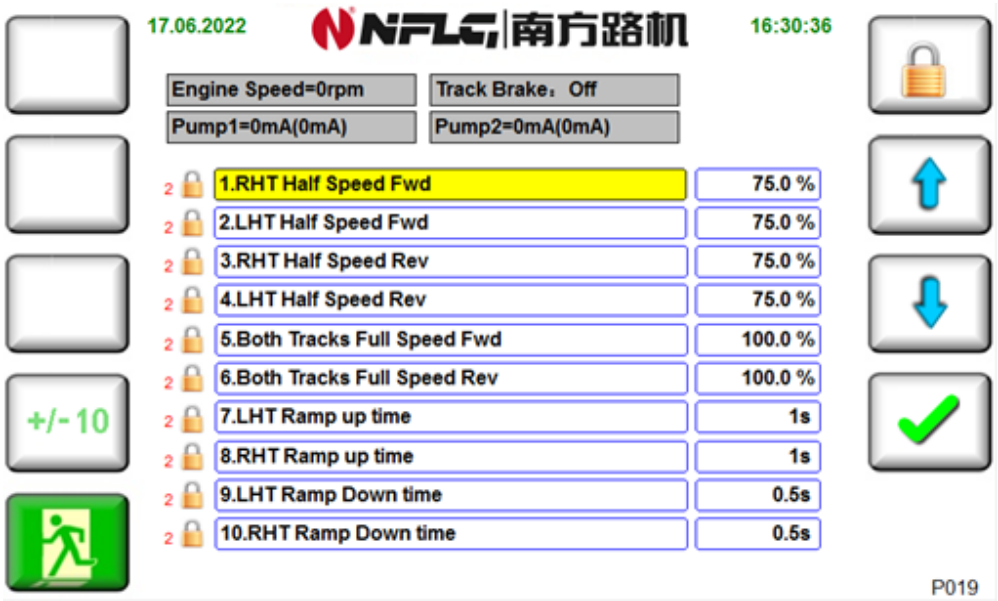
(6) Upper and lower chain control parameter interface

Parameter setting description:

① Subordinate equipment input function selection - there are "normally open" and "normally closed" options, which can be set according to the signal type connected to the subordinate equipment. It is recommended to activate this function if there is a lower-level device produced in conjunction with this device. In the case of normal external line connection, once the lower-level equipment fails and stops, the system receives the stop signal of the lower-level equipment, and after a certain delay (the second parameter), it will stop the feeder of the equipment, which can prevent continuous supply of material from causing unpredictable impact on the lower equipment.

② Main conveyor belt stack sensor function - This sensor is optional, it is used to measure the height of the material under the main conveyor head, and then control the automatic start or stop of the feeding belt. The parameter settings of the fifth to eighth items are set according to the actual needs of different working conditions.

(7) Tracked walking control parameter interface



Parameter setting description:

① The first to fourth items are the output of the given current percentage when the left/right single tracked moves forward and backward. The size of the given current percentage directly affects the speed of the crawler walking or turning. The fine-tuning of these values can ensure the consistency of the walking speed of the left and right tracks.

②Left and right tracked tracks forward/backward at high speed - press the button on the remote control for simultaneous forward/backward movement of dual tracks, and the system will output the current of the proportional valve of the swash plate

pump with this set value.
③Left/right crawler low-speed walking ramp acceleration time, left and right crawler high-speed walking ramp acceleration time - the time required for the given current of the proportional valve of the swash plate pump to increase from 0mA to the set value, to realize the soft-start function of crawler walking, and at the same time it is also to reduce the impact of instantaneous pressure on proportional valves and related hydraulic systems. This time cannot be set too short, generally the default value can be used.
④Crawler stop ramp down time - the time required for the given current of the proportional valve of the swash plate pump to decrease from the set value to 0mA, to realize the soft stop function of the crawler. Due to the complex and changeable working conditions, from a safety point of view, this setting value is 0.1 seconds. Users cannot make changes without authorization.

⑾ Restore factory default interface

This function is only available when the correct secondary password is entered.
Note: The data for parameter initialization is set by our company based on design principles and experience values, and may not be the best parameter data for the actual operation of the equipment. After the parameter initialization is executed, all the parameters set by the user will be set to the factory default values! Therefore, before executing parameter initialization, please make sure to back up the original data to prevent data loss after initialization. The company is not responsible for equipment damage, software crash, production stagnation and other losses caused by the user's illegal and illegal operations!

6.4 Node information interface

In the "Function Information" interface, select the [I/O Status] button to enter the "Node Information" interface, as shown in Figure 6.4.1.

This interface shows the connection status of all nodes in the entire CANBUS network, and you can choose to enter the node I/O interface you want to view according to your needs. The following points need to be explained:

① "✗" Indicates that the connection between the corresponding node and the PLC is abnormal, "✓" Indicates that the connection between the corresponding node and the PLC is normal.

② In the node I/O status interface, if it is a switch value, the gray light indicates that there is no input or output signal, and the green light indicates that an input or output signal is detected; if it is an analog value, you can view the original value of the signal input.

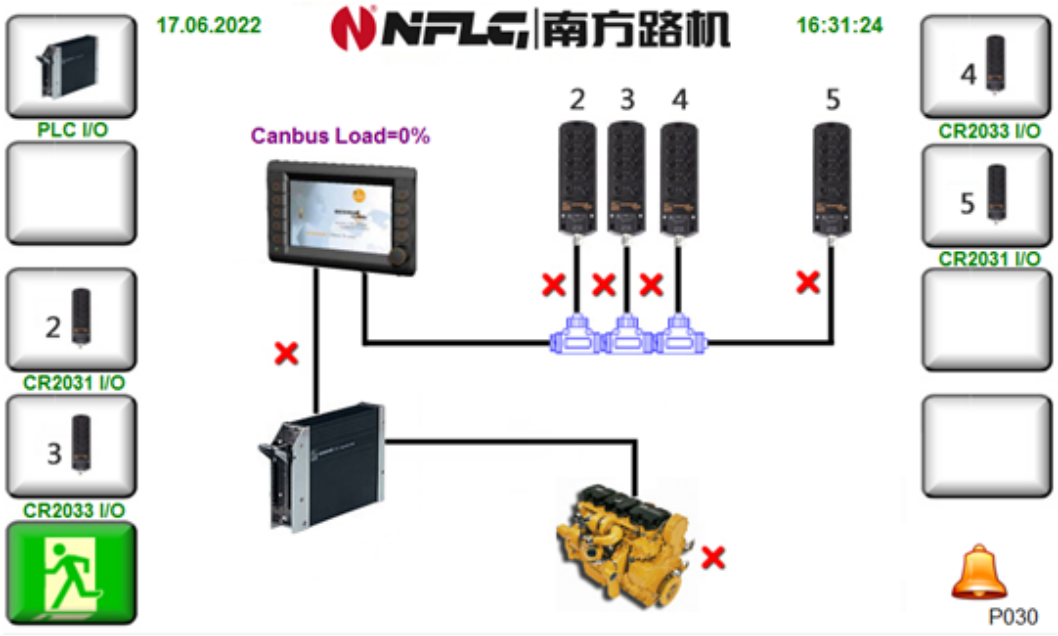


Figure 6.4.1 "Node Information" interface

③ In the "Node Information" interface, select the [PLC I/O] button to enter the "PLC I/O Status" interface, and then select the [PLC Diagnosis] button to enter the "PLC Diagnosis" interface, as shown in Figure 6.4.2 As shown, the "PLC Diagnosis" interface covers all the information of PLC fault codes and fault bits.

Note: After the "PLC fault!" alarm occurs, the key switch cannot be powered off. It is necessary to enter the "PLC Diagnosis" interface to view the specific PLC fault information and record it, which is convenient for subsequent analysis and troubleshooting.

Figure 6.4.2 "PLC Diagnosis" interface

6.5 Tracked walking mode interface

In the "Main Menu" interface, select the [Walking Mode] button (Fig. 6.1.1-3) to enter the "Crawler Walking Mode" interface, as shown in Fig. 6.5.1. When you want to move the device, you must enter this interface to perform related operations.

Figure 6.5.1 Tracked Walking Mode

1—[Wired Remote] selection button

2—[Wireless Remote Control] selection button

3—[Exit] or [Function Information] button

4—Indicates that the main conveyor belt is not in the correct position

(1)Wired remote control operation

① Press the [Wired Remote] button, the button icon will appear "✔", indicates that the wired remote control function has been selected.

② Make sure that the plug of the wired remote control is correctly connected to the corresponding socket of the cabinet, as shown in Figure 6.5.2-F.

③ Press the button of Figure 6.5.2—E, the wired remote control is activated, and the buzzer and the yellow alarm light simultaneously act to start an early warning.

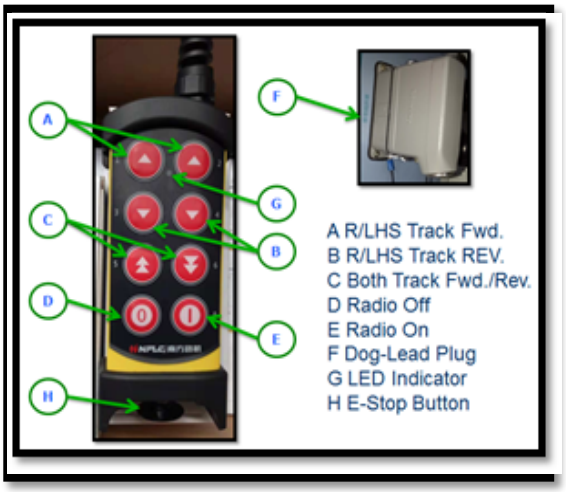


Figure 6.5.2 Wired remote

④ There needs to be an early warning time of 7 seconds. <u>During the early warning process, do not operate the buttons on any wired remote control, otherwise the</u>

warning will be re-timed. After the warning is over, the indicator light 6.5.2-G changes from red to green.

⑤ Control the walking or turning of the crawler by pressing the 6.5.2-A, B or C direction keys in the figure.

⑥ After the operation is completed, press the button in Figure 6.5.2-D to turn off the function of each button of the wired remote control, the indicator light 6.5.2-G changes from green to red, and then press the [Wired Remote Control] button again on the display screen , the icon "✔" disappears, indicating that the wired remote control function is disabled.

(2) Wireless remote control operation

Note: The ON\OFF switch of the back shell of the transmitter should be in the ON position before using it, otherwise the transmitter will not start.

① Start the transmitter (Figure 6.5.3) - within the effective range of the wireless signal, make sure to press the stop button, and then pull out the stop button, the top indicator light on the transmitter lights up (green means the battery is fully charged, red means the battery level insufficient), the indicator lights SL and SR flash (red), press the two power-on buttons at the same time, hold for at least 1 second, then the indicators SL and SR light up (red), and finally release the two power-on buttons at the same time, the indicator light SL And the SR goes out, the top indicator light flashes (green), the transmitter is started up.

② Press the [Wireless Remote] button, "✔" will appear on the button icon, indicating that the wireless remote control function has been selected.

③ Press the button of Figure 6.5.3-E, the function of each button of the wireless remote control is activated, and the buzzer and the alarm light will act at the same time to start an early warning. At the same time, the green alarm light on the device illuminates, indicating that the wireless remote control is active and within range.

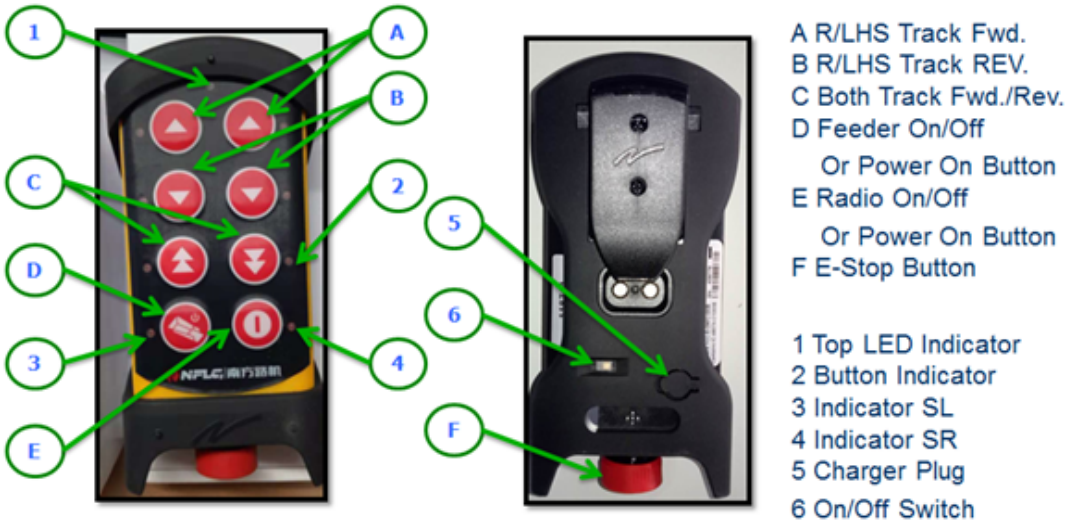


Figure 6.6.3 Wireless transmitter

④There needs to be a 7-second warning. During the warning process, do not operate any buttons on the remote control, otherwise the warning will be re-timed. ⑤After the warning is completed, the walking or turning of the crawler can be controlled by pressing the direction keys of Figure 6.5.3-A or B.

⑥ After the operation is completed, press the button in Figure 6.5.3-E again to disable the function of each button of the wireless remote control, and then press the [Wireless Remote Control] button again on the display screen. At this time, the icon "✔" disappears, indicating that the wireless remote control function is closed.

Note: After the transmitter has not been operated for a period of time (default 6 minutes), it will automatically turn off the transmitter to save power. Therefore, after the operation of the transmitter is completed, please perform the operation of step ⑥ as soon as possible to avoid the false alarm of "emergency stop failure of the wireless remote control", which will cause the equipment to stop. The protection chain of the emergency stop button to the equipment is only effective when the functions of the buttons of the wireless remote control are activated.

⑦ If you do not operate the wireless remote control for a long time, please toggle Figure 6.6.3-6 to the off position to turn off the power of the remote control.

6.6 Automatic production mode interface

In the "Main Menu" interface, select the [Auto Mode] button to enter the "Auto Production Mode" interface, as shown in Figure 6.6.1.

Figure 6.6.1 Automatic production mode

In this mode, each component starts and stops in a certain order and logical relationship. If the upper-level component is not started, the lower-level component will not start; and the stop of any upper-level component will cause all lower-level components to stop.

The following will describe the icon information, status information, and function buttons in this interface in detail, and the numbers correspond to the labels on Figure 6.6.1:

1—The [Auto start and stop] button is responsible for the start and stop of the automatic production process of the equipment. If there is a related fault alarm, please find out and remove the fault before starting the equipment, otherwise the equipment will not start normally (for details of faults, please refer to Chapter 7).

2—"Iron remover status" icon, if the iron remover function is activated, the icon will be highlighted (Figure 6.6.2-2); if the iron remover function is disabled, the icon will be grayed out and marked with a red cross. After the equipment production process is started, this function button is in an invalid state, and no operation is allowed, so the activation or deactivation of the iron remover function operation can only be performed before the equipment production process is not started. When activated, the start and stop of the iron remover will be

linked to the production process of the equipment; when disabled, the system will ignore the control of the iron remover.

3—"Side belt conveyor status" icon, if the side belt conveyor function is activated, the icon will be highlighted (Figure 6.6.2-3); if the side belt conveyor function is disabled, the icon will be grayed out and marked with a red cross. After the equipment production process is started, this function button is in an invalid state, and no operation is allowed, so the activation or deactivation of the side belt conveyor function operation can only be performed before the equipment production process is not started. When activated, the start and stop of the side belt conveyor will be interlocked into the automatic production process of the equipment; when disabled, the system will ignore the control of the opposite side belt conveyor.

4—[Exit] or [Function Information] button, if the automatic production process of the equipment is not started, you can exit this interface and go to the "main menu" interface. When the automatic production process of the equipment starts, this button will automatically change to the [Function Information] button (Figure 6.6.2-1), from which you can enter the "Function Information" page (see Section 6.2).

5—Press the [Wireless Remote] button, "✔" will appear on the button icon, indicating that the wireless remote control function has been selected. Press the button of Figure 6.6.3—E, the function of each button of the wireless transmitter is activated, and the green alarm light on the device lights up, indicating that the wireless remote control is active and within the effective range. In this mode, the wireless transmitter allows remote control of the start and stop of the feeder and the adjustment of the feeder speed. Please refer to chapter 6.5 for the start of the wireless transmitter and the introduction of each button.

6—[Feeder start and stop] button to manually control the start and stop of the feeder. Even in automatic production mode, the feeder must be started manually. The gray display of the button icon indicates that the starting conditions of the feeding belt have not been met (Figure 6.6.1-5), and manual start is only allowed under the premise of fully satisfying the relevant interlock. At this time, the button icon changes to Figure 6.6.2-4 shown.

7—[Application selection] button, you can enter the "application selection" interface, for different stone properties, the system has seven applications to choose from.

8—[Clog Clear] button to enter the interface of "Block Clear Mode", see chapter 6.9 for details.

9—Display the relevant information of the crushing machine - the turning direction of the main machine (can be switched in the "blocking removal mode" interface), the upper Bar value is the crushing machine pressure, and the lower Bar is the main machine tie rod pressure.

10—Crushing host speed adjustment bar, there are ten speeds to choose from. Corresponding to the parameter values of SS1 to SS10 under the interface of "Crushing host control parameters".

11—Automatically suspend the crushing main pressure value of the feeder, and choose different applications (seven options), this value is also different.

12—Feeder speed adjustment. When the working mode of the feeder is "manual", the speed can be adjusted freely manually through the knob. If the working mode is "Auto", only the automatic adjustment range of the feeder speed can be set, 0% corresponds to the ninth item of the "Feeder control parameters" interface, and 100% corresponds to the tenth item of the "Feeder control parameters" interface. Parameter value setting, please refer to chapter 6.3.

1	2	3	4

Figure 6.6.2

6.7 Manual production mode interface

In the "Main Menu" interface, select the [Auto Mode] button to enter the "Manual Production Mode" interface, as shown in Figure 6.7.1.

Figure 6.7.1 Manual production mode

1—[Main conveyor belt start and stop] button
2—[Iron remover start and stop] button
6—[Side belt conveyor start and stop] button
7—[Broken host start and stop] button
8—[Feeder start and stop] button

For the introduction of other icon information, status information, and function buttons, please refer to the content of the "Automatic Production Mode" interface, see chapter 6.6 for details. In this mode, the start and stop of each component need to be manually operated, and there is a certain sequence and logical relationship. If the upper-level component is not started, the lower-level component will not start; and the stop of any upper-level component will cause all lower-level components to stop. The starting sequence of each component is: main conveyor belt -> iron remover (can be ignored when not in use) -> side belt conveyor (can be ignored when not in use) -> crushing main machine -> feeder, and the stop sequence is just opposite to the start sequence.

6.8 Blockage clearing mode interface

In the interface of "automatic production mode" or "manual production mode", select the button of [blocking clear] to enter the interface of "blocking clearing mode", as shown in 6.8.1.

The function of this blockage removal is not a conventional method to clear the blockage in the crusher chamber, but only an auxiliary method. Under certain conditions, through the repeated operation of the forward and reverse jog of the host, the blockage in the crusher chamber can be quickly and effectively removed.

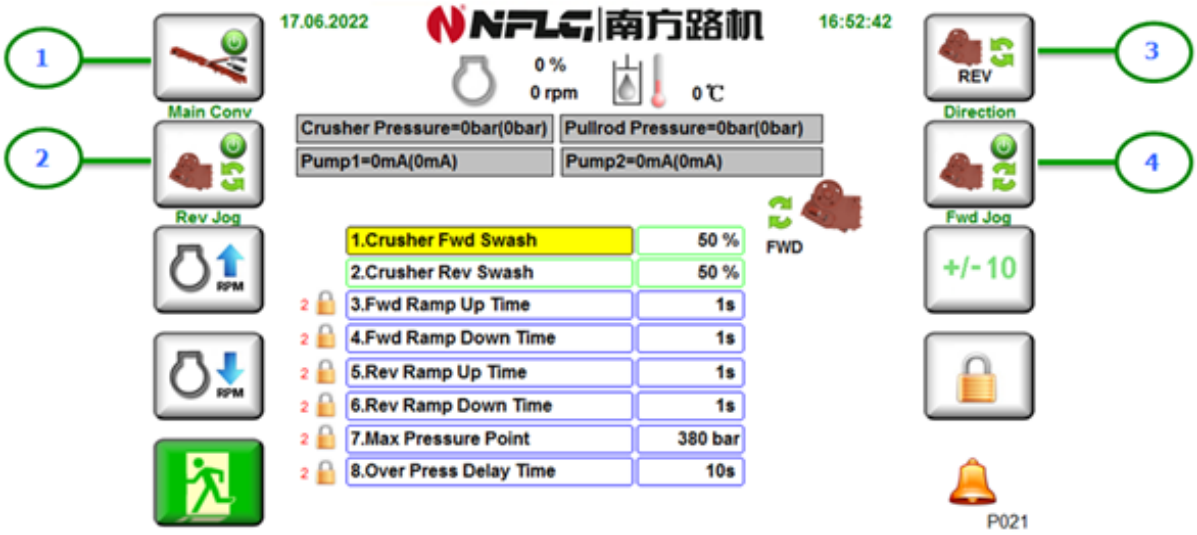


Figure 6.8.1 Blockage clearing mode

1—[Main conveyor belt start and stop] button can start or stop the output of the main conveyor belt, which is only valid in this interface. The main conveyor belt must be started before the forward and reverse jog operation of the crusher.

2—[Crusher reverse jog] button, press and hold the button to start the crusher in reverse (counterclockwise), release the button and stop the crusher.

3—[Crusher steering switch] button to determine the steering of the crusher in automatic production mode or manual production mode, which can be switched through this button.

4—[Crusher forward jog] button, press and hold the button to start the crusher forward (clockwise), release the button and stop the crusher.

Note: Before performing the blockage removal operation in this mode, the stones accumulated between the main conveyor belt and the bottom chamber of the crusher should be cleaned first to ensure that there is no stone stuck between the two, so as to avoid causing unpredictable damage to the main conveyor belt after the main conveyor belt is started.

Parameter setting description:

①The forward rotation and reverse rotation of the crusher - the opening of the swash plate - these two parameter values are only used for the jog operation

of the crusher. During the process of the crusher blocking and cleaning, it can be adjusted reasonably according to the needs.

②Forward and reverse ramp up time - when the crusher starts jogging, the time required for the proportional valve that controls the opening of the swash plate pump to rise from 0mA to a given mA value, to realize the soft start of the crusher, only suitable for the jog operation of the crusher.

③Forward and reverse ramp down time - when the crusher starts jogging, the time required for the proportional valve that controls the opening of the swash plate pump to drop from a given mA value to 0mA to achieve a soft stop of the crusher, only applicable to the jog operation of the crusher.

④The maximum pressure setting value of the crusher - this parameter is not applicable to the application under the "blocking material removal mode" interface, and is only valid under the "automatic production mode" or "manual production mode" interface. When the actual monitoring pressure of the crushing machine is greater than this set value, and after a certain time delay (parameter 8), the system will pop up an alarm of "high pressure of the crushing machine", and the production process will automatically stop in sequence.

6.9 Folding mode interface

Under the "Main Menu" interface, select the [Folding Mode] button (Figure 6.1.1-6), and it will enter the "Folding Mode" interface, as shown in Figure 6.9.1.

Figure 6.9.1 Folding mode interface

This interface can realize the ascending and descending operation of the tail of the main conveyor belt. Before the crawler runs, the tail of the main conveyor belt must be raised to a high position; and before the equipment is produced, the tail of the main conveyor belt must be lowered to a low position.

6.10 Fault information interface

After any fault occurs, no matter which interface it is currently in, it will directly jump to the "Fault Information" interface, as shown in Figure 6.10.1. After exiting this interface, you can enter the "Fault Information" interface again through the [Alarm Information] button under the "Function Information" interface.

Figure 6.10.1 Fault information interface

It should be noted:

① The entry displayed in the fault information interface is the current fault state of the system. The elimination of each fault entry must satisfy two conditions—the external factors that caused the fault have been eliminated and the fault information needs to be confirmed manually.

There are two states for the elimination of the fault entry - when a fault occurs, if the external factor that caused the fault has been eliminated, the displayed alarm bell icon will change from "🔔" to "🔕", at this time, manual confirmation will be performed(the operation of manual confirmation—>turn the circular knob on the display screen, select the fault, and then press the circular knob), and the fault entry will disappear automatically; when a fault occurs, if a manual confirmation operation is performed first, the fault entry will appear the icon "✔" at the bottom right, indicates that the fault entry has been known to the user. Once the external factors causing the fault are eliminated, the fault entry will disappear automatically.

②[PLC Diagnosis] button will only be displayed when the “PLC fault” alarm occurs in the system. Through this button, you can enter the “PLC Diagnosis” interface to view the fault code or fault information of the current PLC.

6.11 Fault record interface

If there is an unresolved fault currently, you can select the [Fault Record] button in the "Fault Information" interface to enter the "Fault Record" interface. If there is no fault currently, you can select [Alarm Information] in the "Function Information" interface to directly enter the "Fault Record" interface, as shown in Figure 6.11.1. The "fault record" interface displays the latest 200 historical fault information of the system, which will be saved for later inspection or maintenance.

Figure 6.11.1 Fault record interface

6.12 Engine information interface

In the "Function Information" interface, select the [Engine Information] button to enter the "Engine Information" interface, as shown in Figure 6.12.1. The engine ECU transmits the basic data to the on-board controller through J1939 communication. When the on-board controller communicates with the display screen normally, the data displayed on this interface will be synchronized with the data monitored by the engine ECU.

Figure 6.12.1 Engine information interface

The meanings of the icons in the interface are from left to right, and from top to bottom are engine speed RPM, engine operating hours hrs, engine load %, ECU voltage VDC, engine coolant temperature DegC, intake manifold temperature DegC, engine instantaneous fuel consumption L/Hr, oil pressure Kpa, intake manifold pressure Kpa.

6.13 Engine failure history interface

In the "Engine Information" interface, select the [Fault Record] button to enter the "Engine Fault Record" interface, as shown in Figure 6.13.1. If an "engine failure" alarm occurs in the system, you should enter this interface to check the specific failure information of the engine. You can search the engine maintenance manual through the SPN code and FMI code to obtain more detailed failure descriptions. If necessary, please contact the engine manufacturer for handling.

Figure 6.13.1 Engine failure history interface

6.14 Application selection interface

In the "Automatic Production Mode" or "Manual Production Mode" interface, select the [Application Selection] button to enter the "Application Selection" interface, as shown in 6.14.1.

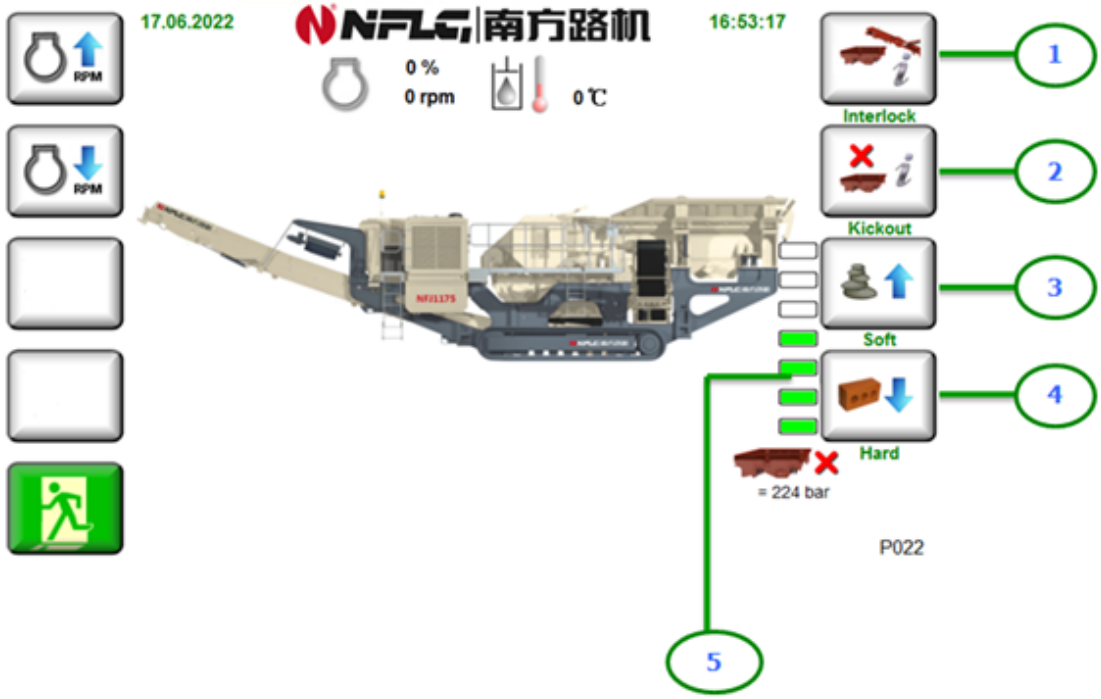


Figure 6.14.1 Application selection interface

1—Enter the interface of "upper and lower chain control parameters".

2—Enter the interface of "Pause Parameters of Feeder", see Figure 6.3.5.

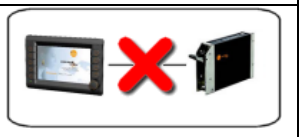
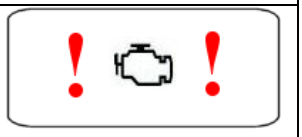
3—[Soft material] selection button 4—[Hard material] selection button

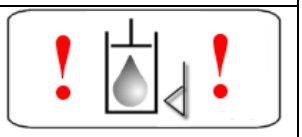
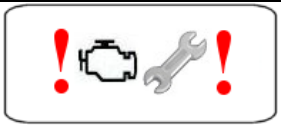
5—Status display of the selected material application type

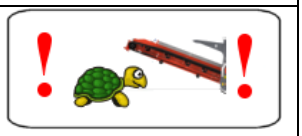
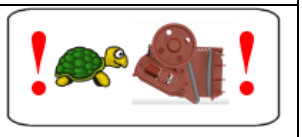
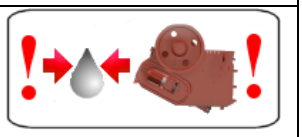
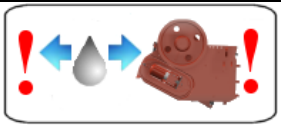
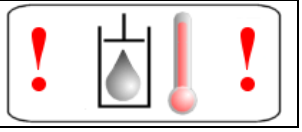
For different stone characteristics and application occasions, the system has seven application types to choose from. For the corresponding control parameter setting values, please refer to the above-mentioned "Feeder Pause Parameters" interface parameter setting description. It can be seen from the status display of the application type: the softer the material, the lower the pressure of the crushing host of the suspended feeder, on the contrary, the harder the material, the greater the pressure of the crushing host of the suspended feeder. The operator can choose according to the actual situation of the site to achieve the best performance of the equipment.

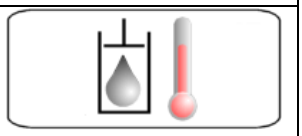
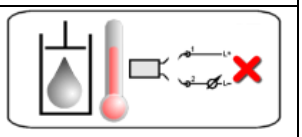
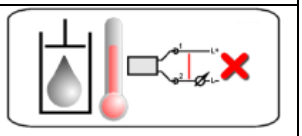
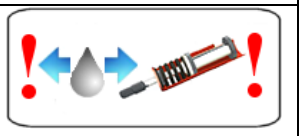
7 Fault tracing

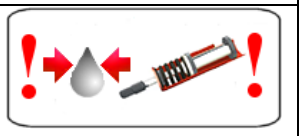
The following table lists some common fault items of equipment, and gives possible causes of faults and reference solutions. For further assistance, please contact our company.

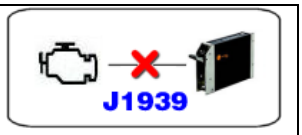
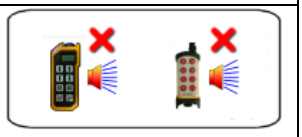
Error code	Possible reason	Solution	Fault Icon & Description
1	• The communication line between the PLC and the display screen is incorrectly wired or not connected.	✓ Check the wiring to make sure that CANH and CANL are properly connected.	 The connection between the display screen and the PLC is interrupted!
	• The CANBUS network terminating resistor is not connected or installed incorrectly.	✓ Check the resistance value on the CAN network in the power-off state, it should be 60 ohms.	
	• The PLC is faulty or the program is not running.	✓ Check the PLC status light, normally it flashes green at 2Hz.	
2	• The engine ECU has detected some kind of malfunction in itself.	✓ Check the specific SPN code and FMI code of the engine failure to obtain more detailed failure information, and contact the manufacturer in time.	 Engine failure!
3	• The CANBUS bus plug on the side of the cabinet is not connected or loose.	✓ Check the CAN bus plug on the side of the cabinet to make sure it is connected properly.	CANBUS bus error!
	• The CANBUS network terminating resistor is not connected or installed incorrectly.	✓ Check the resistance value on the CAN network in the power-off state, it should be 60 ohms.	
	• A node on the network has an incorrect baud rate setting.	✓Check the baud rate settings of each node to ensure that the baud rates are consistent.	
5	• The emergency stop button of the equipment is pressed.	✓ Check the condition of all emergency stop buttons on the equipment, and turn right to reset.	Equipment emergency stop failure!
	• The emergency stop button contacts are damaged or have poor contact.	✓ In the power-off state, check the on-off status of each emergency stop button contact separately, and replace the emergency stop button contact of the same type if necessary.	

	• Damaged safety relay or poor contact.	✓ Replace the safety relay of the same type.	
6	•The wired remote control plug or emergency plug is not connected.	✓ Check the wired remote control plug or emergency plug on the side of the cabinet to make sure it is properly connected.	Wired remote control emergency stop failure!
	• The emergency stop button of the wired remote control is damaged.	✓ Replace the emergency stop button of the same model.	
	• The emergency stop button of the wired remote control is pressed.	✓ Check the emergency stop button on the remote control and rotate to reset.	
7	• The emergency stop button of the wireless remote control is damaged.	✓ Please send the wireless remote control back to the manufacturer for processing.	Wireless remote control emergency stop failure!
	• The emergency stop button is pressed while the wireless remote control is active.	✓ Check the emergency stop button on the remote control, pull it out to reset.	
	• The wireless remote automatically powers off when activated.	✓ When the wireless remote control is not in use, please turn off the wireless remote control function in time.	
8	• The hydraulic oil level is low.	✓ Add suitable hydraulic oil in time.	 The hydraulic oil level is low!
	• Damaged hydraulic oil level sensor.	✓ Replace the lubricating oil level sensor of the same type.	
	• Hydraulic oil level sensor wiring is loose or not connected.	✓ Check the wiring to make sure the hydraulic oil sensor wiring is correct and secure.	
10	• Engine speed is below 300rpm.	✓ It may be caused by too much external load. Please check carefully and find out the reason according to the actual situation on site.	Engine emergency stop!
11	• The engine has been running for 250 hours without maintenance.	✓ Please strictly abide by the requirements of the Caterpillar engine maintenance manual, and carry out comprehensive maintenance and	 Engine maintenance expired!

		maintenance of the engine on a regular basis.	
12	♦The speed wheel sensor wiring is loose or not connected.	✓Check the wiring to make sure the speed wheel sensor wiring is correct and secure.	 Main conveyor speed is low!
	♦Damaged speed wheel sensor.	✓Replace the speed sensor of the same specification.	
	♦The main belt conveys too much material or the main belt is stuck.	✓Remove material from the belt; check the main belt mechanics for the cause of the jam.	
	♦The speed pulley is not in proper contact with the belt.	✓Adjust the installation position of the speed pulley to ensure that the speed pulley is in correct contact with the belt.	
13	♦The host speed proximity switch is damaged.	✓Replace the speed proximity switch of the crusher of the same specification.	 The speed of the crusher is low!
	♦The crushing cavity is blocked or there are unbreakable materials.	✓Avoid over-sized materials or unbreakable materials from entering the crusher.	
	♦The speed proximity switch is not installed correctly and no signal is detected.	✓Adjust the installation position of the proximity switch, the detection distance should be less than 6mm.	
14	♦The crusher chamber is blocked or there are unbreakable materials.	✓Avoid over-sized materials or unbreakable materials from entering the crusher.	 The crushing host has high pressure!
	♦If there is no material in the chamber, there may be interference with the relevant mechanical structure, or there may be a problem with the hydraulic system.	✓Check the relevant mechanical or hydraulic parts to find out the cause.	
15	♦The oil replenishment pump is damaged and cannot replenish oil to the hydraulic circuit of the crusher, and the pressure is lower than 15Bar	✓Replace the charge pump as soon as possible.	 Crusher low pressure!
18	♦Hydraulic oil temperature is above critical setpoint.	✓Stop and wait until the hydraulic oil temperature drops. If this fault occurs	

		frequently, it must be checked mechanically or hydraulically. ✓ Check whether the hydraulic oil cooling fan is working normally and whether the filter element is blocked.	High hydraulic oil temperature shutdown!
19	• The hydraulic oil temperature is higher than the alarm set value.	✓ This alarm only prompts the operator and does not do any chain control. If it is found that the temperature rises too fast, you should pay attention and stop the machine to check the mechanical or hydraulic parts if necessary. ✓ Check whether the hydraulic oil cooling fan is working normally and whether the filter element is blocked.	 Hydraulic oil temperature high alarm!
20	• The wiring of the hydraulic oil temperature sensor is loose or not connected, and the analog signal is lower than 3.5mA.	✓ Check the sensor wiring to make sure it is correct and secure.	 Hydraulic oil temperature sensor open circuit!
	• The hydraulic oil temperature sensor is damaged, and the analog signal is lower than 3.5mA.	✓ Replace the temperature sensor with the same specification.	
21	• The hydraulic oil temperature sensor is damaged, and the analog signal is higher than 21mA.	✓ Replace the temperature sensor with the same specification.	 Hydraulic oil temperature sensor short circuit!
	• The hydraulic oil temperature sensor circuit is short-circuited, and the analog signal value is greater than 21mA.	✓ Check the wiring and replace the entire casting line if necessary.	
22	• There is a leak in the tie rod cylinder or in the connected oil line.	✓ Check the relevant hydraulic parts to make sure there are no oil leaks.	 Low lever pressure! (The troubleshooting of this fault also applies to fault code 25)
	• The plug of the filling solenoid valve is energized, and the valve core cannot be opened normally.	✓ Check the solenoid valve spool or the entire tie rod hydraulic valve block.	
	• The throttle valve of the tie rod valve block is not closed tightly.	✓ Close the throttle valve on the tie rod valve block as required.	

	♦The fill solenoid valve is not energized.	✓Check the wiring to make sure that the wiring is correct and secure, and that the cables are not damaged.	
24	♦The filling solenoid valve is de-energized and the spool cannot close properly.	✓ Check the solenoid valve spool or the entire tie rod hydraulic valve block.	 Tie rod pressure is high!
	♦The fill solenoid valve remains energized continuously.	✓ Make sure that the upper limit of the tie rod charging pressure is set correctly.	
	♦The hydraulic flow is too large, and too much oil enters the tie rod instantly.	✓Please check and adjust the hydraulic related parts.	
25	♦The throttle valve of the tie rod valve block is not closed tightly.	✓Close the throttle valve on the tie rod valve block as required.	Tie rod filling time out! (The troubleshooting of this fault also applies to fault code 22)
	♦The relief valve pressure setting of the tie rod valve block is incorrect.	✓Set the relief valve pressure of the tie rod valve block as required.	
	♦There is a leak in the tie rod cylinder or in the connected oil line.	✓Check the relevant hydraulic parts to make sure there are no oil leaks.	
	♦The hydraulic flow is too small, and too little oil enters the tie rod within the specified time.	✓Please check and adjust the hydraulic related parts.	
26	♦The bus plug of the node module is not connected or is loose.	✓Check the corresponding bus plug on the T-adapter to make sure the connection is normal and secure.	Node module 2 connection interrupted! (Refer to this section for troubleshooting other node module connection interruptions)
	♦Node module is not powered.	✓Check the wiring or power supply, the node module requires DC24V power supply.	
	♦The address of the node module is not set correctly.	✓ The node module address must be consistent with the electrical schematic.	
	♦A channel cable failure or input/output failure on the node module.	✓Remove the plugs in sequence from channels 1-8 of the node module, attempting to reset the fault each time one is removed.	
30	♦The engine ECU is not powered or the power supply is	✓Check the power wiring of the ECU to ensure that the	

	abnormal.	ECU plug is securely connected.	 J1939 connection interrupted!
	♦The communication wiring between the engine and the PLC is wrong or not connected.	✓Check the communication wiring to make sure that CAN2H and CAN2L are connected correctly.	
	♦The J1939 network terminating resistor is not connected or installed incorrectly.	✓When the power is off, check the resistance value on the network, it should be 60 ohms.	
35	♦The wired remote control and wireless remote control functions are selected and activated at the same time.	✓Wired remote and wireless remote cannot be used at the same time, you must deselect both remotes and re-enable a single remote.	 Do not use wired and wireless remote control at the same time!
36	♦VBBR power failure.	✓Check system voltage.	PLC failure!
	♦I/O failure.	✓Determine which channel is faulty based on the information from the PLC diagnostic interface, check and find out the cause of the fault.	

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